

A Living Review of Quantum Information Science in High Energy Physics

Pamela Pajarillo, So Chigusa, Jesse Thaler

*Center for Theoretical Physics - a Leinweber Institute, Massachusetts Institute of Technology
Cambridge, MA 02139, USA*

E-mail: `pampaja@mit.edu`, `schigusa@mit.edu`, `jthaler@mit.edu`

ABSTRACT: Inspired by “A Living Review of Machine Learning for Particle Physics”¹, the goal of this document is to provide a nearly comprehensive list of citations for those developing and applying quantum information approaches to experimental, phenomenological, or theoretical analyses. Applications of quantum information science to high energy physics is a relatively new field of research. As a living document, it will be updated as often as possible with the relevant literature with the latest developments. Suggestions are most welcome.

¹See <https://github.com/iml-wg/HEPML-LivingReview>.

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1 Introduction

The purpose of this note is to collect references for quantum information science as applied to particle and nuclear physics. The papers are listed in reverse chronological order. In order to be as useful as possible, this document will continually change. Please check back ² regularly. You can simply download the .bib file to get all of the latest references. Suggestions are most welcome.

2 Nuclear and Particle Physics in Quantum Information Science

2.1 Reviews and Whitepapers and Proceedings

2.1.1 Reviews

2.1.1.1 Lattice gauge theory simulations in the quantum information era [1]

- **Authors:** M. Dalmonte, S. Montangero
- **Posted on arXiv:** 11 February 2016
- **Published in Contemporary Physics 57 388 (2016):** 09 March 2016

2.1.1.2 Search for New Physics with Atoms and Molecules [2]

- **Authors:** M.S. Safronova, D. Budker, D. DeMille, Derek F. Jackson Kimball, A. Derevianko, C.W. Clark
- **Posted on arXiv:** 04 October 2017
- **Published in Reviews of Modern Physics, 90, 025008 (2018):** 30 June 2018

²See <https://github.com/PamelaPajarillo/NUPAQIS-LivingReview>.

2.1.1.3 Quantum Machine Learning in High Energy Physics [3]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

2.1.1.4 Quantum Computing Applications in Future Colliders [4]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

2.1.1.5 Review on Quantum Computing for Lattice Field Theory [5]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn
- **Posted on arXiv:** 01 February 2023
- **Published in nan:** 01 February 2023

2.1.1.6 Machine learning for anomaly detection in particle physics [6]

- **Authors:** Vasilis Belis, Patrick Odagiu, Thea Klaeboe Aarrestad
- **Posted on arXiv:** 20 December 2023
- **Published in Reviews in Physics, vol. 12, 2024, 100091:** 18 January 2024

2.1.1.7 Quantum entanglement and Bell inequality violation at colliders [7]

- **Authors:** Alan J. Barr, Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 12 February 2024
- **Published in Prog.Part.Nucl.Phys.:** 19 July 2024

2.1.2 Whitepapers and Proceedings

2.1.2.1 Quantum Sensing for High Energy Physics [8]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowering, Davide Braga, Karen Byrum, Gustavo Cencelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle,

James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O’Brien, Roger O’Brien, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi

- **Posted on arXiv:** 29 March 2018

2.1.2.2 Mechanical quantum sensing in the search for dark matter [9]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao

- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

2.1.2.3 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [10]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

2.1.2.4 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [11]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

2.1.2.5 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [12]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

2.1.2.6 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [13]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borrás, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

2.1.2.7 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.1.2.8 New Horizons: Scalar and Vector Ultralight Dark Matter [15]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz,

S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O’Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

2.1.2.9 Quantum Networks for High Energy Physics [16]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

2.1.2.10 Quantum Simulation for High-Energy Physics [17]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti

- **Posted on arXiv:** 07 April 2022

- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

2.1.2.11 Quantum computing hardware for HEP algorithms and sensing [18]

- **Authors:** M. Sohaib Alam, Sergey Belomestnykh, Nicholas Bornman, Gustavo Canceledo, Yu-Chiu Chao, Mattia Checchin, Vinh San Dinh, Anna Grassellino, Erik J. Gustafson, Roni Harnik, Corey Rae Harrington McRae, Ziwen Huang, Keshav Kapoor, Taeyoon Kim, James B. Kowalkowski, Matthew J. Kramer, Yulia Krasnikova, Prem Kumar, Doga Murat Kurkcuoglu, Henry Lamm, Adam L. Lyon, Despina Milathianaki, Akshay Murthy, Josh Mutus, Ivan Nekrashevich, JinSu Oh, A. BarışÖzgüler, Gabriel Nathan Perdue, Matthew Reagor, Alexander Romanenko, James A. Sauls, Leandro Stefanazzi, Norm M. Tubman, Davide Venturelli, Changqing Wang, Xinyuan You, David M.T. van Zanten, Lin Zhou, Shaojiang Zhu, Silvia Zorzetti

- **Posted on arXiv:** 18 April 2022

2.1.2.12 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [19]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

2.1.2.13 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.1.2.14 Quantum Sensors for High Energy Physics [21]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

2.2 Anomaly Detection

2.2.1 Quantum Computing Paradigms - Continuous Variable Quantum Computing

2.2.1.1 Unsupervised event classification with graphs on classical and photonic quantum computers [22]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

2.2.2 Quantum Computing Paradigms - Quantum Annealing

2.2.2.1 A Quantum Algorithm for Model-Independent Searches for New Physics [23]

- **Authors:** Konstantin T. Matchev, Prasanth Shyamsundar, Jordan Smolinsky
- **Posted on arXiv:** 04 March 2020
- **Published in LHEP:** 09 April 2023

2.2.3 Quantum Machine Learning - Supervised Methods

2.2.3.1 Unravelling physics beyond the standard model with classical and quantum anomaly detection [24]

- **Authors:** Julian Schuhmacher, Laura Boggia, Vasilis Belis, Ema Puljak, Michele Grossi, Maurizio Pierini, Sofia Vallecorsa, Francesco Tacchino, Panagiotis Barkoutsos, Ivano Tavernelli
- **Posted on arXiv:** 25 January 2023
- **Published in Mach.Learn.Sci.Tech.:** 16 November 2023

2.2.4 Quantum Machine Learning - Unsupervised Methods

2.2.4.1 Quantum anomaly detection in the latent space of proton collision events at the LHC [25]

- **Authors:** Vasilis Belis, Kinga Anna Woźniak, Ema Puljak, Panagiotis Barkoutsos, Günther Dissertori, Michele Grossi, Maurizio Pierini, Florentin Reiter, Ivano Tavernelli, Sofia Vallecorsa
- **Posted on arXiv:** 25 January 2023
- **Published in Commun.Phys.:** 14 October 2024

2.2.5 Quantum Machine Learning - Variational Quantum Algorithms

2.2.5.1 Anomaly detection in high-energy physics using a quantum autoencoder [26]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky, Michihisa Takeuchi
- **Posted on arXiv:** 09 December 2021
- **Published in Phys. Rev. D 105, 095004, 2022:** 01 May 2022

2.2.5.2 Quantum anomaly detection for collider physics [27]

- **Authors:** Sulaiman Alvi, Christian W. Bauer, Benjamin Nachman
- **Posted on arXiv:** 16 June 2022
- **Published in JHEP:** 22 February 2023

2.2.6 Crosslists

2.2.6.1 Machine learning for anomaly detection in particle physics [6]

- **Authors:** Vasilis Belis, Patrick Odagiu, Thea Klæboe Aarrestad
- **Posted on arXiv:** 20 December 2023
- **Published in Reviews in Physics, vol. 12, 2024, 100091:** 18 January 2024

2.3 Beyond the Standard Model

2.3.1 Quantum Algorithms - Grover's Search Algorithm

2.3.1.1 Application of a Quantum Search Algorithm to High- Energy Physics Data at the Large Hadron Collider [28]

- **Authors:** Anthony E. Armenakas, Oliver K. Baker
- **Posted on arXiv:** 01 October 2020

2.3.2 Quantum Entanglement and Bell Inequalities

2.3.2.1 Flavor patterns of fundamental particles from quantum entanglement? [29]

- **Authors:** Jesse Thaler, Sokratis Trifinopoulos
- **Posted on arXiv:** 30 October 2024
- **Published in Phys.Rev.D:** 01 March 2025

2.3.3 Quantum Machine Learning - Variational Quantum Algorithms

2.3.3.1 Hybrid Quantum-Classical Graph Convolutional Network [30]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 15 January 2021

2.3.4 Crosslists

2.3.4.1 Search for New Physics with Atoms and Molecules [2]

- **Authors:** M.S. Safronova, D. Budker, D. DeMille, Derek F. Jackson Kimball, A. Derevianko, C.W. Clark
- **Posted on arXiv:** 04 October 2017
- **Published in Reviews of Modern Physics, 90, 025008 (2018):** 30 June 2018

2.3.4.2 Event Classification with Quantum Machine Learning in High-Energy Physics [31]

- **Authors:** Koji Terashi, Michiru Kaneda, Tomoe Kishimoto, Masahiko Saito, Ryu Sawada, Junichi Tanaka
- **Posted on arXiv:** 23 February 2020
- **Published in Comput. Softw. Big Sci. 5, 2 (2021):** 03 January 2021

2.3.4.3 Quantum Machine Learning for Particle Physics using a Variational Quantum Classifier [32]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 14 October 2020
- **Published in JHEP:** 2021

2.3.4.4 Quantum convolutional neural networks for high energy physics data analysis [33]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

2.3.4.5 Quantum algorithm for the classification of supersymmetric top quark events [34]

- **Authors:** Pedrame Bargassa, Timothée Cabos, Samuele Cavinato, Artur Cordeiro Oudot Choi, Timothée Hessel
- **Posted on arXiv:** 31 May 2021
- **Published in Phys. Rev. D 104 (2021) 096004:** 01 November 2021

2.3.4.6 Unravelling physics beyond the standard model with classical and quantum anomaly detection [24]

- **Authors:** Julian Schuhmacher, Laura Boggia, Vasilis Belis, Ema Puljak, Michele Grossi, Maurizio Pierini, Sofia Vallecorsa, Francesco Tacchino, Panagiotis Barkoutsos, Ivano Tavernelli
- **Posted on arXiv:** 25 January 2023
- **Published in Mach.Learn.Sci.Tech.:** 16 November 2023

2.3.4.7 Quantum anomaly detection in the latent space of proton collision events at the LHC [25]

- **Authors:** Vasilis Belis, Kinga Anna Woźniak, Ema Puljak, Panagiotis Barkoutsos, Günther Dissertori, Michele Grossi, Maurizio Pierini, Florentin Reiter, Ivano Tavernelli, Sofia Vallecorsa
- **Posted on arXiv:** 25 January 2023
- **Published in Commun.Phys.:** 14 October 2024

2.3.4.8 Quantum Sensors for High Energy Physics [21]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

2.3.4.9 Quantum Metric Learning for New Physics Searches at the LHC [35]

- **Authors:** A. Hammad, Kyoungchul Kong, Myeonghun Park, Soyoung Shim
- **Posted on arXiv:** 28 November 2023

2.4 Dark Matter - Particle-like Dark Matter

2.4.1 Quantum Algorithms - Quantum Simulations

2.4.1.1 Quantum simulations of dark sector showers [36]

- **Authors:** So Chigusa, Masahito Yamazaki
- **Posted on arXiv:** 26 April 2022
- **Published in Phys.Lett.B:** 26 September 2022

2.4.2 Quantum Sensors - Atomic/Molecular/Nuclear Sensors

2.4.2.1 Trapped Electrons and Ions as Particle Detectors [37]

- **Authors:** Daniel Carney, Hartmut Häffner, David C. Moore, Jacob M. Taylor
- **Posted on arXiv:** 12 April 2021
- **Published in Phys Rev Lett, Vol. 127, No. 6, 2021. "Editor's suggestion":** 05 August 2021

2.4.2.2 Millicharged Dark Matter Detection with Ion Traps [38]

- **Authors:** Dmitry Budker, Peter W. Graham, Harikrishnan Ramani, Ferdinand Schmidt-Kaler, Christian Smorra, Stefan Ulmer
- **Posted on arXiv:** 11 August 2021
- **Published in PRX Quantum:** 01 February 2022

2.4.3 Quantum Sensors - Optomechanical Sensors

2.4.3.1 Search for Kilogram-scale Dark Matter with Precision Displacement Sensors [39]

- **Authors:** Akio Kawasaki
- **Posted on arXiv:** 26 August 2018
- **Published in Phys. Rev. D 99, 023005 (2019):** 07 January 2019

2.4.3.2 Proposal for gravitational direct detection of dark matter [40]

- **Authors:** Daniel Carney, Sohitri Ghosh, Gordan Krnjaic, Jacob M. Taylor
- **Posted on arXiv:** 01 March 2019
- **Published in Phys.Rev.D:** 14 October 2020

2.4.3.3 Search for composite dark matter with optically levitated sensors [41]

- **Authors:** Fernando Monteiro, Gadi Afek, Daniel Carney, Gordan Krnjaic, Jiaxiang Wang, David C. Moore
- **Posted on arXiv:** 23 July 2020
- **Published in Phys. Rev. Lett. 125, 181102 (2020):** 29 October 2020

2.4.3.4 Coherent Scattering of Low Mass Dark Matter from Optically Trapped Sensors [42]

- **Authors:** Gadi Afek, Daniel Carney, David C. Moore
- **Posted on arXiv:** 05 November 2021
- **Published in Phys.Rev.Lett.:** 09 March 2022

2.4.3.5 Models of ultraheavy dark matter visible to macroscopic mechanical sensing arrays [43]

- **Authors:** Carlos Blanco, Bahaa Elshimy, Rafael F. Lang, Robert Orlando
- **Posted on arXiv:** 29 December 2021
- **Published in Phys.Rev.D:** 01 June 2022

2.4.3.6 Optomechanical dark matter instrument for direct detection [44]

- **Authors:** Christopher G. Baker, Warwick P. Bowen, Peter Cox, Matthew J. Dolan, Maxim Goryachev, Glen Harris
- **Posted on arXiv:** 16 June 2023
- **Published in Phys. Rev. D 110, 043005 (2024):** 02 August 2024

2.4.3.7 Probing levitodynamics with multi-stochastic forces and the simple applications on the dark matter detection in optical levitation experiment [45]

- **Authors:** Xi Cheng, Ji-Heng Guo, Wenyu Wang, Bin Zhu
- **Posted on arXiv:** 07 December 2023
- **Published in Nucl. Phys. B 1010 (2025) 116780:** 19 December 2024

2.4.3.8 Dark Matter Searches with Levitated Sensors [46]

- **Authors:** Eva Kilian, Markus Rademacher, Jonathan M.H. Gosling, Julian H. Iacoponi, Fiona Alder, Marko Toroš, Antonio Pontin, Chamkaur Ghag, Sougato Bose, Tania S. Monteiro, P.F. Barker
- **Posted on arXiv:** 31 January 2024
- **Published in AVS Quantum Sci. 6, 030503 (2024):** 2024

2.4.3.9 Probing light particles with optically trapped sensors through nucleon scattering [47]

- **Authors:** Bhaskar Dutta, Dilip Kumar Ghosh, Sk Jeesun
- **Posted on arXiv:** 31 January 2025
- **Published in Phys.Rev.D:** 01 January 2026

2.4.3.10 Mechanical sensors for ultraheavy dark matter searches via long-range forces [48]

- **Authors:** Juehang Qin, Dorian W.P. Amaral, Sunil A. Bhave, Erqian Cai, Daniel Carney, Rafael F. Lang, Shengchao Li, Alberto M. Marino, Giacomo Marocco, Claire Marvinney, Jared R. Newton, Jacob M. Taylor, Christopher Tunnell
- **Posted on arXiv:** 14 March 2025
- **Published in Phys.Rev.D 112 (2025) 7, 072003:** 01 October 2025

2.4.4 Quantum Sensors - Solid State Sensors

2.4.4.1 Dark matter direct detection with quantum dots [49]

- **Authors:** Carlos Blanco, Rouven Essig, Marivi Fernandez-Serra, Harikrishnan Ramani, Oren Slone
- **Posted on arXiv:** 10 August 2022
- **Published in Phys.Rev.D:** 01 May 2023

2.4.4.2 Dark Matter Induced Power in Quantum Devices [50]

- **Authors:** Anirban Das, Noah Kurinsky, Rebecca K. Leane
- **Posted on arXiv:** 17 October 2022
- **Published in Phys. Rev. Lett. 132, 121801 (2024):** 22 March 2024

2.4.4.3 Transmon Qubit constraints on dark matter-nucleon scattering [51]

- **Authors:** Anirban Das, Noah Kurinsky, Rebecca K. Leane
- **Posted on arXiv:** 30 April 2024
- **Published in JHEP:** 25 July 2024

2.4.4.4 Quantum parity detectors: A qubit-based particle-detection scheme with meV thresholds for rare-event searches [52]

- **Authors:** Karthik Ramanathan, Brandon J. Sandoval, John E. Parker, Lalit M. Joshi, Andrew D. Beyer, Pierre M. Echtertnach, Serge Rosenblum, Sunil R. Golwala
- **Posted on arXiv:** 27 May 2024
- **Published in APS Open Science:** 01 May 2026

2.4.5 Crosslists

2.4.5.1 Quantum Sensing for High Energy Physics [8]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi
- **Posted on arXiv:** 29 March 2018

2.4.5.2 Mechanical quantum sensing in the search for dark matter [9]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhawe, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L.

Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao

- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol.** 6 024002, 2021 (invited): 18 January 2021

2.4.5.3 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [12]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

2.4.5.4 Quantum Networks for High Energy Physics [16]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie
- **Posted on arXiv:** 31 March 2022

2.4.5.5 Quantum Sensors for High Energy Physics [21]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cencelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

2.5 Dark Matter - Wave-like Dark Matter

2.5.1 Quantum Sensors - Atomic/Molecular/Nuclear Sensors

2.5.1.1 Intensity interferometry for ultralight bosonic dark matter detection [53]

- **Authors:** Hector Masia-Roig, Nataniel L. Figueroa, Ariday Bordon, Joseph A. Smiga, Yevgeny V. Stadnik, Dmitry Budker, Gary P. Centers, Alexander V. Gramolin, Paul S. Hamilton, Sami Khamis, Christopher A. Palm, Szymon Pustelny, Alexander O. Sushkov, Arne Wickenbrock, Derek F. Jackson Kimball

- **Posted on arXiv:** 05 February 2022

- **Published in Phys. Rev. D 108, 015003 (2023):** 01 July 2023

2.5.1.2 One-Electron Quantum Cyclotron as a Milli-eV Dark-Photon Detector [54]

- **Authors:** Xing Fan, Gerald Gabrielse, Peter W. Graham, Roni Harnik, Thomas G. Myers, Harikrishnan Ramani, Benedict A.D. Sukra, Samuel S.Y. Wong, Yawen Xiao

- **Posted on arXiv:** 12 August 2022

- **Published in Phys. Rev. Lett. 129, 261801 (2022):** 23 December 2022

2.5.1.3 Detection of bosonovae with quantum sensors on Earth and in space [55]

- **Authors:** Jason Arakawa, Joshua Eby, Marianna S. Safronova, Volodymyr Takhistov, Muhammad H. Zaheer

- **Posted on arXiv:** 28 June 2023

- **Published in Phys.Rev.D:** 01 October 2024

2.5.1.4 Rydberg-atom-based single-photon detection for haloscope axion searches [56]

- **Authors:** Eleanor Graham, Sumita Ghosh, Yuqi Zhu, Xiran Bai, Sidney B. Cahn, Elsa Durcan, Michael J. Jewell, Danielle H. Speller, Sabrina M. Zacarias, Laura T. Zhou, Reina H. Maruyama

- **Posted on arXiv:** 23 October 2023

- **Published in Phys.Rev.D:** 01 February 2024

2.5.1.5 Quantum entanglement of ions for light dark matter detection [57]

- **Authors:** Asuka Ito, Ryuichiro Kitano, Wakutaka Nakano, Ryoto Takai

- **Posted on arXiv:** 20 November 2023

- **Published in JHEP:** 16 February 2024

2.5.1.6 Nuclear interferometer for ultralight dark matter detection [58]

- **Authors:** Hannah Banks, Elina Fuchs, Matthew McCullough

- **Posted on arXiv:** 15 July 2024

- **Published in Phys. Rev. D, 2026:** 01 July 2026

2.5.1.7 Highly excited electron cyclotron for QCD axion and dark-photon detection [59]

- **Authors:** Xing Fan, Gerald Gabrielse, Peter W. Graham, Harikrishnan Ramani, Samuel S.Y. Wong, Yawen Xiao
- **Posted on arXiv:** 07 October 2024
- **Published in Phys.Rev.D 111 (2025) 7, 075022:** 01 April 2025

2.5.1.8 Eliminating Incoherent Noise: A Coherent Quantum Approach in Multi-Sensor Dark Matter Detection [60]

- **Authors:** Jing Shu, Bin Xu, Yuan Xu
- **Posted on arXiv:** 29 October 2024

2.5.1.9 Sensitively searching for microwave dark photons with atomic ensembles [61]

- **Authors:** Suirong He, De He, Yufen Li, Li Gao, Xianing Feng, Hao Zheng, L.F. Wei
- **Posted on arXiv:** 01 December 2024
- **Published in Phys.Rev.D:** 15 September 2025

2.5.1.10 Searches for exotic spin-dependent interactions with spin sensors [62]

- **Authors:** Min Jiang, Haowen Su, Yifan Chen, Man Jiao, Ying Huang, Yuanhong Wang, Xing Rong, Xinhua Peng, Jiangfeng Du
- **Posted on arXiv:** 04 December 2024
- **Published in Rep. Prog. Phys. 88 (2025) 016401:** 13 December 2024

2.5.1.11 Spin squeezing of macroscopic nuclear spin ensembles [63]

- **Authors:** Eric Boyers, Garry Goldstein, Alexander O. Sushkov
- **Posted on arXiv:** 19 February 2025
- **Published in Phys.Rev.D:** 01 March 2025

2.5.1.12 Detecting Dark Matter Using Optically Trapped Rydberg Atom Tweezer Arrays [64]

- **Authors:** So Chigusa, Taiyo Kasamaki, Toshi Kusano, Takeo Moroi, Kazunori Nakayama, Naoya Ozawa, Yoshiro Takahashi, Atsuhiko Umemoto, Amar Vutha
- **Posted on arXiv:** 17 July 2025
- **Published in Phys. Rev. Lett. 136, 151801 (2026):** 14 April 2026

2.5.1.13 On the Speed-up of Wave-like Dark Matter Searches with Entangled Qubits [65]

- **Authors:** Arushi Bodas, Sohriti Ghosh, Roni Harnik
- **Posted on arXiv:** 13 October 2025

2.5.1.14 Symmetric Dicke States as Optimal Probes for Wave-Like Dark Matter [66]

- **Authors:** Ping He, Jing Shu, Bin Xu, Jincheng Xu
- **Posted on arXiv:** 16 December 2025

2.5.2 Quantum Sensors - Optical/Photonic Sensors

2.5.2.1 Analysis of single-photon and linear amplifier detectors for microwave cavity dark matter axion searches [67]

- **Authors:** S.K. Lamoreaux, K.A. van Bibber, K.W. Lehnert, G. Carosi
- **Posted on arXiv:** 15 June 2013
- **Published in Phys.Rev.D:** 23 August 2013

2.5.2.2 Laser Interferometers as Dark Matter Detectors [68]

- **Authors:** Evan D. Hall, Rana X. Adhikari, Valery V. Frolov, Holger Müller, Maxim Pospelov, Rana X Adhikari
- **Posted on arXiv:** 03 May 2016
- **Published in Phys.Rev.D:** 23 October 2018

2.5.2.3 Accelerating dark-matter axion searches with quantum measurement technology [69]

- **Authors:** Huaixiu Zheng, Matti Silveri, R.T. Brierley, S.M. Girvin, K.W. Lehnert
- **Posted on arXiv:** 08 July 2016

2.5.2.4 Squeezed vacuum used to accelerate the search for a weak classical signal [70]

- **Authors:** M. Malnou, D.A. Palken, B.M. Brubaker, Leila R. Vale, Gene C. Hilton, K.W. Lehnert
- **Posted on arXiv:** 17 September 2018
- **Published in Phys.Rev.X:** 04 May 2019

2.5.2.5 A quantum-enhanced search for dark matter axions [71]

- **Authors:** K.M. Backes, D.A. Palken, S. Al Kenany, B.M. Brubaker, S.B. Cahn, A. Droster, Gene C. Hilton, Sumita Ghosh, H. Jackson, S.K. Lamoreaux, A.F. Leder, K.W. Lehnert, S.M. Lewis, M. Malnou, R.H. Maruyama, N.M. Rapidis, M. Simanovskaia, Sukhman Singh, D.H. Speller, I. Urdinaran, Leila R. Vale, E.C. van Assendelft, K. van Bibber, H. Wang
- **Posted on arXiv:** 04 August 2020
- **Published in Nature:** 10 February 2021

2.5.2.6 Searching for Dark Matter with a Superconducting Qubit [72]

- **Authors:** Akash V. Dixit, Srivatsan Chakram, Kevin He, Ankur Agrawal, Ravi K. Naik, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 28 August 2020
- **Published in Phys. Rev. Lett. 126, 141302 (2021):** 09 April 2021

2.5.2.7 Cavity Entanglement and State Swapping to Accelerate the Search for Axion Dark Matter [73]

- **Authors:** K. Wurtz, B.M. Brubaker, Y. Jiang, E.P. Ruddy, D.A. Palken, K.W. Lehnert
- **Posted on arXiv:** 08 July 2021
- **Published in PRX Quantum:** 10 December 2021

2.5.2.8 Entangled Sensor-Networks for Dark-Matter Searches [74]

- **Authors:** Anthony J. Brady, Christina Gao, Roni Harnik, Zhen Liu, Zheshen Zhang, Quntao Zhuang
- **Posted on arXiv:** 10 March 2022
- **Published in PRX Quantum 3, 030333 (2022):** 01 September 2022

2.5.2.9 Detecting Hidden Photon Dark Matter Using the Direct Excitation of Transmon Qubits [75]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 07 December 2022
- **Published in Phys.Rev.Lett.:** 22 November 2023

2.5.2.10 New results from HAYSTAC’s phase II operation with a squeezed state receiver [76]

- **Authors:** M.J. Jewell, A.F. Leder, K.M. Backes, Xiran Bai, K. van Bibber, B.M. Brubaker, S.B. Cahn, A. Droster, Maryam H. Esmat, Sumita Ghosh, Eleanor Graham, Gene C. Hilton, H. Jackson, Claire Laffan, S.K. Lamoreaux, K.W. Lehnert, S.M. Lewis, M. Malnou, R.H. Maruyama, D.A. Palken, N.M. Rapidis, E.P. Ruddy, M. Simanovskaia, Sukhman Singh, D.H. Speller, Leila R. Vale, H. Wang, Yuqi Zhu
- **Posted on arXiv:** 23 January 2023
- **Published in Phys.Rev.D:** 01 April 2023

2.5.2.11 Stimulated Emission of Signal Photons from Dark Matter Waves [77]

- **Authors:** Ankur Agrawal, Akash V. Dixit, Tanay Roy, Srivatsan Chakram, Kevin He, Ravi K. Naik, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 05 May 2023
- **Published in Phys. Rev. Lett. 132, 140801 (2024):** 03 April 2024

2.5.2.12 Quantum measurements in fundamental physics: a user’s manual [78]

- **Authors:** Jacob Beckey, Daniel Carney, Giacomo Marocco
- **Posted on arXiv:** 13 November 2023
- **Published in SciPost Phys. Rev. 2 (2026):** 2026

2.5.2.13 Quantum Enhancement in Dark Matter Detection with Quantum Computation [79]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 17 November 2023
- **Published in Phys.Rev.Lett.:** 08 July 2024

2.5.2.14 Quantum-Enhanced Sensing of Axion Dark Matter with a Transmon-Based Single Microwave Photon Counter [80]

- **Authors:** C. Braggio, L. Balembois, R. Di Vora, Z. Wang, J. Travesedo, L. Pallegoix, G. Carugno, A. Ortolan, G. Ruoso, U. Gambardella, D. D’Agostino, P. Bertet, E. Flurin
- **Posted on arXiv:** 04 March 2024
- **Published in Phys.Rev.X:** 01 April 2025

2.5.2.15 Search for QCD axion dark matter with transmon qubits and quantum circuit [81]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 29 July 2024
- **Published in Phys.Rev.D:** 01 December 2024

2.5.2.16 Axion bounds from quantum technology [82]

- **Authors:** Martin Bauer, Sreemanti Chakraborti, Guillaume Rostagni
- **Posted on arXiv:** 12 August 2024
- **Published in JHEP:** 05 May 2025

2.5.2.17 Quantum-enhanced dark matter detection with in-cavity control: mitigating the Rayleigh curse [83]

- **Authors:** Haowei Shi, Anthony J. Brady, Wojciech Górecki, Lorenzo Maccone, Roberto Di Candia, Quntao Zhuang
- **Posted on arXiv:** 06 September 2024
- **Published in npj Quantum Information 11, 48 (2025):** 17 March 2025

2.5.2.18 Transmon qubit modeling and characterization for Dark Matter search [84]

- **Authors:** Roberto Moretti, Danilo Labranca, Pietro Campana, Rodolfo Carobene, Marco Gobbo, Manuel A. Castellanos-Beltran, David Olaya, Peter F. Hopkins, Leonardo Banchi, Matteo Borghesi, Alessandro Candido, Stefano Carrazza, Hervé Atsè Corti, Alessandro D’Elia, Marco Faverzani, Elena Ferri, Angelo Nucciotti, Luca Origo, Andrea Pasquale, Alex Stephane Piedjou Komnang, Alessio Rettaroli, Simone Tocci, Claudio Gatti, Andrea Giachero
- **Posted on arXiv:** 09 September 2024
- **Published in IEEE Trans.Quantum Eng.:** 2026

2.5.2.19 Dark Matter Axion Search with HAYSTAC Phase II [85]

- **Authors:** Xiran Bai, M.J. Jewell, J. Echevers, K. van Bibber, A. Droster, Maryam H. Esmat, Sumita Ghosh, Eleanor Graham, H. Jackson, Claire Laffan, S.K. Lamoreaux, A.F. Leder, K.W. Lehnert, S.M. Lewis, R.H. Maruyama, R.D. Nath, N.M. Rapidis, E.P. Ruddy, M. Silva-Feaver, M. Simanovskaia, Sukhman Singh, D.H. Speller, Sabrina Zacarias, Yuqi Zhu
- **Posted on arXiv:** 13 September 2024
- **Published in Phys. Rev. Lett. 134, 151006 (2025):** 18 April 2025

2.5.2.20 Flux-Tunable Cavity for Dark Matter Detection [86]

- **Authors:** Fang Zhao, Ziqian Li, Akash V. Dixit, Tanay Roy, Andrei Vrajitoarea, Riju Banerjee, Alexander Anferov, Kan-Heng Lee, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 12 January 2025
- **Published in Phys.Rev.Lett.:** 13 November 2025

2.5.2.21 Quantum Enhanced Dark-Matter Search with Entangled Fock States in High-Quality Cavities [87]

- **Authors:** Benjamin Freiman, Xinyuan You, Andy C.Y. Li, Raphael Cervantes, Taeyoon Kim, Anna Grasselino, Roni Harnik, Yao Lu
- **Posted on arXiv:** 30 October 2025

2.5.2.22 Quantum-Error-Correction-Inspired Noise Mitigation for Wavelike Dark Matter Searches with Quantum Sensors [88]

- **Authors:** Hajime Fukuda, Takeo Moroi, Thanaporn Sichanugrist
- **Posted on arXiv:** 05 November 2025
- **Published in Phys.Rev.Lett.:** 08 July 2026

2.5.3 Quantum Sensors - Optomechanical Sensors

2.5.3.1 Sound of Dark Matter: Searching for Light Scalars with Resonant-Mass Detectors [89]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Ken Van Tilburg
- **Posted on arXiv:** 07 August 2015
- **Published in Phys.Rev.Lett.:** 22 January 2016

2.5.3.2 Dark Matter Direct Detection with Accelerometers [90]

- **Authors:** Peter W. Graham, David E. Kaplan, Jeremy Mardon, Surjeet Rajendran, William A. Terrano
- **Posted on arXiv:** 18 December 2015
- **Published in Phys.Rev.D:** 20 April 2016

2.5.3.3 Ultralight dark matter detection with mechanical quantum sensors [91]

- **Authors:** Daniel Carney, Anson Hook, Zhen Liu, Jacob M. Taylor, Yue Zhao
- **Posted on arXiv:** 13 August 2019
- **Published in New Journal of Physics, Volume 23, February 2021:** 02 March 2021

2.5.3.4 Searching for scalar dark matter with compact mechanical resonators [92]

- **Authors:** Jack Manley, Russell Stump, Dalziel Wilson, Daniel Grin, Swati Singh
- **Posted on arXiv:** 16 October 2019
- **Published in Phys. Rev. Lett. 124, 151301 (2020):** 17 April 2020

2.5.3.5 Ultrasensitive optomechanical detection of an axion-mediated force based on a sharp peak emerging in probe absorption spectrum [93]

- **Authors:** Lei Chen, Jian Liu, Kadi Zhu
- **Posted on arXiv:** 28 November 2019
- **Published in Opt.Communic.: 15 January 2021**

2.5.3.6 Searching for vector dark matter with an optomechanical accelerometer [94]

- **Authors:** Jack Manley, Mitul Dey Chowdhury, Daniel Grin, Swati Singh, Dalziel J. Wilson
- **Posted on arXiv:** 09 July 2020
- **Published in Phys. Rev. Lett. 126, 061301 (2021):** 11 February 2021

2.5.3.7 Entanglement-enhanced optomechanical sensor array with application to dark matter searches [95]

- **Authors:** Anthony J. Brady, Xin Chen, Kewen Xiao, Yi Xia, Jack Manley, Mitul Dey Chowdhury, Zhen Liu, Roni Harnik, Dalziel J. Wilson, Zheshen Zhang, Quntao Zhuang
- **Posted on arXiv:** 13 October 2022
- **Published in Commun.Phys.: 01 September 2023**

2.5.3.8 Axion detection with optomechanical cavities [96]

- **Authors:** Clara Murgui, Yikun Wang, Kathryn M. Zurek
- **Posted on arXiv:** 15 November 2022
- **Published in Phys.Lett.B: 27 June 2024**

2.5.3.9 Superfluid helium ultralight dark matter detector [97]

- **Authors:** M. Hirschel, V. Vadakkumbatt, N.P. Baker, F.M. Schweizer, J.C. Sankey, S. Singh, J.P. Davis
- **Posted on arXiv:** 14 September 2023
- **Published in Physical Review D 109, 095011 (2024):** 01 May 2024

2.5.3.10 Vector wave dark matter and terrestrial quantum sensors [98]

- **Authors:** Dorian W.P. Amaral, Mudit Jain, Mustafa A. Amin, Christopher Tunnell
- **Posted on arXiv:** 04 March 2024
- **Published in JCAP:** 21 June 2024

2.5.4 Quantum Sensors - Solid State Sensors

2.5.4.1 Searching for an exotic spin-dependent interaction with a single electron-spin quantum sensor [99]

- **Authors:** Xing Rong, Mengqi Wang, Jianpei Geng, Xi Qin, Maosen Guo, Man Jiao, Yijin Xie, Pengfei Wang, Pu Huang, Fazhan Shi, Yi-Fu Cai, Chongwen Zou, Jiangfeng Du
- **Posted on arXiv:** 12 June 2017
- **Published in Nature Commun.:** 21 February 2018

2.5.4.2 Axion search with quantum nondemolition detection of magnons [100]

- **Authors:** Tomonori Ikeda, Asuka Ito, Kentaro Miuchi, Jiro Soda, Hisaya Kurashige, Yutaka Shikano
- **Posted on arXiv:** 17 February 2021
- **Published in Phys. Rev. D 105, 102004 (2022):** 15 May 2022

2.5.4.3 Proposal for the search for new spin interactions at the micrometer scale using diamond quantum sensors [101]

- **Authors:** P.-H. Chu, N. Ristoff, J. Smits, N. Jackson, Y.J. Kim, I. Savukov, V.M. Acosta
- **Posted on arXiv:** 29 December 2021
- **Published in Physical Review Research 4, 023162 (2022):** 31 May 2022

2.5.4.4 Ultimate precision limit of noise sensing and dark matter search [102]

- **Authors:** Haowei Shi, Quntao Zhuang
- **Posted on arXiv:** 29 August 2022
- **Published in npj Quantum Inf.:** 20 March 2023

2.5.4.5 Light dark matter search with nitrogen-vacancy centers in diamonds [103]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 24 February 2023
- **Published in JHEP:** 12 March 2025

2.5.4.6 Axion detection via superfluid ^3He ferromagnetic phase and quantum measurement techniques [104]

- **Authors:** So Chigusa, Dan Kondo, Hitoshi Murayama, Risshin Okabe, Hiroyuki Sudo
- **Posted on arXiv:** 17 September 2023
- **Published in JHEP:** 26 September 2024

2.5.4.7 Nuclear spin metrology with nitrogen vacancy center in diamond for axion dark matter detection [105]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Yuichiro Matsuzaki, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 09 July 2024
- **Published in Phys. Rev. D 111 (2025) 075028:** 01 April 2025

2.5.4.8 Listening for new physics with quantum acoustics [106]

- **Authors:** Ryan Linehan, Tanner Trickle, Christopher R. Conner, Sohriti Ghosh, Tongyan Lin, Mukul Sholapurkar, Andrew N. Cleland
- **Posted on arXiv:** 22 October 2024
- **Published in Phys.Rev.D:** 01 December 2025

2.5.4.9 Entanglement-enhanced ac magnetometry in the presence of Markovian noise [107]

- **Authors:** Thanaporn Sichanugrist, Hajime Fukuda, Takeo Moroi, Kazunori Nakayama, So Chigusa, Norikazu Mizuochi, Masashi Hazumi, Yuichiro Matsuzaki
- **Posted on arXiv:** 28 October 2024
- **Published in Phys.Rev.A:** 07 April 2025

2.5.4.10 Probing Sub-eV Dark Photon, Scalar and Axion-like Particle Dark Matters with Transmon Qubits [108]

- **Authors:** Wei Chao, Yu Gao, Ming-Jie Jin, Xiao-Sheng Liu, Xi-Lei Sun
- **Posted on arXiv:** 30 December 2024
- **Published in Phys.Dark Univ. 50 (2025) 102171:** 15 November 2025

2.5.4.11 Piezoelectric bulk acoustic resonators for dark photon detection [109]

- **Authors:** Tanner Trickle
- **Posted on arXiv:** 09 January 2025
- **Published in Phys.Rev.D:** 01 September 2025

2.5.4.12 Background suppression in quantum sensing of dark matter via collective entangled-state projection [110]

- **Authors:** Shion Chen, Hajime Fukuda, Yutaro Iiyama, Yuya Mino, Takeo Moroi, Mikio Nakahara, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 02 October 2025
- **Published in Phys.Rev.D:** 01 March 2026

2.5.4.13 Hybrid-spin decoupling for noise-resilient DC quantum sensing [111]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Yuichiro Matsuzaki, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 20 November 2025

2.5.5 Crosslists

2.5.5.1 Search for New Physics with Atoms and Molecules [2]

- **Authors:** M.S. Safronova, D. Budker, D. DeMille, Derek F. Jackson Kimball, A. Derevianko, C.W. Clark
- **Posted on arXiv:** 04 October 2017
- **Published in Reviews of Modern Physics, 90, 025008 (2018):** 30 June 2018

2.5.5.2 Quantum Sensing for High Energy Physics [8]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowering, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi
- **Posted on arXiv:** 29 March 2018

2.5.5.3 Quantum sensor networks as exotic field telescopes for multi-messenger astronomy [112]

- **Authors:** Conner Dailey, Colin Bradley, Derek F. Jackson Kimball, Ibrahim A. Sulai, Szymon Pustelny, Arne Wickenbrock, Andrei Derevianko
- **Posted on arXiv:** 11 February 2020
- **Published in Nature Astron.:** 02 November 2020

2.5.5.4 Mechanical quantum sensing in the search for dark matter [9]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao

- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

2.5.5.5 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [12]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

2.5.5.6 New Horizons: Scalar and Vector Ultralight Dark Matter [15]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibbrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O’Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou
- **Posted on arXiv:** 28 March 2022

2.5.5.7 Quantum Networks for High Energy Physics [16]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

2.5.5.8 Electromagnetic cavities as mechanical bars for gravitational waves [113]

- **Authors:** Asher Berlin, Diego Blas, Raffaele Tito D’Agnolo, Sebastian A.R. Ellis, Roni Harnik, Yonatan Kahn, Jan Schütte-Engel, Michael Wentzel
- **Posted on arXiv:** 02 March 2023
- **Published in Phys.Rev.D:** 15 October 2023

2.5.5.9 Quantum sensing for particle physics [114]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

2.5.5.10 Quantum Sensors for High Energy Physics [21]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Canelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

2.5.5.11 Superradiant interactions of the cosmic neutrino background, axions, dark matter, and reactor neutrinos [115]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Marios Galanis
- **Posted on arXiv:** 07 August 2024
- **Published in Phys.Rev.D:** 01 March 2025

2.5.5.12 Multimessenger astronomy beyond the Standard Model: New window from quantum sensors [116]

- **Authors:** Jason Arakawa, Muhammad H. Zaheer, Volodymyr Takhistov, Marianna S. Safronova, Joshua Eby, Charles Cheung
- **Posted on arXiv:** 12 February 2025
- **Published in JCAP:** 10 February 2026

2.5.5.13 Superradiant interactions for relic detection with entangled nuclear spins [117]

- **Authors:** Marios Galanis, Onur Hosten, Asimina Arvanitaki, Savas Dimopoulos
- **Posted on arXiv:** 28 August 2025
- **Published in Phys.Rev.A:** 10 June 2026

2.6 Detector Technologies and Simulations

2.6.1 Quantum Computing Paradigms - Continuous Variable Quantum Computing

2.6.1.1 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [118]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

2.6.2 Quantum Computing Paradigms - Quantum Annealing

2.6.2.1 Unfolding measurement distributions via quantum annealing [119]

- **Authors:** Kyle Cormier, Riccardo Di Sipio, Peter Wittek
- **Posted on arXiv:** 22 August 2019
- **Published in JHEP:** 22 November 2019

2.6.3 Quantum Machine Learning - Unsupervised Methods

2.6.3.1 Dual-Parameterized Quantum Circuit GAN Model in High Energy Physics [120]

- **Authors:** Su Yeon Chang, Steven Herbert, Sofia Vallecorsa, Elías F. Combarro, Ross Duncan
- **Posted on arXiv:** 29 March 2021
- **Published in EPJ Web Conf.:** 2021

2.6.3.2 Running the Dual-PQC GAN on noisy simulators and real quantum hardware [121]

- **Authors:** Su Yeon Chang, Edwin Agnew, Elías F. Combarro, Michele Grossi, Steven Herbert, Sofia Vallecorsa
- **Posted on arXiv:** 30 May 2022
- **Published in nan:** 2023

2.6.3.3 Precise image generation on current noisy quantum computing devices [122]

- **Authors:** Florian Rehm, Sofia Vallecorsa, Kerstin Borrás, Dirk Krücker, Michele Grossi, Valle Varo
- **Posted on arXiv:** 11 July 2023
- **Published in IOP Quantum Science and Technology (October 2023):** 30 October 2023

2.6.4 Quantum Sensors - Atomic/Molecular/Nuclear Sensors

2.6.4.1 Quantum Systems for Enhanced High Energy Particle Physics Detectors [123]

- **Authors:** M. Doser, E. Auffray, F.M. Brunbauer, I. Frank, H. Hillemanns, G. Orlandini, G. Kornakov
- **Published in nan:** 24 June 2022

2.6.4.2 Quantum sensing for particle physics [114]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

2.6.5 Crosslists

2.6.5.1 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.6.5.2 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borrás, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel

J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.6.5.3 Quantum-centric Supercomputing for Physics Research [124]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

2.7 Effective Field Theories

2.7.1 Quantum Algorithms - Quantum Simulations

2.7.1.1 Simulating Collider Physics on Quantum Computers Using Effective Field Theories [125]

- **Authors:** Christian W. Bauer, Marat Freytsis, Benjamin Nachman
- **Posted on arXiv:** 09 February 2021
- **Published in Phys.Rev.Lett.:** 18 November 2021

2.7.2 Quantum Entanglement and Bell Inequalities

2.7.2.1 Minimal entanglement and emergent symmetries in low-energy QCD [126]

- **Authors:** Qiaofeng Liu, Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 October 2022
- **Published in Phys.Rev.C 107, 025204 (2023):** 14 February 2023

2.7.3 Crosslists

2.7.3.1 Cloud Quantum Computing of an Atomic Nucleus [127]

- **Authors:** E.F. Dumitrescu, A.J. McCaskey, G. Hagen, G.R. Jansen, T.D. Morris, T. Papanbrock, R.C. Pooser, D.J. Dean, P. Lougovski
- **Posted on arXiv:** 11 January 2018
- **Published in Phys. Rev. Lett. 120, 210501 (2018):** 24 May 2018

2.7.3.2 Quantum Computing for Neutrino-Nucleus Scattering [128]

- **Authors:** Alessandro Roggero, Andy C.Y. Li, Joseph Carlson, Rajan Gupta, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019
- **Published in Phys. Rev. D** **101**, 074038 (2020): 28 April 2020

2.7.3.3 Improving Hamiltonian encodings with the Gray code [129]

- **Authors:** Olivia Di Matteo, Anna McCoy, Peter Gysbers, Takayuki Miyagi, R.M. Woloshyn, Petr Navrátil
- **Posted on arXiv:** 11 August 2020
- **Published in Phys. Rev. A** **103**, 042405 (2021): 03 April 2021

2.7.3.4 Quantum SMEFT tomography: Top quark pair production at the LHC [130]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 10 March 2022
- **Published in Phys.Rev.D:** 01 September 2022

2.7.3.5 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [11]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

2.7.3.6 Quantum Simulation for High-Energy Physics [17]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum** **4**, 027001, 2023: 01 May 2023

2.7.3.7 Subatomic Many-Body Physics Simulations on a Quantum Frequency Processor [131]

- **Authors:** Hsuan-Hao Lu, Natalie Klco, Joseph M. Lukens, Titus D. Morris, Aaina Bansal, Andreas Ekström, Gaute Hagen, Thomas Papenbrock, Andrew M. Weiner, Martin J. Savage, Pavel Lougovski
- **Published in nan:** 2019

2.8 Event Classification

2.8.1 Quantum Algorithms - Grover's Search Algorithm

2.8.1.1 Implementation and analysis of quantum computing application to Higgs boson reconstruction at the large Hadron Collider [132]

- **Authors:** Anthony Alexiades Armenakas, Oliver K. Baker
- **Published in Sci.Rep.:** 24 November 2021

2.8.2 Quantum Computing Paradigms - Continuous Variable Quantum Computing

2.8.2.1 Continuous-variable photonic quantum extreme learning machines for fast collider-data selection [133]

- **Authors:** Benedikt Maier, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 15 October 2025
- **Published in EPJ Quant.Technol.:** 27 April 2026

2.8.3 Quantum Computing Paradigms - Quantum Annealing

2.8.3.1 Solving a Higgs optimization problem with quantum annealing for machine learning [134]

- **Authors:** Alex Mott, Joshua Job, Jean Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Published in Nature:** 18 October 2017

2.8.3.2 Quantum adiabatic machine learning by zooming into a region of the energy surface [135]

- **Authors:** Alexander Zlokapa, Alex Mott, Joshua Job, Jean-Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 13 August 2019
- **Published in Phys. Rev. A 102, 062405 (2020):** 05 December 2020

2.8.3.3 Restricted Boltzmann Machines for galaxy morphology classification with a quantum annealer [136]

- **Authors:** João Caldeira, Joshua Job, Steven H. Adachi, Brian Nord, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019

2.8.3.4 Quantum algorithm for the classification of supersymmetric top quark events [34]

- **Authors:** Pedrame Bargassa, Timothée Cabos, Samuele Cavinato, Artur Cordeiro Oudot Choi, Timothée Hessel
- **Posted on arXiv:** 31 May 2021
- **Published in Phys. Rev. D 104 (2021) 096004:** 01 November 2021

2.8.3.5 Completely quantum neural networks [137]

- **Authors:** Steve Abel, Juan C. Criado, Michael Spannowsky
- **Posted on arXiv:** 23 February 2022
- **Published in Phys.Rev.A:** 01 August 2022

2.8.4 Quantum Machine Learning - Supervised Methods

2.8.4.1 Quantum Support Vector Machines for Continuum Suppression in B Meson Decays [138]

- **Authors:** Jamie Heredge, Charles Hill, Lloyd Hollenberg, Martin Sevier
- **Posted on arXiv:** 22 March 2021
- **Published in Comput.Softw.Big Sci.:** 30 November 2021

2.8.4.2 Application of quantum machine learning using the quantum kernel algorithm on high energy physics analysis at the LHC [139]

- **Authors:** Sau Lan Wu, Shaojun Sun, Wen Guan, Chen Zhou, Jay Chan, Chi Lung Cheng, Tuan Pham, Yan Qian, Alex Zeng Wang, Rui Zhang, Miron Livny, Jennifer Glick, Panagiotis Kl. Barkoutsos, Stefan Woerner, Ivano Tavernelli, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 11 April 2021
- **Published in Phys. Rev. Research 3, 033221 (2021):** 08 September 2021

2.8.4.3 Higgs analysis with quantum classifiers [140]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in nan:** 2021

2.8.4.4 Enforcing exact permutation and rotational symmetries in the application of quantum neural networks on point cloud datasets [141]

- **Authors:** Zhelun Li, Lento Nagano, Koji Terashi
- **Posted on arXiv:** 17 May 2024
- **Published in Phys.Rev.Res.:** 10 October 2024

2.8.4.5 From Reachability to Learnability: Geometric Design Principles for Quantum Neural Networks [142]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky
- **Posted on arXiv:** 03 March 2026

2.8.5 Quantum Machine Learning - Unsupervised Methods

2.8.5.1 Guided graph compression for quantum graph neural networks [143]

- **Authors:** Mikel Casals, Vasilis Belis, Elias F. Combarro, Eduard Alarcón, Sofia Vallecorsa, Michele Grossi
- **Posted on arXiv:** 11 June 2025
- **Published in Mach.Learn.Sci.Tech.:** 12 September 2025

2.8.6 Quantum Machine Learning - Variational Quantum Algorithms

2.8.6.1 Event Classification with Quantum Machine Learning in High-Energy Physics [31]

- **Authors:** Koji Terashi, Michiru Kaneda, Tomoe Kishimoto, Masahiko Saito, Ryu Sawada, Junichi Tanaka
- **Posted on arXiv:** 23 February 2020
- **Published in Comput. Softw. Big Sci. 5, 2 (2021):** 03 January 2021

2.8.6.2 Quantum Machine Learning for Particle Physics using a Variational Quantum Classifier [32]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 14 October 2020
- **Published in JHEP:** 2021

2.8.6.3 Application of quantum machine learning using the quantum variational classifier method to high energy physics analysis at the LHC on IBM quantum computer simulator and hardware with 10 qubits [144]

- **Authors:** Sau Lan Wu, Jay Chan, Wen Guan, Shaojun Sun, Alex Wang, Chen Zhou, Miron Livny, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 21 December 2020
- **Published in J.Phys.G:** 26 October 2021

2.8.6.4 Hybrid actor-critic algorithm for quantum reinforcement learning at CERN beam lines [145]

- **Authors:** Michael Schenk, Elías F. Combarro, Michele Grossi, Verena Kain, Kevin Shing Bruce Li, Mircea-Marian Popa, Sofia Vallecorsa
- **Posted on arXiv:** 22 September 2022
- **Published in Quantum Sci.Technol.:** 21 February 2024

2.8.6.5 Fitting a collider in a quantum computer: tackling the challenges of quantum machine learning for big datasets [146]

- **Authors:** Miguel Caçador Peixoto, Nuno Filipe Castro, Miguel Crispim Romão, Maria Gabriela Jordão Oliveira, Inês Ochoa
- **Posted on arXiv:** 06 November 2022
- **Published in Front. Artif. Intell. 6 (2023) 1268852:** 20 November 2023

2.8.6.6 Quantum Metric Learning for New Physics Searches at the LHC [35]

- **Authors:** A. Hammad, Kyoungchul Kong, Myeonghun Park, Soyoung Shim
- **Posted on arXiv:** 28 November 2023

2.8.6.7 Hybrid Quantum Vision Transformers for Event Classification in High Energy Physics [147]

- **Authors:** Eyup B. Unlu, Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhongtian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva
- **Posted on arXiv:** 01 February 2024
- **Published in Axioms v. 13, no 3, (2024) 187:** 13 March 2024

2.8.6.8 Quantum Vision Transformers for Quark–Gluon Classification [148]

- **Authors:** Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhongtian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva, Eyup B. Unlu
- **Posted on arXiv:** 16 May 2024
- **Published in Axioms 2024, 13(5), 323:** 13 May 2024

2.8.7 Crosslists

2.8.7.1 Quantum Machine Learning in High Energy Physics [3]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

2.8.7.2 Unsupervised event classification with graphs on classical and photonic quantum computers [22]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

2.8.7.3 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.8.7.4 Quantum Computing Applications in Future Colliders [4]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

2.8.7.5 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.8.7.6 Quantum-centric Supercomputing for Physics Research [124]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

2.9 Event Generation

2.9.1 Quantum Algorithms - Grover's Search Algorithm

2.9.1.1 Quantum algorithm for Feynman loop integrals [149]

- **Authors:** Selomit Ramírez-Uribe, Andr  s E. Renter  a-Olivo, Germ  n Rodrigo, German F.R. Sborlini, Luiz Vale Silva
- **Posted on arXiv:** 18 May 2021
- **Published in JHEP:** 16 May 2022

2.9.1.2 Loop Feynman integration on a quantum computer [150]

- **Authors:** Jorge J. Mart  nez de Lejarza, Leandro Cieri, Michele Grossi, Sofia Vallecorsa, Germ  n Rodrigo
- **Posted on arXiv:** 05 January 2024
- **Published in Phys. Rev. D 110, 074031 (2024):** 01 October 2024

2.9.2 Quantum Algorithms - Quantum Simulations

2.9.2.1 Quantum Algorithm for High Energy Physics Simulations [151]

- **Authors:** Christian W. Bauer, Wibe A. de Jong, Benjamin Nachman, Davide Provasoli
- **Posted on arXiv:** 05 April 2019
- **Published in Phys. Rev. Lett. 126, 062001 (2021):** 11 February 2021

2.9.2.2 Towards a quantum computing algorithm for helicity amplitudes and parton showers [152]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 13 October 2020
- **Published in Phys. Rev. D 103, 076020 (2021):** 27 April 2021

2.9.2.3 Improving quantum simulation efficiency of final state radiation with dynamic quantum circuits [153]

- **Authors:** Plato Deliyannis, James Sud, Diana Chamaki, Zoë Webb-Mack, Christian W. Bauer, Benjamin Nachman
- **Posted on arXiv:** 18 March 2022
- **Published in Physical Review D 106, 036007 (2022):** 01 August 2022

2.9.2.4 Collider events on a quantum computer [154]

- **Authors:** Gösta Gustafson, Stefan Prestel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 21 July 2022
- **Published in JHEP:** 07 November 2022

2.9.2.5 Quantum parton shower with kinematics [155]

- **Authors:** Christian W. Bauer, So Chigusa, Masahito Yamazaki
- **Posted on arXiv:** 30 October 2023
- **Published in Phys.Rev.A:** 26 March 2024

2.9.3 Quantum Algorithms - Quantum Walks

2.9.3.1 Quantum walk approach to simulating parton showers [156]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 28 September 2021
- **Published in Phys. Rev. D 106 (2022) 056002:** 01 September 2022

2.9.4 Quantum Machine Learning - Unsupervised Methods

2.9.4.1 Style-based quantum generative adversarial networks for Monte Carlo events [157]

- **Authors:** Carlos Bravo-Prieto, Julien Baglio, Marco Cè, Anthony Francis, Dorota M. Grabowska, Stefano Carrazza
- **Posted on arXiv:** 13 October 2021
- **Published in Quantum 6, 777 (2022):** 17 August 2022

2.9.4.2 Quantum integration of elementary particle processes [158]

- **Authors:** Gabriele Agliardi, Michele Grossi, Mathieu Pellen, Enrico Prati
- **Posted on arXiv:** 05 January 2022
- **Published in Phys.Lett.B:** 07 June 2022

2.9.4.3 Generative invertible quantum neural networks [159]

- **Authors:** Armand Rousselot, Michael Spannowsky
- **Posted on arXiv:** 24 February 2023
- **Published in SciPost Phys. 16, 146 (2024):** 04 June 2024

2.9.4.4 Quantum-centric Supercomputing for Physics Research [124]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

2.9.5 Quantum Machine Learning - Variational Quantum Algorithms

2.9.5.1 Partonic collinear structure by quantum computing [160]

- **Authors:** Tianyin Li, Xingyu Guo, Wai Kin Lai, Xiaohui Liu, Enke Wang, Hongxi Xing, Dan-Bo Zhang, Shi-Liang Zhu
- **Posted on arXiv:** 07 June 2021
- **Published in Phys.Rev.D:** 01 June 2022

2.9.5.2 Unsupervised quantum circuit learning in high energy physics [161]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton
- **Posted on arXiv:** 07 March 2022
- **Published in Phys.Rev.D:** 01 November 2022

2.9.5.3 Quantum Wishlist: Lessons from Parton Showers [162]

- **Authors:** Masahito Yamazaki
- **Posted on arXiv:** 25 February 2025
- **Published in nan:** 24 October 2025

2.9.6 Crosslists

2.9.6.1 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.9.6.2 Quantum Simulation for High-Energy Physics [17]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

2.9.6.3 Quantum Computing Applications in Future Colliders [4]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

2.9.6.4 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [19]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

2.9.6.5 Quantum data learning for quantum simulations in high-energy physics [163]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

2.9.6.6 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.9.6.7 Quantum Algorithms in Particle Physics [164]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

2.10 Gravitational Waves

2.10.1 Quantum Sensors - Optical/Photonic Sensors

2.10.1.1 Quantum sensor networks as exotic field telescopes for multi-messenger astronomy [112]

- **Authors:** Conner Dailey, Colin Bradley, Derek F. Jackson Kimball, Ibrahim A. Sulai, Szymon Pustelny, Arne Wickenbrock, Andrei Derevianko
- **Posted on arXiv:** 11 February 2020
- **Published in Nature Astron.:** 02 November 2020

2.10.1.2 Electromagnetic cavities as mechanical bars for gravitational waves [113]

- **Authors:** Asher Berlin, Diego Blas, Raffaele Tito D’Agnolo, Sebastian A.R. Ellis, Roni Harnik, Yonatan Kahn, Jan Schütte-Engel, Michael Wentzel
- **Posted on arXiv:** 02 March 2023
- **Published in Phys.Rev.D:** 15 October 2023

2.10.1.3 Multimessenger astronomy beyond the Standard Model: New window from quantum sensors [116]

- **Authors:** Jason Arakawa, Muhammad H. Zaheer, Volodymyr Takhistov, Marianna S. Safronova, Joshua Eby, Charles Cheung
- **Posted on arXiv:** 12 February 2025
- **Published in JCAP:** 10 February 2026

2.10.2 Quantum Sensors - Optomechanical Sensors

2.10.2.1 Quantum interactions between a laser interferometer and gravitational waves [165]

- **Authors:** Belinda Pang, Yanbei Chen
- **Posted on arXiv:** 28 August 2018
- **Published in Phys. Rev. D 98, 124006 (2018):** 11 December 2018

2.10.2.2 Superconducting Levitated Detector of Gravitational Waves [166]

- **Authors:** Daniel Carney, Gerard Higgins, Giacomo Marocco, Michael Wentzel
- **Posted on arXiv:** 02 August 2024
- **Published in Carney (2025) Phys. Rev. Lett. 134, 181402:** 06 May 2025

2.11 Jet Reconstruction

2.11.1 Quantum Algorithms - Grover’s Search Algorithm

2.11.1.1 Quantum Algorithms for Jet Clustering [167]

- **Authors:** Annie Y. Wei, Preksha Naik, Aram W. Harrow, Jesse Thaler
- **Posted on arXiv:** 23 August 2019
- **Published in Phys. Rev. D 101, 094015 (2020):** 15 May 2020

2.11.1.2 Quantum Algorithms in Particle Physics [164]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

2.11.2 Quantum Computing Paradigms - Quantum Annealing

2.11.2.1 Adiabatic quantum algorithm for multijet clustering in high energy physics [168]

- **Authors:** Diogo Pires, Yasser Omar, João Seixas
- **Posted on arXiv:** 28 December 2020
- **Published in Phys.Lett.B:** 13 June 2023

2.11.2.2 Leveraging Quantum Annealer to identify an Event-topology at High Energy Colliders [169]

- **Authors:** Minho Kim, Pyungwon Ko, Jae-hyeon Park, Myeonghun Park
- **Posted on arXiv:** 15 November 2021

2.11.2.3 Quantum annealing for jet clustering with thrust [170]

- **Authors:** Andrea Delgado, Jesse Thaler
- **Posted on arXiv:** 05 May 2022
- **Published in Phys.Rev.D:** 01 November 2022

2.11.2.4 Degeneracy engineering for classical and quantum annealing: A case study of sparse linear regression in collider physics [171]

- **Authors:** Eric R. Anschuetz, Lena Funcke, Patrick T. Komiske, Serhii Kryhin, Jesse Thaler
- **Posted on arXiv:** 20 May 2022
- **Published in Phys. Rev. D 106, 056008 (2022):** 01 September 2022

2.11.3 Quantum Machine Learning - Supervised Methods

2.11.3.1 Quantum-inspired machine learning on high-energy physics data [172]

- **Authors:** Timo Felser, Marco Trenti, Lorenzo Sestini, Alessio Gianelle, Davide Zuliani, Donatella Lucchesi, Simone Montangero
- **Posted on arXiv:** 28 April 2020
- **Published in npj Quantum Inf.:** 15 July 2021

2.11.3.2 Quantum-inspired event reconstruction with Tensor Networks: Matrix Product States [173]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 15 June 2021
- **Published in JHEP 08 (2021) 112:** 23 August 2021

2.11.3.3 Classical versus quantum: Comparing tensor-network-based quantum circuits on Large Hadron Collider data [174]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 21 February 2022
- **Published in Phys. Rev. A 106 (Dec, 2022) 062423:** 19 December 2022

2.11.4 Quantum Machine Learning - Unsupervised Methods

2.11.4.1 A Digital Quantum Algorithm for Jet Clustering in High-Energy Physics [175]

- **Authors:** Diogo Pires, Pedrame Bargassa, João Seixas, Yasser Omar
- **Posted on arXiv:** 11 January 2021

2.11.4.2 Quantum clustering and jet reconstruction at the LHC [176]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 13 April 2022
- **Published in Physical Review D 106, 036021 (2022):** 01 August 2022

2.11.5 Quantum Machine Learning - Variational Quantum Algorithms

2.11.5.1 Quantum Machine Learning for b-jet charge identification [177]

- **Authors:** Alessio Gianelle, Patrick Koppenburg, Donatella Lucchesi, Davide Nicotra, Eduardo Rodrigues, Lorenzo Sestini, Jacco de Vries, Davide Zuliani
- **Posted on arXiv:** 28 February 2022
- **Published in JHEP:** 01 August 2022

2.11.6 Crosslists

2.11.6.1 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.11.6.2 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borrás, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.12 Lattice Field Theories - Scalar/Fermion Theories

2.12.1 Quantum Algorithms - Quantum Simulations

2.12.1.1 Quantum Computation of Scattering in Scalar Quantum Field Theories [178]

- **Authors:** Stephen P. Jordan, Keith S.M. Lee, John Preskill
- **Posted on arXiv:** December 2011

2.12.1.2 Quantum Algorithms for Fermionic Quantum Field Theories [179]

- **Authors:** Stephen P. Jordan, Keith S. M. Lee, John Preskill
- **Posted on arXiv:** 28 April 2014

2.12.1.3 Digitization of scalar fields for quantum computing [180]

- **Authors:** Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 30 August 2018
- **Published in Phys. Rev. A 99, 052335 (2019):** 23 May 2019

2.12.1.4 Scalar Quantum Field Theories as a Benchmark for Near-Term Quantum Computers [181]

- **Authors:** Kubra Yeter-Aydeniz, Eugene F. Dumitrescu, Alex J. McCaskey, Ryan S. Bennink, Raphael C. Pooser, George Siopsis
- **Posted on arXiv:** 29 November 2018
- **Published in Phys. Rev. A 99, 032306 (2019):** 04 March 2019

2.12.1.5 Real-Time Scattering on Quantum Computers via Hamiltonian Truncation [182]

- **Authors:** James Ingoldby, Michael Spannowsky, Timur Sypchenko, Simon Williams, Matthew Wingate
- **Posted on arXiv:** 06 May 2025

2.12.2 Quantum Computing Paradigms - Continuous Variable Quantum Computing

2.12.2.1 Simulating quantum field theories on continuous-variable quantum computers [183]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 15 March 2024
- **Published in Phys. Rev. A 110 (2024) 1, 012607:** 08 July 2024

2.12.2.2 Real-time scattering processes with continuous-variable quantum computers [184]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 03 February 2025
- **Published in Phys. Rev. A 112 (2025), 012614:** 16 July 2025

2.12.2.3 Qumode Tensor Networks for False Vacuum Decay in Quantum Field Theory [185]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 20 June 2025

2.13 Lattice Field Theories - Gauge Theories

2.13.1 Quantum Algorithms - Quantum Simulations

2.13.1.1 Simulating lattice gauge theories on a quantum computer [186]

- **Authors:** Tim Byrnes, Yoshihisa Yamamoto
- **Posted on arXiv:** October 2005
- **Published in Phys.Rev.A:** 2006

2.13.1.2 Ultracold Quantum Gases and Lattice Systems: Quantum Simulation of Lattice Gauge Theories [187]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 07 May 2013
- **Published in Annalen Phys.:** 2013

2.13.1.3 Towards Quantum Simulating QCD [188]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 25 September 2014
- **Published in Nucl.Phys.A:** 28 September 2014

2.13.1.4 Quantum Simulations of Lattice Gauge Theories using Ultracold Atoms in Optical Lattices [189]

- **Authors:** Erez Zohar, J. Ignacio Cirac, Benni Reznik
- **Posted on arXiv:** 08 March 2015
- **Published in Rept.Prog.Phys.:** 18 December 2015

2.13.1.5 Real-time dynamics of lattice gauge theories with a few-qubit quantum computer [190]

- **Authors:** E.A. Martinez, C.A. Muschik, P. Schindler, D. Nigg, A. Erhard, M. Heyl, P. Hauke, M. Dalmonte, T. Monz, P. Zoller, R. Blatt
- **Posted on arXiv:** 15 May 2016
- **Published in Nature:** 22 June 2016

2.13.1.6 General Methods for Digital Quantum Simulation of Gauge Theories [191]

- **Authors:** Henry Lamm, Scott Lawrence, Yukari Yamauchi
- **Posted on arXiv:** 19 March 2019
- **Published in Phys. Rev. D 100, 034518 (2019):** 29 August 2019

2.13.1.7 SU(2) non-Abelian gauge field theory in one dimension on digital quantum computers [192]

- **Authors:** Natalie Klco, Jesse R. Stryker, Martin J. Savage
- **Posted on arXiv:** 19 August 2019
- **Published in Phys. Rev. D 101, 074512 (2020):** 22 April 2020

2.13.1.8 Simulating Lattice Gauge Theories within Quantum Technologies [193]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

2.13.1.9 Toward scalable simulations of Lattice Gauge Theories on quantum computers [194]

- **Authors:** Simon V. Mathis, Guglielmo Mazzola, Ivano Tavernelli
- **Posted on arXiv:** 20 May 2020
- **Published in Phys. Rev. D 102, 094501 (2020):** 04 November 2020

2.13.1.10 Role of boundary conditions in quantum computations of scattering observables [195]

- **Authors:** Raúl A. Briceño, Juan V. Guerrero, Maxwell T. Hansen, Alexandru M. Sturzu
- **Posted on arXiv:** 01 July 2020
- **Published in Phys. Rev. D 103, 014506 (2021):** 07 January 2021

2.13.1.11 Quantum simulation of gauge theory via orbifold lattice [196]

- **Authors:** Alexander J. Buser, Hrant Gharibyan, Masanori Hanada, Masazumi Honda, Junyu Liu
- **Posted on arXiv:** 12 November 2020
- **Published in JHEP 2109 (2021) 034:** 06 September 2021

2.13.1.12 Lattice renormalization of quantum simulations [197]

- **Authors:** Marcela Carena, Henry Lamm, Ying-Ying Li, Wanqiang Liu
- **Posted on arXiv:** 02 July 2021
- **Published in Phys.Rev.D:** 01 November 2021

2.13.1.13 Lattice Quantum Chromodynamics and Electrodynamics on a Universal Quantum Computer [198]

- **Authors:** Angus Kan, Yunseong Nam
- **Posted on arXiv:** 27 July 2021

2.13.1.14 Efficient representation for simulating U(1) gauge theories on digital quantum computers at all values of the coupling [199]

- **Authors:** Christian W. Bauer, Dorota M. Grabowska
- **Posted on arXiv:** 15 November 2021
- **Published in Phys.Rev.D:** 01 February 2023

2.13.1.15 Quantum error correction with gauge symmetries [200]

- **Authors:** Abhishek Rajput, Alessandro Roggero, Nathan Wiebe
- **Posted on arXiv:** 09 December 2021
- **Published in npj Quantum Inf.:** 25 April 2023

2.13.1.16 Preparations for quantum simulations of quantum chromodynamics in $1+1$ dimensions. I. Axial gauge [201]

- **Authors:** Roland C. Farrell, Ivan A. Chernyshev, Sarah J.M. Powell, Nikita A. Zemlevskiy, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 04 July 2022
- **Published in Phys. Rev. D 107, 054512 (2023):** 01 March 2023

2.13.1.17 Measurement-based quantum simulation of Abelian lattice gauge theories [202]

- **Authors:** Hiroki Sueno, Takuya Okuda
- **Posted on arXiv:** 19 October 2022
- **Published in SciPost Phys.** 14, 129 (2023): 25 May 2023

2.13.1.18 General quantum algorithms for Hamiltonian simulation with applications to a non-Abelian lattice gauge theory [203]

- **Authors:** Zohreh Davoudi, Alexander F. Shaw, Jesse R. Stryker
- **Posted on arXiv:** 28 December 2022
- **Published in Quantum** 7, 1213 (2023): 20 December 2023

2.13.1.19 Simulating $\langle \mathcal{Z} \rangle$ in lattice gauge theory on a quantum computer [204]

- **Authors:** Clement Charles, Erik J. Gustafson, Elizabeth Hardt, Florian Herren, Norman Hogan, Henry Lamm, Sara Starecheski, Ruth S. Van de Water, Michael L. Wagman
- **Posted on arXiv:** 03 May 2023
- **Published in Phys.Rev.E:** 26 January 2024

2.13.1.20 Quantum simulation of fundamental particles and forces [205]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 09 April 2024
- **Published in Nature Rev.Phys.:** 21 June 2023

2.13.1.21 Toward QCD on quantum computer: orbifold lattice approach [206]

- **Authors:** Georg Bergner, Masanori Hanada, Enrico Rinaldi, Andreas Schafer
- **Posted on arXiv:** 22 January 2024
- **Published in JHEP:** 20 May 2024

2.13.1.22 Scattering wave packets of hadrons in gauge theories: Preparation on a quantum computer [207]

- **Authors:** Zohreh Davoudi, Chung-Chun Hsieh, Saurabh V. Kadam
- **Posted on arXiv:** 01 February 2024
- **Published in Quantum:** 11 November 2024

2.13.1.23 Quantum Simulation of SU(3) Lattice Yang-Mills Theory at Leading Order in Large- N_c Expansion [208]

- **Authors:** Anthony N. Ciavarella, Christian W. Bauer
- **Posted on arXiv:** 15 February 2024
- **Published in Phys.Rev.Lett.:** 09 September 2024

2.13.1.24 Scalable cold-atom quantum simulator of a 3 + 1D U(1) lattice gauge theory with dynamical matter [209]

- **Authors:** Simone Orlando, Guo-Xian Su, Bing Yang, Jad C. Halimeh
- **Posted on arXiv:** 07 January 2026

2.13.2 Quantum Computing Paradigms - Quantum Annealing

2.13.2.1 A regression algorithm for accelerated lattice QCD that exploits sparse inference on the D-Wave quantum annealer [210]

- **Authors:** Nga T.T. Nguyen, Garrett T. Kenyon, Boram Yoon
- **Posted on arXiv:** 14 November 2019
- **Published in Sci Rep 10, 10915 (2020):** 02 July 2020

2.13.2.2 SU(2) lattice gauge theory on a quantum annealer [211]

- **Authors:** Sarmed A Rahman, Randy Lewis, Emanuele Mendicelli, Sarah Powell
- **Posted on arXiv:** 15 March 2021
- **Published in Phys. Rev. D 104, 034501 (2021):** 01 August 2021

2.13.3 Quantum Machine Learning - Supervised Methods

2.13.3.1 Quantum data learning for quantum simulations in high-energy physics [163]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

2.13.4 Quantum Machine Learning - Variational Quantum Algorithms

2.13.4.1 SU(2) hadrons on a quantum computer via a variational approach [212]

- **Authors:** Yasar Y. Atas, Jinglei Zhang, Randy Lewis, Amin Jahanpour, Jan F. Haase, Christine A. Muschik
- **Posted on arXiv:** 17 February 2021
- **Published in Nature Communications 2021:** 11 November 2021

2.13.5 Crosslists

2.13.5.1 Quantum simulation of a lattice Schwinger model in a chain of trapped ions [213]

- **Authors:** Philipp Hauke, David Marcos, Marcello Dalmonte, Peter Zoller
- **Posted on arXiv:** 10 June 2013
- **Published in Phys.Rev.X:** 22 November 2013

2.13.5.2 Lattice gauge theory simulations in the quantum information era [1]

- **Authors:** M. Dalmonte, S. Montangero
- **Posted on arXiv:** 11 February 2016
- **Published in Contemporary Physics 57 388 (2016):** 09 March 2016

2.13.5.3 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [10]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

2.13.5.4 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [13]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borrás, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

2.13.5.5 Quantum Simulation for High-Energy Physics [17]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

2.13.5.6 Quantum computing hardware for HEP algorithms and sensing [18]

- **Authors:** M. Sohaib Alam, Sergey Belomestnykh, Nicholas Bornman, Gustavo Cancellato, Yu-Chiu Chao, Mattia Checchin, Vinh San Dinh, Anna Grassellino, Erik J. Gustafson, Roni Harnik, Corey Rae Harrington McRae, Ziwen Huang, Keshav Kapoor, Taeyoon Kim, James B. Kowalkowski, Matthew J. Kramer, Yulia Krasnikova, Prem Kumar, Doga Murat Kurkcuoglu, Henry Lamm, Adam L. Lyon, Despina Milathianaki, Akshay Murthy, Josh Mutus, Ivan Nekrashevich, JinSu Oh, A. BarışÖzgüler, Gabriel Nathan Perdue, Matthew Reagor, Alexander Romanenko, James A. Sauls, Leandro Stefanazzi, Norm M. Tubman, Davide Venturelli, Changqing Wang, Xinyuan You, David M.T. van Zanten, Lin Zhou, Shaojiang Zhu, Silvia Zorzetti
- **Posted on arXiv:** 18 April 2022

2.13.5.7 Review on Quantum Computing for Lattice Field Theory [5]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn
- **Posted on arXiv:** 01 February 2023
- **Published in nan:** 01 February 2023

2.13.5.8 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borrás, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico

Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.13.5.9 Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits [214]

- **Authors:** Roland C. Farrell, Marc Illa, Anthony N. Ciavarella, Martin J. Savage
- **Posted on arXiv:** 08 August 2023
- **Published in PRX Quantum 5 (2024) 2, 020315:** 01 April 2024

2.13.5.10 Steps toward quantum simulations of hadronization and energy loss in dense matter [215]

- **Authors:** Roland C. Farrell, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 10 May 2024
- **Published in Phys.Rev.C 111 (2025) 1, 015202:** 14 January 2025

2.13.5.11 Pathfinding quantum simulations of neutrinoless double- decay [216]

- **Authors:** Ivan A. Chernyshev, Roland C. Farrell, Marc Illa, Martin J. Savage, Andrii Maksymov, Felix Tripier, Miguel Angel Lopez-Ruiz, Andrew Arrasmith, Yvette de Sereville, Aharon Brodutch, Claudio Girotto, Ananth Kaushik, Martin Roetteler
- **Posted on arXiv:** 06 June 2025
- **Published in Nature Commun.:** 23 January 2026

2.14 Lattice Field Theories - Schwinger Model

2.14.1 Quantum Algorithms - Quantum Simulations

2.14.1.1 Quantum simulation of a lattice Schwinger model in a chain of trapped ions [213]

- **Authors:** Philipp Hauke, David Marcos, Marcello Dalmonte, Peter Zoller
- **Posted on arXiv:** 10 June 2013
- **Published in Phys.Rev.X:** 22 November 2013

2.14.1.2 Quantum Algorithms for Simulating the Lattice Schwinger Model [217]

- **Authors:** Alexander F. Shaw, Pavel Lougovski, Jesse R. Stryker, Nathan Wiebe
- **Posted on arXiv:** 25 February 2020
- **Published in Quantum 4, 306 (2020):** 10 August 2020

2.14.1.3 Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits [214]

- **Authors:** Roland C. Farrell, Marc Illa, Anthony N. Ciavarella, Martin J. Savage
- **Posted on arXiv:** 08 August 2023
- **Published in PRX Quantum 5 (2024) 2, 020315:** 01 April 2024

2.14.1.4 Steps toward quantum simulations of hadronization and energy loss in dense matter [215]

- **Authors:** Roland C. Farrell, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 10 May 2024
- **Published in Phys.Rev.C 111 (2025) 1, 015202:** 14 January 2025

2.14.1.5 Enhancing quantum field theory simulations on NISQ devices with Hamiltonian truncation [218]

- **Authors:** James Ingoldby, Michael Spannowsky, Timur Sypchenko, Simon Williams
- **Posted on arXiv:** 26 July 2024
- **Published in Phys. Rev. D 110 (2024), 096016:** 01 November 2024

2.14.2 Quantum Machine Learning - Variational Quantum Algorithms

2.14.2.1 Quantum-classical computation of Schwinger model dynamics using quantum computers [219]

- **Authors:** N. Klco, E.F. Dumitrescu, A.J. McCaskey, T.D. Morris, R.C. Pooser, M. Sanz, E. Solano, P. Lougovski, M.J. Savage
- **Posted on arXiv:** 08 March 2018
- **Published in Phys. Rev. A 98, 032331 (2018):** 29 September 2018

2.14.2.2 Quantum-Classical Simulation of Quantum Field Theory by Quantum Circuit Learning [220]

- **Authors:** Kazuki Ikeda
- **Posted on arXiv:** 27 November 2023
- **Published in Annalen Phys.:** 01 June 2025

2.14.3 Crosslists

2.14.3.1 Quantum data learning for quantum simulations in high-energy physics [163]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

2.15 Neutrino Physics

2.15.1 Quantum Sensors - Atomic/Molecular/Nuclear Sensors

2.15.1.1 Superradiant interactions for relic detection with entangled nuclear spins [117]

- **Authors:** Marios Galanis, Onur Hosten, Asimina Arvanitaki, Savas Dimopoulos
- **Posted on arXiv:** 28 August 2025
- **Published in Phys.Rev.A:** 10 June 2026

2.15.2 Quantum Sensors - Optical/Photonic Sensors

2.15.2.1 Superradiant interactions of the cosmic neutrino background, axions, dark matter, and reactor neutrinos [115]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Marios Galanis
- **Posted on arXiv:** 07 August 2024
- **Published in Phys.Rev.D:** 01 March 2025

2.15.3 Quantum Sensors - Optomechanical Sensors

2.15.3.1 Requirements on quantum superpositions of macro-objects for sensing neutrinos [221]

- **Authors:** Eva Kilian, Marko Toroš, Frank F. Deppisch, Ruben Saakyan, Sougato Bose
- **Posted on arXiv:** 27 April 2022
- **Published in Phys. Rev. Research 5, 023012 (2023):** 07 April 2023

2.15.3.2 Searches for Massive Neutrinos with Mechanical Quantum Sensors [222]

- **Authors:** Daniel Carney, Kyle G. Leach, David C. Moore
- **Posted on arXiv:** 12 July 2022
- **Published in Phys.Rev.X:** 01 February 2023

2.16 Nuclear Many-Body Physics - Nuclear Structure

2.16.1 Quantum Algorithms - Quantum Simulations

2.16.1.1 Symmetry assisted preparation of entangled many-body states on a quantum computer [223]

- **Authors:** D. Lacroix
- **Posted on arXiv:** 11 June 2020
- **Published in Phys.Rev.Lett.:** 03 December 2020

2.16.1.2 Quantum computation of nuclear observables involving linear combinations of unitary operators [224]

- **Authors:** Pooja Siwach, P. Arumugam
- **Posted on arXiv:** 16 June 2022
- **Published in Phys.Rev.C:** 28 June 2022

2.16.2 Quantum Computing Paradigms - Quantum Annealing

2.16.2.1 Quasiparticle pairing encoding of atomic nuclei for quantum annealing [225]

- **Authors:** Emanuele Costa, Axel Pèrez-Obiol, Javier Menéndez, Arnau Rios, Artur García-Sáez, Bruno Juliá-Díaz
- **Posted on arXiv:** 11 October 2025
- **Published in Phys.Lett.B:** 30 November 2025

2.16.3 Quantum Machine Learning - Variational Quantum Algorithms

2.16.3.1 Cloud Quantum Computing of an Atomic Nucleus [127]

- **Authors:** E.F. Dumitrescu, A.J. McCaskey, G. Hagen, G.R. Jansen, T.D. Morris, T. Papenbrock, R.C. Pooser, D.J. Dean, P. Lougovski
- **Posted on arXiv:** 11 January 2018
- **Published in Phys. Rev. Lett. 120, 210501 (2018):** 24 May 2018

2.16.3.2 Improving Hamiltonian encodings with the Gray code [129]

- **Authors:** Olivia Di Matteo, Anna McCoy, Peter Gysbers, Takayuki Miyagi, R.M. Woloshyn, Petr Navrátil
- **Posted on arXiv:** 11 August 2020
- **Published in Phys. Rev. A 103, 042405 (2021):** 03 April 2021

2.16.3.3 Lipkin model on a quantum computer [226]

- **Authors:** Michael J. Cervia, A.B. Balantekin, S.N. Coppersmith, Calvin W. Johnson, Peter J. Love, C. Poole, K. Robbins, M. Saffman
- **Posted on arXiv:** 08 November 2020
- **Published in Phys. Rev. C 104, 024305 (2021):** 03 August 2021

2.16.3.4 Variational approaches to constructing the many-body nuclear ground state for quantum computing [227]

- **Authors:** I. Stetcu, A. Baroni, J. Carlson
- **Posted on arXiv:** 12 October 2021
- **Published in Phys. Rev. C 105, 064308 (2022):** 21 June 2022

2.16.3.5 Simulating excited states of the Lipkin model on a quantum computer [228]

- **Authors:** Manqoba Q. Hlatshwayo, Yinu Zhang, Herlik Wibowo, Ryan LaRose, Denis Lacroix, Elena Litvinova
- **Posted on arXiv:** 02 March 2022
- **Published in Phys.Rev.C:** 18 August 2022

2.16.3.6 Solving nuclear structure problems with the adaptive variational quantum algorithm [229]

- **Authors:** A.M. Romero, J. Engel, Ho Lun Tang, Sophia E. Economou
- **Posted on arXiv:** 03 March 2022
- **Published in Phys. Rev. C 105, 064317 (2022):** 27 June 2022

2.16.3.7 Quantum computing of the ${}^6\text{Li}$ nucleus via ordered unitary coupled cluster [230]

- **Authors:** Oriel Kiss, Michele Grossi, Pavel Lougovski, Federico Sanchez, Sofia Vallecorsa, Thomas Papenbrock
- **Posted on arXiv:** 02 May 2022
- **Published in Phys. Rev. C 106, 034325 (2022):** 29 September 2022

2.16.3.8 QCSH: A full quantum computer nuclear shell-model package [231]

- **Authors:** Peng Lv, Shi-Jie Wei, Hao-Nan Xie, Gui-Lu Long
- **Posted on arXiv:** 24 May 2022
- **Published in Sci.China Phys.Mech.Astron.:** 06 March 2023

2.16.3.9 Subatomic Many-Body Physics Simulations on a Quantum Frequency Processor [131]

- **Authors:** Hsuan-Hao Lu, Natalie Klco, Joseph M. Lukens, Titus D. Morris, Aaina Bansal, Andreas Ekström, Gaute Hagen, Thomas Papenbrock, Andrew M. Weiner, Martin J. Savage, Pavel Lougovski
- **Published in nan:** 2019

2.16.3.10 Nuclear shell-model simulation in digital quantum computers [232]

- **Authors:** A. Pèrez-Obiol, A.M. Romero, J. Menéndez, A. Rios, A. García-Sáez, B. Juliá-Díaz
- **Posted on arXiv:** 07 February 2023
- **Published in Sci.Rep.:** 29 July 2023

2.16.3.11 Deep quantum circuit simulations of low-energy nuclear states [233]

- **Authors:** Ang Li, Alessandro Baroni, Ionel Stetcu, Travis S. Humble
- **Posted on arXiv:** 26 October 2023
- **Published in Eur.Phys.J.A:** 14 May 2024

2.16.3.12 Exploiting symmetries in nuclear Hamiltonians for ground state preparation [234]

- **Authors:** Joe Gibbs, Zoë Holmes, Paul Stevenson
- **Posted on arXiv:** 15 February 2024
- **Published in Quantum Machine Intelligence:** 30 January 2025

2.16.3.13 Comparison of variational quantum eigensolvers in light nuclei [235]

- **Authors:** Miquel Carrasco-Codina, Emanuele Costa, Antonio Márquez Romero, Javier Menéndez, Arnau Rios
- **Posted on arXiv:** 18 July 2025
- **Published in Phys.Rev.C:** 23 February 2026

2.16.3.14 Toward scalable quantum computations of atomic nuclei [236]

- **Authors:** Chenyi Gu, Matthias Heinz, Oriel Kiss, Thomas Papenbrock
- **Posted on arXiv:** 19 July 2025
- **Published in Phys. Rev. C 113, 034321 (2026):** 24 March 2026

2.16.3.15 Bridging quantum computing and nuclear structure: Atomic nuclei on a trapped-ion quantum computer [237]

- **Authors:** Sota Yoshida, Takeshi Sato, Takumi Ogata, Masaaki Kimura
- **Posted on arXiv:** 25 September 2025
- **Published in Phys. Rev. Research 8, 013134 (2026):** 06 February 2026

2.16.4 Quantum Measurement and Tomography

2.16.4.1 Shadow-based quantum subspace algorithm for the nuclear shell model [238]

- **Authors:** Ruyu Yang, Tianren Wang, Bing-Nan Lu, Ying Li, Xiaosi Xu
- **Posted on arXiv:** 15 June 2023
- **Published in Phys.Rev.A:** 21 January 2025

2.17 Nuclear Many-Body Physics - Nuclear Reactions and Response

2.17.1 Quantum Algorithms - Quantum Phase Estimation

2.17.1.1 Dynamic linear response quantum algorithm [239]

- **Authors:** Alessandro Roggero, Joseph Carlson
- **Posted on arXiv:** 04 April 2018
- **Published in Phys. Rev. C 100, 034610 (2019):** 14 September 2019

2.17.2 Quantum Algorithms - Quantum Simulations

2.17.2.1 Quantum Computing for Neutrino-Nucleus Scattering [128]

- **Authors:** Alessandro Roggero, Andy C.Y. Li, Joseph Carlson, Rajan Gupta, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019
- **Published in Phys. Rev. D 101, 074038 (2020):** 28 April 2020

2.17.2.2 Quantum simulation of nuclear inelastic scattering [240]

- **Authors:** Weijie Du, James P. Vary, Xingbo Zhao, Wei Zuo
- **Posted on arXiv:** 01 June 2020
- **Published in Phys. Rev. A 104, 012611 (2021):** 23 July 2021

2.17.2.3 Imaginary-time propagation on a quantum chip [241]

- **Authors:** Francesco Turro, Alessandro Roggero, Valentina Amitrano, Piero Luchi, Kyle A. Wendt, Jonathan L. DuBois, Sofia Quaglioni, Francesco Pederiva
- **Posted on arXiv:** 24 February 2021
- **Published in Phys.Rev.A:** 28 February 2022

2.17.2.4 Nuclear two point correlation functions on a quantum computer [242]

- **Authors:** Alessandro Baroni, Joseph Carlson, Rajan Gupta, Andy C.Y. Li, Gabriel N. Perdue, Alessandro Roggero
- **Posted on arXiv:** 04 November 2021
- **Published in Phys.Rev.D:** 01 April 2022

2.17.2.5 A quantum algorithm for the linear response of nuclei [243]

- **Authors:** Nifeeya Singh, P. Arumugam
- **Posted on arXiv:** 17 October 2022
- **Published in Indian J.Phys.:** 31 May 2025

2.17.2.6 Faster spectral density calculation using energy moments [244]

- **Authors:** Jeremy Hartse, Alessandro Roggero
- **Posted on arXiv:** 01 November 2022
- **Published in Eur.Phys.J.A:** 09 March 2023

2.17.2.7 Quantum error mitigation for Fourier moment computation [245]

- **Authors:** Oriel Kiss, Michele Grossi, Alessandro Roggero
- **Posted on arXiv:** 23 January 2024
- **Published in Phys. Rev. D 111, 034504 (2025):** 01 February 2025

2.17.2.8 Nuclear scattering via quantum computing [246]

- **Authors:** Peiyan Wang, Weijie Du, Wei Zuo, James P. Vary
- **Posted on arXiv:** 30 January 2024
- **Published in Phys.Rev.C 109 6, 064623 (2024):** 26 June 2024

2.17.2.9 Solving reaction dynamics with quantum computing algorithms [247]

- **Authors:** Ronen Weiss, Alessandro Baroni, Joseph Carlson, Ionel Stetcu
- **Posted on arXiv:** 29 March 2024
- **Published in Physical Review C 111, 064004 (2025):** 25 June 2025

2.17.2.10 Quantum computing for extracting nuclear resonances [248]

- **Authors:** Hantao Zhang, Dong Bai, Zhongzhou Ren
- **Posted on arXiv:** 10 September 2024
- **Published in Phys.Lett.B:** 12 December 2024

2.17.2.11 Pathfinding quantum simulations of neutrinoless double- decay [216]

- **Authors:** Ivan A. Chernyshev, Roland C. Farrell, Marc Illa, Martin J. Savage, Andrii Maksymov, Felix Tripier, Miguel Angel Lopez-Ruiz, Andrew Arrasmith, Yvette de Sereville, Aharon Brodutch, Claudio Girotto, Ananth Kaushik, Martin Roetteler
- **Posted on arXiv:** 06 June 2025
- **Published in Nature Commun.:** 23 January 2026

2.17.3 Quantum Machine Learning - Variational Quantum Algorithms

2.17.3.1 Quantum simulations of nuclear resonances with variational methods [249]

- **Authors:** Ashutosh Singh, Pooja Siwach, P. Arumugam
- **Posted on arXiv:** 15 April 2025
- **Published in Phys.Rev.C:** 25 August 2025

2.18 Quantum Chromodynamics

2.18.1 Quantum Algorithms - Quantum Simulations

2.18.1.1 Quantum simulation of quantum field theory in the light-front formulation [250]

- **Authors:** Michael Kreshchuk, William M. Kirby, Gary Goldstein, Hugo Beauchemin, Peter J. Love
- **Posted on arXiv:** 10 February 2020
- **Published in Phys. Rev. A 105, 032418 (2022):** 08 March 2022

2.18.1.2 Quantum simulation of colour in perturbative quantum chromodynamics [251]

- **Authors:** Herschel A. Chawdhry, Mathieu Pellen
- **Posted on arXiv:** 08 March 2023
- **Published in SciPost Phys. 15, 205 (2023):** 24 November 2023

2.18.1.3 Quantum simulation of scattering amplitudes and interferences in perturbative QCD [252]

- **Authors:** Herschel A. Chawdhry, Mathieu Pellen, Simon Williams
- **Posted on arXiv:** 09 July 2025

2.18.1.4 Quantum simulating multi-particle processes in high energy nuclear physics: dijet production and color (de)coherence [253]

- **Authors:** João Barata, Meijian Li, Wenyang Qian, Carlos A. Salgado, João M. Silva
- **Posted on arXiv:** 13 April 2026
- **Published in JHEP:** 08 July 2026

2.18.2 Quantum Entanglement and Bell Inequalities

2.18.2.1 Symmetry from entanglement suppression [254]

- **Authors:** Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 April 2021
- **Published in Phys.Rev.D:** 01 October 2021

2.18.3 Crosslists

2.18.3.1 Partonic collinear structure by quantum computing [160]

- **Authors:** Tianyin Li, Xingyu Guo, Wai Kin Lai, Xiaohui Liu, Enke Wang, Hongxi Xing, Dan-Bo Zhang, Shi-Liang Zhu
- **Posted on arXiv:** 07 June 2021
- **Published in Phys.Rev.D:** 01 June 2022

2.18.3.2 Minimal entanglement and emergent symmetries in low-energy QCD [126]

- **Authors:** Qiaofeng Liu, Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 October 2022
- **Published in Phys.Rev.C 107, 025204 (2023):** 14 February 2023

2.19 Quantum Correlations at Colliders

2.19.1 Quantum Entanglement and Bell Inequalities

2.19.1.1 Entanglement and quantum tomography with top quarks at the LHC [255]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 04 March 2020
- **Published in Eur. Phys. J. Plus (2021) 136:907:** 03 September 2021

2.19.1.2 Testing Bell Inequalities at the LHC with Top-Quark Pairs [256]

- **Authors:** M. Fabbrichesi, R. Floreanini, G. Panizzo
- **Posted on arXiv:** 23 February 2021
- **Published in Phys.Rev.Lett. 127 (2021) 16, 161801:** 15 October 2021

2.19.1.3 Quantum tops at the LHC: from entanglement to Bell inequalities [257]

- **Authors:** Claudio Severi, Cristian Degli Esposti Boschi, Fabio Maltoni, Maximiliano Sioli
- **Posted on arXiv:** 19 October 2021
- **Published in Eur.Phys.J.C:** 02 April 2022

2.19.1.4 Quantum SMEFT tomography: Top quark pair production at the LHC [130]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 10 March 2022
- **Published in Phys.Rev.D:** 01 September 2022

2.19.1.5 Quantum information with top quarks in QCD [258]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 10 March 2022
- **Published in Quantum:** 29 September 2022

2.19.1.6 Improved tests of entanglement and Bell inequalities with LHC tops [259]

- **Authors:** J.A. Aguilar-Saavedra, J.A. Casas
- **Posted on arXiv:** 01 May 2022
- **Published in Eur.Phys.J.C:** 03 August 2022

2.19.1.7 Constraining new physics in entangled two-qubit systems: top-quark, tau-lepton and photon pairs [260]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli
- **Posted on arXiv:** 24 August 2022
- **Published in Eur.Phys.J.C:** 19 February 2023

2.19.1.8 Quantum Discord and Steering in Top Quarks at the LHC [261]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 08 September 2022
- **Published in Phys. Rev. Lett. 130, 221801 (2023):** 31 May 2023

2.19.1.9 Bell inequalities and quantum entanglement in weak gauge boson production at the LHC and future colliders [262]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 01 February 2023
- **Published in Eur.Phys.J.C:** 15 September 2023

2.19.1.10 Three-Body Entanglement in Particle Decays [263]

- **Authors:** Kazuki Sakurai, Michael Spannowsky
- **Posted on arXiv:** 02 October 2023
- **Published in Phys.Rev.Lett.:** 09 April 2024

2.19.1.11 Observation of quantum entanglement with top quarks at the ATLAS detector [264]

- **Authors:** Georges Aad, Braden Keim Abbott, Kira Abeling, Nils Julius Abicht, Haider Abidi, Asmaa Aboulhorma, Halina Abramowicz, Henso Abreu, Yiming Abulaiti, Bobby Samir Acharya, Claire Adam Bourdarios, Leszek Adamczyk, Sagar Addepalli, Matt Addison, Jahred Adelman, Aytul Adiguzel, Tim Adye, A.A. Affolder, Tony Affolder, Yoav Afik, Merve Nazlim Agaras, Jinky Agarwala, Anamika Aggarwal, Catalin Agheorghiesei, Ammara Ahmad, Faig Ahmadov, Waleed Syed Ahmed, Sudha Ahuja, Xiacong Ai, Giulio Aielli, Arya Aikot, Malak Ait Tamlihat, Brahim Aitbenchikh, Iakov Aizenberg, Melike Akbiyik, Torsten Akesson, Andrei Akimov, Daiya Akiyama, Nilima Nilesh Akolkar, Sina Aktas, Konie Al Khoury, Gian Luigi Alberghi, Justin Albert, Pietro Albicocco, Guillaume Lucas Albouy, Sara Alderweireldt, Zackary Lee Alegria, Martin Aleksa, I.N. Aleksandrov, Calin Alexa, Theodoros Alexopoulos, Fabrizio Alfonsi, Malte

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Marie Demers, Mikhail Demichev, Serguei Denisov, Louis D'Eramo, Dominik Karol Derendarz, Frederic Derue, Paul Dervan, Klaus Desch, Christopher Deutsch, Francesco Armando Di Bello, Anna Di Ciaccio, Lucia Di Ciaccio, Antonio Di Domenico, Camilla Di Donato, Alessandro Di Girolamo, Giulia Di Gregorio, Andrea Di Luca, Biagio Di Micco, Roberto Di Nardo, Magda Diamantopoulou, Flavia De Almeida Dias, T.Dias Do Vale, Marco Aurelio Diaz Gutierrez, Federico Guillermo Diaz Capriles, Mariia Didenko, Edward Diehl, Leena Diehl, Sergio Diez Cornell, Carmen Diez Pardos, Christina Dimitriadi, Aleksandra Dimitrievska, Jochen Christian Dingfelder, Ioan-Mihail Dinu, Sebastian Dittmeier, Fido Dittus, Fares Djama, Tamar Djobava, Caterina Doglioni, Adriana Dohnalova, Jiri Dolejsi, Zdenek Dolezal, Kristin Dona, Marisilvia Donadelli, Binbin Dong, Julien Noce Donini, Adelina D'Onofrio, Monica D'Onofrio, Jens Dopke, Alessandra Doria, Nuno Dos Santos Fernandes, Patrick Dougan, Maria Teresa Dova, Tony Doyle, Maxence Draguet, Etienne Dreyer, Ioannis Drivas-Koulouris, Matthew Drnevich, Mariia Drozdova, Dongshuo Du, Tristan Arnoldus Du Pree, Filipp Dubinin, Michal Dubovsky, Ehud Duchovni, Guenter Duckeck, Otilia Anamaria Ducu, Dominik Duda, Alexey Dudarev, Emily Rose Duden, Matteo D'Uffizi, Laurent Duflot, Michael Duehrssen-Debling, Ana Elena Dumitriu, Monica Dunford, Sascha Dungs, Katherine Elaine Dunne, Arnaud Duperrin, H.Duran Yildiz, Michael Johannes Dueren, Archil Durglishvili, Brianna Dwyer, G.I. Dyckes, Mateusz Dyndal, Bartosz Sebastian Dziedzic, Zoe Olivia Earnshaw, Gregor Hieronymus Eberwein, Barbora Eckerova, Stephen Eggebrecht, Edmar Egidio Purcino De Souza, Lukas Ehrke, Gerald Eigen, Kevin Frank Einsweiler, Tord Johan Carl Ekelof, Per Alexander Ekman, Saad El Farkh, Yassine El Ghazali, Hassnae El Jarrari, Ali El Moussaouy, Venugopal Ellajosyula, Mattias Ellert, Frank Ellinghaus, Nick Ellis, Johannes Elmsheuser, Markus Elsing, Dmitry Emeliyanov, Yuji Enari, Irina Ene, Shalini Epari, Paula Agnieszka Erland, Martin Errenst, Marc Escalier, Carlos Escobar Ibanez, Erez Etzion, Guiomar Gaspar De Andrade Evans, Hal Evans, Levi Evans, Meirin Oan Evans, Aleksei Ezhilov, Sanae Ezzarqtouni, Federica Fabbri, Laura Fabbri, Gabriel Facini, Vitaliy Fadeyev, Rinat Fakhrutdinov, Dionysios Fakoudis, Speranza Falciano, L.F. Falda Ulhoa Coelho, Peter Johannes Falke, Jana Faltova, Cunwei Fan, Yunyun Fan, Yaquan Fang, Marcello Fanti, Mohammed Faraj, Zahra Farazpay, Amir Farbin, Ada Farilla, Trisha Farooque, Sinead Farrington, Farida Fassi, Dimitris Fassouliotis, Michele Faucci Giannelli, William James Fawcett, Louis Fayard, Pavol Federic, Pavla Federicova, Oleg Fedin, Gleb Fedotov, Matthew Feickert, Lorenzo Feligioni, Deion Elgin Fellers, Cunfeng Feng, Minyu Feng, Zhuoran Feng, Michael James Fenton, Lars Ferencz, Ruby Alice Molly Ferguson, Sergio Ivan Fernandez Luengo, Pablo Fernandez Martinez, M.J. V. Fernoux, James Ferrando, Arnaud Ferrari, Pamela Ferrari, Roberto Ferrari, Didier Ferrere, Claudio Ferretti, Frank Fiedler, Petr Fiedler, Andrej Filipcic, Emily Filmer, Frank Filthaut, Miguel Castro Nunes Fiolhais, Luca Fiorini, Wade Cameron Fisher, Tobias Fitschen, Peter Michael Fitzhugh, Ivor Fleck, Philipp Fleischmann, Tobias Flick, Marvin Flores, Luis Roberto Flores Castillo, Leyre Flores Sanz De Acedo, Francesco Maria Follega, Nikolai Fomin, Joel Hengwei Foo, Andrea

Formica, Alessandra Forti, Etienne Marie Fortin, Anne Winifred Fortman, Maria Giovanna Foti, Leonidas Fountas, Daniel Fournier, Harald Fox, Paolo Francavilla, Simone Francescato, Stefano Franchellucci, Matteo Franchini, Silvia Franchino, David Francis, Luca Franco, Vinicius Franco Lima, Laura Franconi, Melissa Franklin, Guglielmo Frattari, Arran Charles Freegard, Werner Spolidoro Freund, Yuval Yitzhak Frid, Johnny Friend, Nick Fritzsche, Alexander Froch, Daniel Froidevaux, James Frost, Yao Fu, Sebastian Julio Fuenzalida Garrido, Minori Fujimoto, Kin Yip Fung, Eduardo Furtado De Simas Filho, Marin Furukawa, Juan Fuster Verdu, Alessandro Gabrielli, Andrea Gabrielli, Philipp Gadow, Guido Gagliardi, Louis-Guillaume Gagnon, Elizabeth Gallas, Bruce Joseph Gallop, Kock Kiam Gan, Sanmay Ganguly, Yanyan Gao, Francisca Garay Walls, Berenice Garcia, Carmen Garcia, Andrea Garcia Alonso, Arianna Gemma Garcia Caffaro, Jose Enrique Garcia Navarro, Maurice Garcia-Sciveres, Gwen Llewellyn Gardner, Robert William Gardner, Nicoletta Garelli, Diksha Garg, Rocky Bala Garg, Jack Gargan, Christopher Andrew Garner, Cameron Michael Garvey, Philipp Do Nascimento Gaspar, Veritas Katharina Gassmann, Gabriella Gaudio, Viveka Gautam, Paolo Gauzzi, Igor Gavrilenko, Alexander Gavriliuk, Colin Warren Gay, Goetz Gaycken, Evangelos Gazis, Andrei Alexandru Geanta, Carolyn Gee, Alex Gekow, Claudia Gemme, Marie-Helene Genest, Simonetta Gentile, Andrew Donald Gentry, Simon George, William Frederick George, Theodoros Geralis, Paul Gessinger, Marvin Emin Geyik, Mohammed Ghani, Mazuza Mohammad A. Ghneimat, Keanu Ghorbanian, Arpan Ghosal, Aishik Ghosh, Anindya Ghosh, Benedetto Giacobbe, Stefano Giagu, Tommaso Giani, Paola Giannetti, Antonio Giannini, Stephen Gibson, Matthew Gignac, Damian Gil, Alexander Kevin Gilbert, Benjamin Jacob Gilbert, Dag Gillberg, Geoffrey Gilles, Louis Ginabat, Doug Gingrich, Mapo Giordani, Pierre-Francois Giraud, Gilberto Giugliarelli, Danilo Giugni, Francesco Giuli, Ioannis Gkialas, Leonid Gladilin, Claudia Glasman, Galen Rhodes Gledhill, Gediminas Glemza, Marija Glisic, Ivan Gnesi, Yeonju Go, Maximilian Emanuel Goblirsch-Kolb, Benedikt Gocke, Dominique Godin, Berare Gokturk, Steven Goldfarb, Tobias Golling, Mpho Gift Doctor Gololo, Dmitry Golubkov, Jason Peter Gombas, Agostinho Da Silva Gomes, G.Gomes Da Silva, Antonio Jesus Gomez Delegido, Ricardo Jose Morais Silva Goncalo, Laura Gonella, Alexi Gongadze, Francesco Gonnella, Julia Lynne Gonski, Richards Gonzalez, Santiago Gonzalez De La Hoz, Ricardo Gonzalez Lopez, Cesar Gonzalez Renteria, Marcus Vinicius Gonzalez Rodrigues, Rebeca Gonzalez Suarez, Sergio Gonzalez Sevilla, Galo Rafael Gonzalvo Rodriguez, Luc Goossens, Benedetto Gorini, Edoardo Gorini, Andrej Gorisek, Thomas Christopher Gosart, A.T. Goshaw, Mikhail Gostkin, Soumyananda Goswami, Carlo Alberto Gottardo, Stefanie Andrea Gotz, Mohamed Gouighri, Vincent Goumarre, Anna Goussiou, Nicolin Govender, Iwona Grabowska-Bold, Kevin Robert Graham, Eirik Gramstad, Sergio Grancagnolo, Charles Michael Grant, Paul Gravila, Francesco Giuseppe Gravilli, Heather Gray, Matteo Greco, Christian Grefe, Ingrid Gregor, Philippe Grenier, Simon Gabriel Grewe, Chiara Grieco, Alex Grillo, Kathryn Grimm, Sebastian Grinstein, Jean-Francois Grivaz, Eilam Gross, Joern Grosse-Knetter, James Cameron Grundy, Liang

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 Christian Zeitnitz, Hao Zeng, Jiancong Zeng, Todd Zenger, Oleg Zenin, Tibor Zenis,
 Seth Zenz, Soufiane Zerradi, Dirk Zerwas, Mingjie Zhai, Dengfeng Zhang, Jie Zhang,

Jinlong Zhang, Kaili Zhang, Lei Zhang, Peng Zhang, Rui Zhang, Shuzhou Zhang, Sijing Zhang, Tingyu Zhang, Xiangke Zhang, Xueyao Zhang, Yulei Zhang, Yuwen Ebony Zhang, Yixiang Zhang, Zhicai Zhang, Zhiqing Philippe Zhang, Haoran Zhao, Tongbin Zhao, Yuzhan Zhao, Zhengguo Zhao, Alexey Zhemchugov, Jinchao Zheng, Kai Zheng, Xiangxuan Zheng, Zhi Zheng, Dewen Zhong, Bing Zhou, Hao Zhou, Ning Zhou, Yong Zhou, You Zhou, Chengguang Zhu, Junjie Zhu, Yifan Zhu, Yingchun Zhu, Xuai Zhuang, Konstantin Zhukov, Nikolai Zimine, Joachim Zinsser, Michal Ziolkowski, Lidija Zivkovic, Antonio Zoccoli, Knut Zoch, Theodore Zorbas, Olga Zormpa, Wenkai Zou, Lukasz Zwalinski

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2.19.1.12 Can Bell inequalities be tested via scattering cross-section at colliders ? [265]

- **Authors:** Song Li, Wei Shen, Jin Min Yang
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2.19.1.13 Quantum detection of new physics in top-quark pair production at the LHC [266]

- **Authors:** Fabio Maltoni, Claudio Severi, Simone Tentori, Eleni Vryonidou
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2.19.1.14 Full quantum tomography of top quark decays [267]

- **Authors:** J.A. Aguilar-Saavedra
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2.19.1.15 Observation of quantum entanglement in top quark pair production in proton–proton collisions at $\sqrt{s} = 13$ TeV [268]

- **Authors:** Aram Hayrapetyan, Armen Tumasyan, Wolfgang Adam, Janik Walter Andrejkovic, Thomas Bergauer, Suman Chatterjee, Konstantinos Damanakis, Marko Dragicevic, Andre Hoang, Priya Sajid Hussain, Manfred Jeitler, Natascha Krammer, Ang Li, Dietrich Liko, Ivan Mikulec, Jochen Schieck, Robert Schöfbeck, Dennis Schwarz, Mangesh Sonawane, Sebastian Templ, Wolfgang Waltenberger, Claudia-Elisabeth Wulz,

Tahys Janssen, Thomas Van Laer, Pierre Van Mechelen, Nordin Breugelmans, Jorgen D'Hondt, Soumya Dansana, Alexandre De Moor, Martin Delcourt, Felix Heyen, Steven Lowette, Inna Makarenko, Denise Müller, Stefaan Tavernier, Michael Tytgat, Gerrit Patrick Van Onsem, Senne Van Putte, David Vannerom, Bugra Bilin, Barbara Clerbaux, Alope Kumar Das, Gilles De Lentdecker, Hugues Evard, Laurent Favart, Paraskevas Gianneios, Johny Jaramillo, Ali Khalilzadeh, Fakhri Alam Khan, Kyeongpil Lee, Mostafa Mahdavihorrani, Andrea Malara, Santiago Paredes, Muhammad Aamir Shahzad, Laurent Thomas, Max Vanden Bemden, Catherine Vander Velde, Pascal Vanlaer, Maarten De Coen, Didar Dobur, Gul Gokbulut, Yanwen Hong, Joscha Knolle, Luka Lambrecht, David Marckx, Kevin Mota Amarilo, Amrutha Samalan, Kirill Skovpen, Niels Van Den Bossche, Jan van der Linden, Liam Wezenbeek, Anna Bencke, Agni Bethani, Giacomo Bruno, Claudio Caputo, Jerome De Favereau De Jeneret, Christophe Delaere, Izzeddin Suat Donertas, Andrea Giammanco, Ahmet Oguz Guzel, Sandhya Jain, Vincent Lemaitre, Jindrich Lidrych, Paola Mastrapasqua, Tu Thong Tran, Sébastien Wertz, Gilvan Alves, Miguel Alves Gallo Pereira, Eduardo Coelho, Gilson Correia Silva, Carsten Hensel, Thales Menezes De Oliveira, Clemencia Mora Herrera, Arthur Moraes, Patricia Rebello Teles, Mariana Soeiro, Antonio Vilela Pereira, Walter Luiz Aldá Júnior, Mapse Barroso Ferreira Filho, Helena Brandao Malbouisson, Wagner Carvalho, Jose Chinellato, Eliza Melo Da Costa, Gustavo Gil Da Silveira, Dilson De Jesus Damiao, Sandro Fonseca De Souza, Raphael Gomes De Souza, Matheus Macedo, Jordan Martins, Luiz Mundim, Helio Nogima, Joao Pedro Pinheiro, Alberto Santoro, Andre Sznajder, Mauricio Thiel, Cesar Augusto Bernardes, Luigi Calligaris, Thiago Tomei, Eduardo De Moraes Gregores, Bruno Lopes Da Costa, Isabela Maietto Silverio, Pedro G. Mercadante, Sergio F. Novaes, Breno Orzari, Sandra Padula, Aleksandar Aleksandrov, Georgy Antchev, Roumyana Hadjiiska, Plamen Iaydjiev, Milena Misheva, Mariana Shopova, Georgi Sultanov, Anton Dimitrov, Leander Litov, Borislav Pavlov, Peicho Petkov, Anton Petrov, Elton Shumka, Sumit Keshri, David Nicolas Laroze Navarrete, Shalini Thakur, Tongguang Cheng, Tahir Javaid, Li Yuan, Zhen Hu, Zhengchen Liang, Jinfeng Liu, Kai Yi, Guo-Ming Chen, He-Sheng Chen, Mingshui Chen, Fabio Iemmi, Chun-Hua Jiang, Anshul Kapoor, Hongbo Liao, Zhenan Liu, Ramkrishna Sharma, Jia-ning Song, Junquan Tao, Chu Wang, Jin Wang, Zebin Wang, Huaqiao Zhang, Jingzhou Zhao, Antonis Agapitos, Yong Ban, Sen Deng, Botao Guo, Chuqiao Jiang, Andrew Levin, Congqiao Li, Qiang Li, Yajun Mao, Sitian Qian, Si-Jin Qian, Xuelong Qin, Xiaohu Sun, Dayong Wang, Heng Yang, Licheng Zhang, Yuzhe Zhao, Chen Zhou, Shuai Yang, Zhengyun You, Khawla Jaffel, Nan Lu, Gerry Bauer, Bolin Li, Jingqing Zhang, Xuyang Gao, Yuji Li, Zhen Lin, Chenfeng Lu, Meng Xiao, Carlos Avila, Diego Andres Barbosa Trujillo, Andrés Cabrera, Carlos Florez, Jorge Fraga, Jose Antonio Reyes Vega, Felipe Ramirez, Cèsar Rendón, Manuel Rodriguez, Anderson Alexis Ruales Barbosa, José David Ruiz Alvarez, Duje Giljanovic, Nikola Godinovic, Damir Lelas, Ana Sculac, Marko Kovac, Andro Petkovic, Toni Sculac, Pedrame Bargassa, Vuko Brigljevic, Bhakti Kanulal Chitroda, Dinko Fer-

encek, Kresimir Jakovcic, Saswat Mishra, Andrei Starodumov, Tatjana Susa, Alexandros Attikis, Konstantinos Christoforou, Andreas Hadjiagapiou, Christos Leonidou, Jehad Mousa, Charalambos Nicolaou, Leonidas Paizanos, Fotios Ptochos, Panos A. Razis, Hans Rykaczewski, Halil Saka, Anton Stepennov, Miroslav Finger, Michael Finger, Antonin Kveton, Edgar Carrera Jarrin, Yasser Assran, Basma El-mahdy, Sherif Elgammal, Muhammad Abdullah Abdulfattah Muhammad, Mohammed Mahmoud, Karl Ehataht, Mario Kadastik, Torben Lange, Saswati Nandan, Christine Nielsen, Joosep Pata, Martti Raidal, Laurits Tani, Christian Veelken, Henning Kirschenmann, Kenneth Osterberg, Mikko Voutilainen, Shudhashil Bharthuar, Nurfikri Bin Norjoharuddeen, Erik Brücken, Francisco Ignacio Garcia Fuentes, Patin Inkaew, Kimmo Tuomas Samuli Kallonen, Tapio Lampèn, Kati Lassila-Perini, Sami Lehti, Tomas Lindèn, Laura Martikainen, Mikael Myllymäki, Milla-maarit Rantanen, Hannu Siikonen, Jorma Tuominiemi, Panja-Riina Luukka, Henri Petrow, Marc Besancon, Fabrice Couderc, Marc De Jardin, Daniel Denegri, Jean-Louis Faure, Federico Ferri, Serguei Ganjour, Philippe Gras, Gautier Hamel de Monchenault, Mintu Kumar, Victor Lohezic, Julie Malcles, Francesco Orlandi, Louis Portales, André Rosowsky, Mehmet Özgür Sahin, Aurore Savoy-Navarro, Polina Simkina, Maksym Titov, Marta Tornago, Florian Beaudette, Giacomo Boldrini, Philippe Busson, Alessandra Cappati, Claude Charlot, Marco Chiusi, Florian Damas, Olivier Davignon, Adinda De Wit, Isaac Telford Ehle, Bruno Afonso Fontana Santos Alves, Shamik Ghosh, Andrew Gilbert, Raphael Granier de Cassagnac, Alexandre Hakimi, Bharadwaj Harikrishnan, Lida Kalipoliti, Geliang Liu, Matthew Nguyen, Christophe Ochando, Roberto Salerno, Jean-Baptiste Sauvan, Yves Sirois, Lourdes Urda Gómez, Elena Vernazza, Alexandre Zabi, Amina Zghiche, Jean-Laurent Agram, Jeremy Andrea, Dylan Apparu, Daniel Bloch, Jean-Marie Brom, Eric Christian Chabert, Caroline Collard, Saskia Falke, Ulrich Goerlach, Raphael Haeberle, Anne-Catherine Le Bihan, Meena Meena, Oceane Poncet, Gourab Saha, Mario Antoine Sessini, Pierre Van Hove, Paul Vaucelle, Adriano Di Florio, David Amram, Stephanie Beauceron, Benjamin Blancon, Gaelle Boudoul, Nicolas Chanon, Didier Contardo, Pierre Depasse, Candan Dozen, Houmani El Mamouni, Jean Fay, Susan Gascon, Maxime Gouzevitch, Christopher Greenberg, Gèrald Grenier, Bernard Ille, Elise Jourd’huy, Imad Baptiste Laktineh, Morgan Lethuillier, Laurent Mirabito, Stephane Perries, Arnab Purohit, Muriel Vander Donckt, Patrice Verdier, Jie Xiao, Arsen Khvedelidze, Irakli Lomidze, Zviad Tsamalaidze, Valeria Botta, Sandra Consuegra Rodríguez, Lutz Feld, Katja Klein, Martin Lipinski, Danilo Meuser, Alexander Pauls, Danyer Pérez Adán, Nicolas Röwert, Marius Teroerde, Svenja Diekmann, Alena Dodonova, Niclas Eich, Dmitry Eliseev, Frederic Engelke, Johannes Erdmann, Martin Erdmann, Peter Fackeldey, Benjamin Fischer, Thomas Hebbeker, Kerstin Hoepfner, Francesco Ivone, Alexander Jung, Ming-yan Lee, Florian Mausolf, Markus Merschmeyer, Arnd Meyer, Swagata Mukherjee, Dennis Noll, Fabian Nowotny, Andrey Pozdnyakov, Yannik Rath, Wahid Redjeb, Florian Rehm, Hans Reithler, Valentina Sarkisovi, Alexander Schmidt, Archana Sharma, Jan Lukas Spah, Annika Stein, Felipe Torres Da Silva De Araujo, Sebastian

Wiedenbeck, Shawn Zaleski, Christian Dziwok, Günter Flügge, Thomas Kress, Andreas Nowack, Oliver Pooth, Achim Stahl, Tim Ziemons, Alexander Zotz, Henriette Aarup Petersen, Maria Aldaya Martin, Juliette Alimena, Simone Amoroso, Ying An, Jorn Bach, Samuel Baxter, Maryam Bayatmakou, Hugo Becerril Gonzalez, Olaf Behnke, Alberto Belvedere, Freya Blekman, Kerstin Borrás, Alan Campbell, Andrea Cardini, Chun Cheng, Federica Colombina, Malinda De Silva, Guenter Eckerlin, Doris Eckstein, Luis Ignacio Estevez Banos, Oleg Filatov, Elisabetta Gallo, Achim Geiser, Valentina Guglielmi, Moritz Guthoff, Andreas Hinzmann, Laurids Jeppe, Benno Kaech, Matthias Kasemann, Claus Kleinwort, Roman Kogler, Matthias Komm, Dirk Krücker, Wolfgang Lange, Daina Leyva Pernia, Katerina Lipka, Wolfgang Lohmann, Florian Lorkowski, Rainer Mankel, Isabell-Alissandra Melzer-Pellmann, Mikel Mendizabal Morentin, Andreas Bernhard Meyer, Gabriele Milella, Keila Moral Figueroa, Andreas Mussgiller, Lakshmi Priya Nair, Jeremi Niedziela, Andreas Nürnberg, Younes Otari, Jiwon Park, Evan Ranken, Alexei Raspereza, Daniil Rastorguev, Jonas Rübenach, Lovisa Rygaard, Alessia Saggio, Moritz Scham, Simon Schnake, Paul Schütze, Christian Schwanenberger, Daria Selivanova, Konstantin Sharko, Mykyta Shchedrolosiev, Dominic Stafford, Federico Vazzoler, Ana Ventura Barroso, Roberval Walsh, Di Wang, Qun Wang, Yiwen Wen, Katarzyna Wichmann, Lucas Wiens, Christoph Wissing, Yewon Yang, André Zimmermann Castro Santos, Anna Albrecht, Steffen Albrecht, Massimiliano Antonello, Samuel Bein, Lisa Benato, Sven Bollweg, Matteo Bonanomi, Patrick Connor, Karim El Morabit, Yannick Fischer, Erika Garutti, Alexander Grohsjean, Johannes Haller, Henrik Ronald Jabusch, Gregor Kasieczka, Philip Keicher, Robert Klanner, William Korcari, Tobias Kramer, Chin-chia Kuo, Viktor Kutzner, Finn Labe, Johannes Lange, Artur Lobanov, Christopher Matthies, Louis Moureaux, Malte Mrowietz, Aliya Nigamova, Yuval Nissan, Alexander Paasch, Karla Josefina Pena Rodriguez, Tobias Quadfasel, Bianca Raciti, Marcel Rieger, Daniel Savoiu, Jörg Schindler, Peter Schleper, Matthias Schröder, Joern Schwandt, Manuel Sommerhalder, Hartmut Stadie, Georg Steinbrück, Alexandra Tews, Moritz Wolf, Sebastian Brommer, Maximilian Burkart, Erik Butz, Thorsten Chwalek, Alexander Dierlamm, Alexander Droll, Umut Elicabuk, Nils Faltermann, Manuel Giffels, Artur Gottmann, Frank Hartmann, Robin Hofsaess, Maximilian Horzela, Ulrich Husemann, Jan Kieseler, Markus Klute, Roland Koppenhöfer, Jay Mathew Lawhorn, Marco Link, Adelina Lintuluoto, Benedikt Maier, Stefan Maier, Soureek Mitra, Michele Mormile, Thomas Müller, Max Neukum, Minseok Oh, Emanuel Pfeffer, Matteo Presilla, Gunter Quast, Klaus Rabbertz, Brendan Regnery, Nikita Shadskiy, Ivan Shvetsov, Hans-Jürgen Simonis, Lars Sowa, Lea Stockmeier, Komal Taukeer, Maria Toms, Nicolò Trevisani, Ralf Florian Von Cube, Michael Wassmer, Sebastian Wieland, Florian Wittig, Roger Wolf, Xunwu Zuo, Georgios Anagnostou, Georgios Daskalakis, Aristotelis Kyriakis, Alkiviadis Papadopoulos, Anna Stakia, Pantelis Kontaxakis, Georgios Melachroinos, Zacharias Painesis, Ioanna Papavergou, Ioannis Paraskevas, Niki Saoulidou, Konstantinos Theofilatos, Eirini Tziaferi, Konstantinos Velidis, Ilias Zisopoulos, Georgios Bakas, Theodoros Chatzistavrou, Georgia Karapostoli,

Konstantinos Kousouris, Ioannis Papakrivopoulos, Eirini Siamarkou, Georgios Tsipolitis, Anna Zacharopoulou, Kosmas Adamidis, Ioannis Bestintzanos, Ioannis Evangelou, Costas Foudas, Christos Kamtsikis, Panagiotis Katsoulis, Panagiotis Kokkas, Polidamas Georgios Kosmoglou Kioseoglou, Nikolaos Manthos, Ioannis Papadopoulos, John Strologas, Csaba Hajdu, Dezso Horvath, Krisztina Márton, Attila József Rádl, Ferenc Sikler, Viktor Veszpremi, Mate Csanad, Krisztián Farkas, Anna Fehérkúti, Mahmud Moussa Abdelkhalek Gadallah, Ármín Kadlecsek, Péter Major, Gabriella Pasztor, Gabor Istvan Veres, Balázs Ujvári, Gyula Zilizi, György Bencze, Sándor Czellar, József Molnár, Zoltan Szillasi, Frigyes Nemes, Tamas Novak, Sunil Bansal, Suman Bala Beri, Vipin Bhatnagar, Geetanjali Chaudhary, Sushil Chauhan, Nitish Dhingra, Amandeep Kaur, Anureet Kaur, Harjot Kaur, Manjit Kaur, Sunil Kumar, Kaur Sandeep, Tanvi Sheokand, Jasbir Singh, Aayushi Singla, Asar Ahmed, Ashutosh Bhardwaj, Arjun Chhetri, Brajesh C. Choudhary, Ashok Kumar, Arun Kumar, Mohammad Naimuddin, Kirti Ranjan, Mahesh Kumar Saini, Saumya Saumya, Sweta Baradia, Soumyadip Barman, Satyaki Bhattacharya, Suman Das Gupta, Shubham Dutta, Suchandra Dutta, Subir Sarkar, Mohammad Mobassir Ameen, Prafulla Kumar Behera, Subash Chandra Behera, Sayan Chatterjee, Ganapati Dash, Pranati Jana, Pritam Kalbhor, Samadhan Kamble, Jyothsna Rani Komaragiri, Deepak Kumar, Prabhat Ranjan Pujahari, Nihar Ranjan Saha, Ashish Sharma, Anup Kumar Sikdar, Rohit Kumar Singh, Pranjal Verma, Sadhana Verma, Anusree Vijay, Shashikant Dugad, Gagan Bihari Mohanty, Bibhuti Parida, Mukund Shelake, Pruthvi Suryadevara, Anirban Bala, Sudeshna Banerjee, Rajdeep Mohan Chatterjee, Monoranjan Guchait, Shilpi Jain, Atul Jaiswal, Sanjeev Kumar, Gobinda Majumder, Kajari Mazumdar, Shubhi Parolia, Aravind Thachayath, Seema Bahinipati, Chandiprasad Kar, Dipak Maity, Prolay Mal, Tribeni Mishra, Vinaya Krishnan Muralleedharan Nair Bindhu, Kousik Naskar, Aruna Nayak, Samarendra Nayak, Kuldeep Pal, Priyanka Sadangi, Sanjay Kumar Swain, Sanu Varghese, Diwakar Vats, Shriniketan Acharya, Alpana Alpana, Sourabh Dube, Bhawna Gomber, Prachurjya Hazarika, Bhumika Kansal, Arnab Laha, Bisnupriya Sahu, Seema Sharma, Kumar Yash Vaish, Hamed Bakhshiansohi, Abideh Jafari, Maryam Zeinali, Sima Bashiri, Shirin Chenarani, Seyed Mohsen Etesami, Yasaman Hosseini, Mohsen Khakzad, Elham Khazaie, Mojtaba Mohammadi Najafabadi, Seddigheh Tizchang, Marta Felcini, Martin Grunewald, Marcello Abbrescia, Anna Colaleo, Donato Creanza, Brunella D'Anzi, Nicola De Filippis, Mauro De Palma, Walaa Elmetenawee, Luigi Fiore, Giuseppe Iaselli, Luigi Longo, Magdy Louka, Giorgio Maggi, Marcello Maggi, Ilirjan Margjeka, Vincenzo Mastrapasqua, Salvatore My, Salvatore Nuzzo, Antonello Pellecchia, Alexis Pompili, Gabriella Pugliese, Raffaella Radogna, Dayron Ramos, Antonio Ranieri, Lucia Silvestris, Federica Maria Simone, Ümit Sözbilir, Anna Stamerra, Donato Troiano, Rosamaria Venditti, Piet Verwilligen, Angela Zaza, Giovanni Abbiendi, Carlo Battilana, Daniele Bonacorsi, Paolo Capiluppi, Andrea Castro, Francesca Romana Cavallo, Marco Cuffiani, Gaetano-Marco Dallavalle, Tommaso Diotallevi, Fabrizio Fabbri, Alessandra Fanfani, Daniele Fasanella, Paolo Giacomelli, Luca Giommi, Claudio Grandi, Luigi Guiducci, Sergio Lo

Meo, Marco Lorusso, Leonardo Lunerti, Stefano Marcellini, Gianni Masetti, Francesco Navarra, Giulia Paggi, Andrea Perrotta, Federica Primavera, Antonio Rossi, Simone Rossi Tisbeni, Tiziano Rovelli, Gian Piero Siroli, Salvatore Costa, Alessandro Di Mattia, Alessandro Lapertosa, Renato Potenza, Alessia Tricomi, Cristina Tuve, Panagiotis Assiouras, Giuseppe Barbagli, Giulio Bardelli, Benedetta Camaiani, Antonio Cassese, Rudy Ceccarelli, Vitaliano Ciulli, Carlo Civinini, Raffaello D'Alessandro, Ettore Focardi, Tomas Kello, Giuseppe Latino, Piergiulio Lenzi, Mattia Lizzo, Marco Meschini, Simone Paoletti, Alkis Papanastassiou, Giacomo Sguazzoni, Lorenzo Viliani, Luigi Benussi, Stefano Bianco, Sabino Meola, Davide Piccolo, Pierre Chatagnon, Fabrizio Ferro, Enrico Robutti, Silvano Tosi, Andrea Benaglia, Francesco Brivio, Flavia Cetorelli, Federico De Guio, Mauro Emanuele Dinardo, Paolo Dini, Simone Gennai, Raffaele Gerosa, Alessio Ghezzi, Pietro Govoni, Luca Guzzi, Marco Toliman Lucchini, Martina Malberti, Sandra Malvezzi, Andrea Massironi, Dario Menasce, Luigi Moroni, Marco Paganoni, Simona Palluotto, Daniele Pedrini, Aurora Perego, Bianca Sofia Pinolini, Giorgio Pizzati, Stefano Ragazzi, Tommaso Tabarelli de Fatis, Salvatore Buontempo, Antimo Cagnotta, Francesco Carnevali, Nicola Cavallo, Francesco Fabozzi, Alberto Orso Maria Iorio, Luca Lista, Pierluigi Paolucci, Biagio Rossi, Rocco Ardino, Patrizia Azzi, Nicola Bacchetta, Pierluigi Bortignon, Gabriele Bortolato, Alberto Bragagnolo, Andrea Claudio Maria Bulla, Roberto Carlin, Paolo Checchia, Tommaso Dorigo, Ugo Gasparini, Franco Gonella, Andrea Gozzelino, Enrico Lusiani, Martino Margoni, Anna Teresa Meneguzzo, Matteo Migliorini, Jacopo Pazzini, Paolo Ronchese, Roberto Rossin, Franco Simonetto, Mia Tosi, Andrea Triossi, Sandro Ventura, Marco Zanetti, Pierluigi Zotto, Alberto Zucchetta, Gianni Zumerle, Chiara Aimè, Alessandro Braghieri, Simone Calzaferri, Davide Fiorina, Paolo Montagna, Valerio Re, Cristina Riccardi, Paola Salvini, Ilaria Vai, Paolo Vitulo, Sehar Ajmal, Maria Elena Ascioti, Gian Mario Bilei, Costanza Carrivale, Diego Ciangottini, Livio Fanò, Matteo Magherini, Valentina Mariani, Mauro Menichelli, Francesco Moscatelli, Alessandro Rossi, Attilio Santocchia, Daniele Spiga, Tommaso Tedeschi, Cristina Andreea Alexe, Patrick Asenov, Paolo Azzurri, Giuseppe Bagliesi, Rajarshi Bhattacharya, Lorenzo Bianchini, Tommaso Boccali, Edoardo Bossini, Davide Bruschini, Rino Castaldi, Maria Agnese Ciocci, Marco Cipriani, Valeria D'Amante, Roberto Dell'Orso, Silvio Donato, Alessandro Giassi, Franco Ligabue, Andrea Carlo Marini, Diego Matos Figueiredo, Alberto Messineo, Marco Musich, Fabrizio Palla, Andrea Rizzi, Gigi Rolandi, Suvankar Roy Chowdhury, Tanmay Sarkar, Angelo Scribano, Paolo Spagnolo, Roberto Tenchini, Guido Tonelli, Nicola Turini, Francesco Vaselli, Andrea Venturi, Piero Giorgio Verdini, Cristian Baldenegro Barrera, Patrizia Barria, Chiara Basile, Mattia Campana, Francesca Cavallari, Leticia Cunqueiro Mendez, Daniele Del Re, Emanuele Di Marco, Marcella Diemoz, Filippo Errico, Egidio Longo, Jelena Mijuskovic, Giovanni Organtini, Francesco Pandolfi, Riccardo Paramatti, Claudio Quaranta, Shahram Rahatlou, Chiara Rovelli, Francesco Santanastasio, Livia Soffi, Nicola Amapane, Roberta Arcidiacono, Stefano Argiro, Michele Arneodo, Nazar Bartosik, Riccardo Bellan, Andrea Bellora, Cristina Biino, Cecilia Borca, Nicolo Cartiglia,

Marco Costa, Roberto Covarelli, Natale Demaria, Linda Finco, Michael Grippo, Bilal Kiani, Federica Legger, Fabio Luongo, Chiara Mariotti, Lazar Markovic, Silvia Maselli, Alberto Mecca, Luca Menzio, Paolo Meridiani, Ernesto Migliore, Marco Monteno, Roberto Mulargia, Maria Margherita Obertino, Giacomo Ortona, Luca Pacher, Nadia Pastrone, Mario Pelliccioni, Marta Ruspa, Federico Siviero, Valentina Sola, Ada Solano, Amedeo Staiano, Cristiano Tarricone, Daniele Trocino, Giulio Umoret, Robert White, Jyoti Babbar, Stefano Belforte, Vieri Candelise, Massimo Casarsa, Fabio Cossutti, Ksenia De Leo, Giuseppe Della Ricca, Sunil Dogra, Jieun Hong, Changgi Huh, Bobae Kim, Jiwoong Kim, Dongyub Lee, Hakseong Lee, Seh Wook Lee, Chang-Seong Moon, Young Do Oh, Min Sang Ryu, Sezen Sekmen, Bongho Tae, Yu Chul Yang, Min Suk Kim, Gyeonghwan Bak, Piljun Gwak, Hyunchul Kim, Dong Ho Moon, Ece Asilar, Jieun Choi, Donghyun Kim, Tae Jeong Kim, Jeremie Alexandre Merlin, Yeonsu Ryou, Suyong Choi, Seyoung Han, Byung-Sik Hong, Kisoo Lee, Kyong Sei Lee, Soohwan Lee, Jaehyeok Yoo, Junghwan Goh, Seungjin Yang, Hyunsoo Kim, Yongsun Kim, Seungjun Lee, John Almond, Ji Hwan Bhyun, Junho Choi, Jin Choi, Won Jun, Junho Kim, Sanghyun Ko, Hyejin Kwon, Haneol Lee, Joonbin Lee, Jeongeun Lee, Byung Hun Oh, Sung Bin Oh, Hyonsan Seo, Unki Yang, Inseok Yoon, Woojin Jang, Da Young Kang, Yechan Kang, Seulgi Kim, Byeonghak Ko, Jason Sang Hun Lee, Yunjae Lee, Inkyu Park, Youn Roh, Ian Watson, Seungkyu Ha, Hwi Dong Yoo, Minuk Choi, Mi Ran Kim, Heewon Lee, Younghoon Lee, Intae Yu, Taha Beyrouthy, Yousef Gharbia, Karlis Dreimanis, Antra Gaile, Conrado Munoz Diaz, Dace Osite, Guntis Pikurs, Andris Potrebko, Markus Seidel, Dimitrios Sidiropoulos Kontos, Normunds Ralfs Strautnieks, Marijus Ambrozias, Andrius Juodagalvis, Aurelijus Rinkevicius, Gintautas Tamulaitis, Imran Yusuff, Zukhaimira Zolkapli, Jose F. Benitez, Alfredo Castaneda Hernandez, Hedwin Aaron Encinas Acosta, Luis Gabriel Gallegos Maríñez, Moisés León Coello, Javier Alberto Murillo Quijada, Ashish Sehrawat, Lizardo Valencia Palomo, Gabriel Ayala, Heriberto Castilla-Valdez, Horacio Crotte Ledesma, Eduard De La Cruz-Burelo, Ivan Heredia-De La Cruz, Ricardo Lopez-Fernandez, Jhovanny Mejia Guisao, Cesar Atzin Mondragon Herrera, Alberto Sánchez Hernández, Cristina Oropeza Barrera, Dalia Lucero Ramirez Guadarrama, Mateo Ramírez García, Irais Bautista, Isabel Pedraza, Humberto Antonio Salazar Ibarguen, Cecilia Uribe Estrada, Itana Bujanja, Natasa Raicevic, Philip H. Butler, Ashfaq Ahmad, Muhammad Irfan Asghar, Ali Awais, Muhammad Imran Malik Awan, Hafeez R. Hoorani, Wajid Ali Khan, Valentina Avati, Leszek Grzanka, Maciej Malawski, Helena Bialkowska, Michal Bluj, Maciej Górski, Malgorzata Kazana, Michal Szleper, Piotr Zalewski, Karol Bunkowski, Krzysztof Doroba, Artur Kalinowski, Marcin Konecki, Jan Krolkowski, Krzysztof Pozniak, Wojciech Zabolotny, Mariana Araujo, Diogo Bastos, Cristóvão Beirão Da Cruz E Silva, Alessio Boletti, Marco Bozzo, Tiziano Camporesi, Giacomo Da Molin, Pietro Faccioli, Michele Gallinaro, Jonathan Hollar, Nuno Leonardo, Giovanni Battista Marozzo, Tahereh Sadat Niknejad, Achille Petrilli, Matteo Pisano, Joao Seixas, Joao Varela, Johan Wilfried Wulff, Petar Adzic, Predrag Milenovic, Milos Dordevic, Jovan Milosevic, Vladimir Rekovic, Juan Alcaraz Maestre, Cristina Fernan-

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Matthew Jones, Andreas Werner Jung, Abraham Mathew Koshy, Miaoyuan Liu, Giulia Negro, Norbert Neumeister, Garyfallia Paspalaki, Stefan Piperov, Valerie Scheurer, Jan-Frederik Schulte, Milan Stojanovic, Jason Thieman, Amandeep Kaur, Fuqiang Wang, Andrew Wildridge, Wei Xie, James Dolen, Neeti Parashar, Atanu Pathak, Darin Acosta, Taylor Carnahan, Karl Matthew Ecklund, Pedro Josè Fernández Manteca, Sarah Freed, Parker Gardner, Frank J.M. Geurts, Iason Krommydas, Wei Li, Jiazhao Lin, Osvaldo Miguel Colin, Brian Paul Padley, Radia Redjimi, John Rotter, Efe Yigitbasi, Yousen Zhang, Arie Bodek, Pawel de Barbaro, Regina Demina, Joseph Lynn Dulemba, Aran Garcia-Bellido, Otto Hindrichs, Aleko Khukhunaishvili, Nukulsinh Parmar, Pavel Parygin, Rhys Taus, Brandon Chiarito, John Paul Chou, Steven Vincent Clark, Divya Gadkari, Yuri Gershtein, Eva Halkiadakis, Maximilian Heindl, Connor Houghton, David Jaroslawski, Sotiroulla Konstantinou, Ian Laflotte, Amitabh Lath, Roy Montalvo, Kevin Nash, Joseph Reichert, Hardik Routray, Prafulla Saha, Sevil Salur, Steve Schnetzer, Sunil Somalwar, Robert Stone, Steffie Ann Thayil, Scott Thomas, Jay Vora, Hui Wang, Daniel Ally, Andrès G. Delannoy, Sara Fiorendi, Samuel Higginbotham, Tova Holmes, Ankush Reddy Kanuganti, Nimmitha Karunarathna, Lawrence Lee, Emery Nibigira, Stefan Spanier, Devin Aebi, Muhammad Ahmad, Towsifa Akhter, Othmane Bouhali, Ricardo Eusebi, Jason Gilmore, Tao Huang, Teruki Kamon, Hyunyong Kim, Sifu Luo, Ryan Mueller, David Overton, Denis Rathjens, Alexei Safonov, Nural Akchurin, Jordan Damgov, Niramay Gogate, Vinay Hegde, Adil Hussain, Yelbir Kazhykarim, Kamal Lamichhane, Sung Won Lee, Aaron Mankel, Timo Peltola, Igor Volobouev, Eric Appelt, Yi Chen, Senta Greene, Alfredo Gurrola, Willard Johns, Raghav Kunawalkam Elayavalli, Andrew Melo, Francesco Romeo, Paul Sheldon, Shengquan Tuo, Julia Velkovska, Jussi Viinikainen, Bryan Cardwell, Heewon Chung, Bradley Cox, John Hakala, Robert Hirosky, Alexander Ledovskoy, Christopher Neu, Saptaparna Bhattacharya, Paul Edmund Karchin, Anagha Aravind, Sunanda Banerjee, Kevin Black, Tulika Bose, Sridhara Dasu, Isabelle De Bruyn, Pieter Everaerts, Camilla Galloni, He He, Matthew Herndon, Alain Herve, Charis Kleio Koraka, Armando Lanaro, Richard Loveless, Jithin Madhusudanan Sreekala, Abhishikth Mallampalli, Abdollah Mohammadi, Susmita Mondal, Ganesh Parida, Laurent Pètrè, Deborah Pinna, Alexander Savin, Victor Shang, Varun Sharma, Wesley H. Smith, Dylan Teague, Ho Fung Tsoi, Wren Vetens, Abigail Warden, Serguei Afanasiev, Vadim Alexakhin, Vladimir Andreev, Yuri Andreev, Tagir Aushev, Maksim Azarkin, Anton Babaev, Vladimir Blinov, Edouard Boos, Vladislav Borshch, Dzmitry Budkouski, Viacheslav Bunichev, Vladimir Chekhovsky, Ruslan Chistov, Mikhail Danilov, Alexander Dermenev, Tatyana Dimova, Dmitry Druzhkin, Mikhail Dubinin, Lev Dudko, Alexander Ershov, Genady Gavrilov, Vladimir Gavrilov, Sergei Gninenko, Victor Golovtcov, Nikolai Golubev, Igor Golutvin, Ilya Gorbunov, Yury Ivanov, Vassili Kachanov, Vladimir Karjavine, Anton Karneyeu, Victor Kim, Martin Kirakosyan, Dmitrii Kirpichnikov, Mikhail Kirsanov, Vyacheslav Klyukhin, Dmitri Konstantinov, Vladimir Koronkov, Aleksei Kozyrev, Nikolai Krasnikov, Alexander Lanev, Petr Levchenko, Na-

talia Lychkovskaya, Vladimir Makarenko, Alexander Malakhov, Anastasia Markina, Viktor Matveev, Victor Murzin, Alexander Nikitenko, Stepan Obraztsov, Vadim Oreshkin, Vladimir Palichik, Victor Perelygin, Maxim Perfilov, Sergey Polikarpov, Vladimir Popov, Olesia Radchenko, Maria Savina, Viktor Savrin, Vladislav Shalaev, Sergey Shmatov, Siarhei Shulha, Yuri Skovpen, Sergei Slabospitskii, Vitaly Smirnov, Dmitry Sosnov, Valentin Sulimov, Adel Terkulov, Oleg Teryaev, Irina Tlisova, Alexander Toropin, Lev Uvarov, Andrey Uzunian, Petr Volkov, Alexey Vorobyev, George Vorotnikov, Nikolay Voytishin, Bekhzod S. Yuldashev, Anatoli Zarubin, Ilia Zhizhin, Alexander Zhokin

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2.19.1.16 Measurements of polarization and spin correlation and observation of entanglement in top quark pairs using $\ell^+\ell^- \rightarrow t\bar{t} + \text{jets}$ events from proton-proton collisions at $\sqrt{s} = 13.6$ TeV

- **Authors:** Aram Hayrapetyan, Armen Tumasyan, Wolfgang Adam, Janik Walter Andrejkovic, Lisa Benato, Thomas Bergauer, Sandhya Jain, Muhammad Abdullah Abdulfattah Muhammad, Thomas Müller, Sayan Chatterjee, Konstantinos Damanakis, Marko Dragicevic, Priya Sajid Hussain, Manfred Jeitler, Natascha Krammer, Ang Li, Dietrich Liko, Ivan Mikulec, Jochen Schieck, Robert Schöffbeck, Dennis Schwarz, Mangesh Sonawane, Wolfgang Waltenberger, Claudia-Elisabeth Wulz, Tahys Janssen, Thomas Van Laer, Pierre Van Mechelen, Nordin Breugelmans, Jorgen D'Hondt, Soumya Dansana, Alexandre De Moor, Martin Delcourt, Felix Heyen, Steven Lowette, Inna Makarenko, Denise Müller, Stefaan Tavernier, Michael Tytgat, Gerrit Patrick Van Onsem, Senne Van Putte, David Vannerom, Bugra Bilin, Barbara Clerbaux, Alope Kumar Das, Isabelle De Bruyn, Gilles De Lentdecker, Hugues Evard, Laurent Favart, Paraskevas Gianneios, Johny Jaramillo, Ali Khalilzadeh, Fakhri Alam Khan, Kyeongpil Lee, Andrea Malara, Santiago Paredes, Muhammad Aamir Shahzad, Laurent Thomas, Max Vanden Bemden, Catherine Vander Velde, Pascal Vanlaer, Maarten De Coen, Didar Dobur, Gul Gokbulut, Yanwen Hong, Joscha Knolle, Luka Lambrecht, David Marckx, Kevin Mota Amarilo, Kirill Skovpen, Niels Van Den Bossche, Jan van der Linden, Liam Wezenbeek, Anna Benecke, Agni Bethani, Giacomo Bruno, Claudio Caputo, Jerome De Favereau De Jeneret, Christophe Delaere, Izzeddin Suat Donertas, Andrea Giammanco, Ahmet Oguz Guzel, Sa. Jain, Vincent Lemaitre, Jindrich Lidrych, Paola Mastrapasqua, Tu Thong Tran, Semra Turkcapar, Gilvan Alves, Eduardo Coelho, Gilson Correia Silva, Carsten Hensel, Thales Menezes De Oliveira, Clemencia Mora Herrera, Patricia Rebello Teles, Mariana Soeiro, Edmilson José Tonelli Manganote, Antonio Vilela Pereira, Walter Luiz Aldá Júnior, Mapse Barroso Ferreira Filho, Helena Brandao Malbouisson, Wagner Carvalho, Jose Chinellato, Eliza Melo Da Costa, Gustavo Gil Da Silveira, Dilson De Jesus Damiao, Sandro Fonseca De Souza, Raphael Gomes De Souza, Tulio Laux

Kuhn, Matheus Macedo, Jordan Martins, Luiz Mundim, Helio Nogima, Joao Pedro Pinheiro, Alberto Santoro, Andre Sznajder, Mauricio Thiel, Cesar Augusto Bernardes, Luigi Calligaris, T.R. Fernandez Perez Tomei, Eduardo De Moraes Gregores, Isabela Maietto Silverio, Pedro G. Mercadante, Sergio F. Novaes, Breno Orzari, Sandra S. Padula, Aleksandar Aleksandrov, Georgy Antchev, Roumyana Hadjiiska, Plamen Iaydjiev, Milena Misheva, Mariana Shopova, Georgi Sultanov, Anton Dimitrov, Leander Litov, Borislav Pavlov, Peicho Petkov, Anton Petrov, Elton Shumka, Sumit Keshri, David Nicolas Laroze Navarrete, Shalini Thakur, Tongguang Cheng, Tahir Javaid, Li Yuan, Zhen Hu, Zhengchen Liang, Jinfeng Liu, Guo-Ming Chen, He-Sheng Chen, Mingshui Chen, Fabio Iemmi, Chun-Hua Jiang, Anshul Kapoor, Hongbo Liao, Zhenan Liu, Ramkrishna Sharma, Jia-ning Song, Junquan Tao, Shilpi Jain, Hakseong Lee, Yongsun Kim, Young Wan Kim, Heewon Lee, Christina Wang, Jin Wang, Drew Baden, Zhangqier Wang, Huaqiao Zhang, Jingzhou Zhao, Antonis Agapitos, Yong Ban, Sen Deng, Botao Guo, Chuqiao Jiang, Andrew Levin, Congqiao Li, Qiang Li, Yajun Mao, Sitian Qian, Si-Jin Qian, Xuelong Qin, Xiaohu Sun, Dayong Wang, Heng Yang, Licheng Zhang, Yuzhe Zhao, Chen Zhou, Shuai Yang, Zhengyun You, Khawla Jaffel, Nan Lu, Gerry Bauer, Bolin Li, Kai Yi, Jingqing Zhang, Yuji Li, Zhen Lin, Chenfeng Lu, Meng Xiao, Carlos Avila, Diego Andres Barbosa Trujillo, Andr s Cabrera, Carlos Florez, Jorge Fraga, Jose Antonio Reyes Vega, Felipe Ramirez, C sar Rend n, Manuel Rodriguez, Anderson Alexis Ruales Barbosa, Jos  David Ruiz Alvarez, Duj  Giljanovic, Nikola Godinovic, Damir Lelas, Ana Sculac, Marko Kovac, Andro Petkovic, Toni Sculac, Pedrame Bargassa, Vuko Brigljevic, Bhakti Kanulal Chitroda, Dinko Ferencek, Kresimir Jakovcic, Andrei Starodumov, Tatjana Susa, Alexandros Attikis, Konstantinos Christoforou, Andreas Hadjiagapiou, Christos Leonidou, Jehad Mousa, Charalambos Nicolaou, Leonidas Paizanos, Fotios Ptochos, Panos A. Razis, Hans Rykaczewski, Halil Saka, Anton Stepenov, Miroslav Finger, Michael Finger, Antonin Kveton, Edgar Carrera Jarrin, Yasser Assran, Basma El-mahdy, Sherif Elgammal, M. Abdullah Al-Mashad, Mohammed Mahmoud, Karl Ehataht, Mario Kadastik, Torben Lange, Saswati Nandan, Christine Nielsen, Joosep Pata, Martti Raidal, Laurits Tani, Christian Veelken, Henning Kirschenmann, Kenneth Osterberg, Mikko Voutilainen, Shudhashil Bharthuar, Nurfikri Bin Norjoharuddeen, Erik Br cken, Francisco Ignacio Garcia Fuentes, Patin Inkaew, Kimmo Tuomas Samuli Kallonen, Tapio Lamp n, Kati Lassila-Perini, Sami Lehti, Tomas Lind n, Mikael Myllym ki, Milla-maarit Rantanen, Hannu Siikonen, Jorma Tuominiemi, Panja-Riina Luukka, Henri Petrow, Marc Besancon, Fabrice Couderc, Marc Dejardin, Daniel Denegri, Jean-Louis Faure, Federico Ferri, Serguei Gansjour, Philippe Gras, Gautier Hamel de Monchenault, Mintu Kumar, Victor Lohezic, Julie Malcles, Francesco Orlandi, Louis Portales, Andr  Rosowsky, Mehmet  zg r Sahin, Aurore Savoy-Navarro, Polina Simkina, Maksym Titov, Marta Tornago, Florian Beaudette, Giacomo Boldrini, Philippe Busson, Alessandra Cappati, Claude Charlot, Marco Chiusi, Th o Damien Cuisset, Florian Damas, Olivier Davignon, Adinda De Wit, Isaac Telford Ehle, Bruno Afonso Fontana Santos Alves, Shamik Ghosh, Andrew

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cher, Erika Garutti, Alexander Grohsjean, Johannes Haller, Daniel Hundhausen, Henrik Ronald Jabusch, Gregor Kasieczka, Philip Keicher, Robert Klanner, William Korrari, Tobias Kramer, Chin-chia Kuo, Viktor Kutzner, Finn Labe, Johannes Lange, Artur Lobanov, Christopher Matthies, Louis Moureaux, Malte Mrowietz, Aliya Nigamova, Yuval Nissan, Alexander Paasch, Karla Josefina Pena Rodriguez, Tobias Quadfasel, Bianca Raciti, Marcel Rieger, Daniel Savoiu, Jörg Schindler, Peter Schleper, Matthias Schröder, Joern Schwandt, Manuel Sommerhalder, Hartmut Stadie, Georg Steinbrück, Alexandra Tews, Bogdan Wiederspan, Moritz Wolf, Sebastian Brommer, Erik Butz, Thorsten Chwalek, Alexander Dierlamm, Alexander Droll, Umut Elicabuk, Nils Faltermann, Manuel Giffels, Artur Gottmann, Frank Hartmann, Robin Hofsaess, Maximilian Horzela, Ulrich Husemann, Jan Kieseler, Markus Klute, Roland Koppenhöfer, Olha Lavoryk, Jay Mathew Lawhorn, Marco Link, Adelina Lintuluoto, Stefan Maier, Soureek Mitra, Michele Mormile, Th. Müller, Max Neukum, Minseok Oh, Emanuel Pfeffer, Matteo Presilla, Gunter Quast, Klaus Rabbertz, Brendan Regnery, Nikita Shadskiy, Ivan Shvetsov, Hans-Jürgen Simonis, Lars Sowa, Lea Stockmeier, Komal Tauqeer, Maria Toms, Nicolò Trevisani, Ralf Florian Von Cube, Michael Wassmer, Sebastian Wieland, Florian Wittig, Roger Wolf, Xunwu Zuo, Georgios Anagnostou, Georgios Daskalakis, Aristotelis Kyriakis, Alkiviadis Papadopoulos, Anna Stakia, Pantelis Kontaxakis, Georgios Melachroinos, Zacharias Painesis, Ioanna Papavergou, Ioannis Paraskevas, Niki Saoulidou, Konstantinos Theofilatos, Eirini Tziaferi, Konstantinos Vellidis, Ilias Zisopoulos, Georgios Bakas, Theodoros Chatzistavrou, Georgia Karapostoli, Konstantinos Kousouris, Ioannis Papakrivopoulos, Eirini Siamarkou, Georgios Tsipolitis, Anna Zacharopoulou, Ioannis Bestintzanos, Ioannis Evangelou, Costas Foudas, Christos Kamtsikis, Panagiotis Katsoulis, Panagiotis Kokkas, Polidamas Georgios Kosmoglou Kioseoglou, Nikolaos Manthos, Ioannis Papadopoulos, John Strologas, Csaba Hajdu, Dezso Horvath, Krisztina Márton, Attila József Rádl, Ferenc Sikler, Viktor Veszpremi, Mate Csanad, Krisztián Farkas, Anna Fehérkúti, Mahmud Moussa Abdelkhalek Gadallah, Ármín Kadlecsek, Péter Major, Gabriella Pasztor, Gabor Istvan Veres, Balázs Ujvári, Gyula Zilizi, György Bencze, Sandor Czellar, József Molnar, Zoltan Szillasi, Frigyes Nemes, Tamas Novak, Sunil Bansal, Suman Bala Beri, Vipin Bhatnagar, Geetanjali Chaudhary, Sushil Chauhan, Nitish Dhingra, Amandeep Kaur, Anureet Kaur, Harjot Kaur, Manjit Kaur, Sanjeev Kumar, Tanvi Sheokand, Jasbir Singh, Aayushi Singla, Asar Ahmed, Ashutosh Bhardwaj, Arjun Chhetri, Brajesh C. Choudhary, Ashok Kumar, Arun Kumar, Mohammad Naimuddin, Kirti Ranjan, Mahesh Kumar Saini, Saumya Saumya, Sweta Baradia, Soumyadip Barman, Dylan Lutton, Saptaparna Bhattacharya, Suman Das Gupta, Shubham Dutta, Suchandra Dutta, Subir Sarkar, Mohammad Mobassir Ameen, Prafulla Kumar Behera, Subash Chandra Behera, Suman Chatterjee, Ganapati Dash, Pranati Jana, Pritam Kalbhori, Samadhan Kamble, Jyothsna Rani Komaragiri, Deepak Kumar, Tribeni Mishra, Bibhuti Parida, Prabhat Ranjan Pujahari, Nihar Ranjan Saha, Ashish Sharma, Anup Kumar Sikdar, Rohit Kumar Singh, Pranjal Verma, Sadhana Verma, Anusree Vijay,

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Kiminsu, Yildiray Komurcu, Gulsen Onengut, Kadri Ozdemir, Ayse Polatoz, Bayram Tali, Ufuk Guney Tok, Ebru Uslan, Ibrahim Soner Zorbakir, Gamze Sokmen, Metin Yalvac, Bora Akgun, Ismail Okan Atakisi, Erhan Gulmez, Mithat Kaya, Ozlem Kaya, Sevgi Tekten, Altan Cakir, Kerem Cankocak, Gizem Gul Dincer, Sercan Sen, Orhan Aydilek, Burak Hacisahinoglu, Ilknur Hos, Berkan Kaynak, Suat Ozkorucuklu, Onur Potok, Hale Sert, Cagdas Simsek, Caglar Zorbilmez, Salim Cerci, Bora Isildak, Deniz Sunar Cerci, Taylan Yetkin, Andriy Boyaryntsev, Boris Grynyov, Leonid Levchuk, David Anthony, James John Brooke, Aaron Bundock, Florian Joel J. Bury, Emyr Clement, David Cussans, Henning Flacher, Maciej Glowacki, Joel Goldstein, Helen F. Heath, Mei-li Holmberg, Lukasz Kreczko, Sudarshan Paramesvaran, Liam Robertshaw, Vincent J. Smith, Katie Walkingshaw Pass, Austin Ball, Ken W. Bell, Alexander Belyaev, Christopher Brew, Robert M. Brown, David J.A. Cockerill, Charlotte Cooke, Alison Elliot, Katherine Victoria Ellis, Kristian Harder, Sam Harper, Jacob Linacre, Konstantinos Manolopoulos, Dave M. Newbold, Emmanuel Olaiya, David Petyt, Thomas Reis, Abanti Ranadhir Sahasransu, Giovanna Salvi, Thomas Schuh, Claire Shepherd-Themistocleous, Ian R. Tomalin, Kathleen Charlotte Whalen, Thomas Williams, Irene Andreou, Robert Bainbridge, Philippe Bloch, Christopher Edward Brown, Oliver Buchmuller, Camilo Andres Carrillo Montoya, Gurpreet Singh Chahal, David Colling, Julia Suzana Dancu, Indranil Das, Paul Dauncey, Gavin Davies, Michel Della Negra, Simon Fayer, Giacomo Fedi, Geoffrey Hall, Alexander Howard, Gregory Iles, Charlotte Rose Knight, Paul Krueper, Jonathon Langford, Kai Hong Law, Jaime León Holgado, Louis Lyons, Anne-Marie Magnan, Benedikt Maier, Stavros Mallios, Mikael Mieskolainen, Jordan Nash, Mark Pesaresi, Prijith Babu Pradeep, Benjamin Charles Radburn-Smith, Alexander Richards, Andrew Rose, Klitos Savva, Christopher Seez, Raghunandan Shukla, Alexander Tapper, Kirika Uchida, George Peter Uttley, Tejinder Virdee, Milos Vojinovic, Nicholas Wardle, Daniel Winterbottom, Joanne Cole, Akram Khan, Paul Kyberd, Ivan Reid, Salavat Abdullin, Andrew Brinkerhoff, Evan Collins, Mohamed Rashad Darwish, Jay Dittmann, Kenichi Hatakeyama, Vinay Hegde, Joshua Hiltbrand, Brooks McMaster, Jonathan Samudio, Siddhesh Sawant, Chosila Sutantawibul, Jonathan Wilson, Rachel Bartek, Aaron Dominguez, Ali Eren Simsek, Shin-Shan Yu, Bhim Bam, Axel Buchot Perraguin, Ruchi Chudasama, Seth Cooper, Colin Crovella, Sergei V. Gleyzer, Emma Pearson, Cilicia Uzziel Perez, Paolo Rumerio, Emanuele Usai, Ruole Yi, Alp Akpinar, Christopher Cosby, Gianfranco De Castro, Zeynep Demiragli, Carlos Erice, Caleb Fangmeier, Celia Fernandez Madrazo, Elisa Fontanesi, Daniel Gastler, Frank Golf, Sihyun Jeon, Joshua O'cain, Ian Reed, James Rohlf, Kaitlin Salyer, David Sperka, Daniel Spitzbart, Indara Suarez, Anna Tsatsos, Angelo Giacomo Zecchinelli, Gaetano Barone, Gabriele Benelli, David Cutts, Loukas Gouskos, Mary Hadley, Ulrich Heintz, Ka Wa Ho, Julie Managan Hogan, Taeun Kwon, Greg Landsberg, Ka Tung Lau, Jingyu Luo, Spandan Mondal, Trevor Russell, Sinan Sagir, Xiaohe Shen, Farrah Simpson, Marko Stamenkovic, Nikhilesh Venkatasubramanian, Samantha Abbott, Benjamin Barton, Christopher Brainerd, Richard Breedon, Hong Cai, Manuel Calderon De La

Barca Sanchez, Maxwell Chertok, Matthew Citron, John Conway, Peter Timothy Cox, Robin Erbacher, Frank Jensen, Ota Kukral, Giovanni Mocellin, Michael Mulhearn, Sydney Ostrom, Wei Wei, Saeahram Yoo, Fengwangdong Zhang, Kosmas Adamidis, Michail Bachtis, Delano Campos, Robert Cousins, Abhisek Datta, Gael Flores Avila, Jay Hauser, Mikhail Ignatenko, Muhammad Ansar Iqbal, Tyler Lam, Ya-feng Lo, Elisabetta Manca, Antonett Nunez Del Prado, David Saltzberg, Vyacheslav Valuev, Robert Clare, J. William Gary, Gail Hanson, Anthony Aportela, Aashay Arora, James G. Branson, Sergio Cittolin, Stephane Cooperstein, Daniel Diaz, Javier Duarte, Leonardo Giannini, Yanxi Gu, Jonathan Guiang, Raghav Kansal, Vyacheslav Krutelyov, Robert Lee, James Letts, Mario Masciovecchio, Farouk Mokhtar, Soumya Mukherjee, Marco Pieri, Daniel Primosch, Melissa Quinnan, Balaji Venkat Sathia Narayanan, Vivek Sharma, Matevz Tadel, Emmanouil Vourliotis, Frank Würthwein, Yifan Xiang, Avraham Yagil, Anders Barzdukas, Liam Brennan, Claudio Campagnari, Keegan Downham, Chiara Grieco, Mohd Meraj Hussain, Joe Incandela, Jiwoong Kim, Amina Jinzi Li, Phillip Master-son, Hualin Mei, Jeffrey Richman, Sai Neha Santpur, Ulascan Sarica, Ryan Schmitz, Francesco Setti, James Sheplock, David Stuart, Tamás Álmos Vámi, Sicheng Wang, Xuli Yan, Danyi Zhang, Soham Bhattacharya, Adolf Bornheim, Olmo Cerri, Anthony Latorre, Jiajing Mao, Harvey B. Newman, Guillermo Reales Gutiérrez, Maria Spiropulu, Jean-Roch Vlimant, Chu Wang, Si Xie, Ren-Yuan Zhu, John Alison, Sitong An, Patrick Bryant, Matteo Cremonesi, Valentina Dutta, Thomas Ferguson, Tirso Alejandro Gómez Espinosa, Abhirami Harilal, Aleesha Kallil Tharayil, Chuyuan Liu, Tanmay Mudholkar, Sindhu Murthy, Pritam Palit, Kyungmin Park, Manfred Paulini, Andrew Roberts, Alejandro Sanchez, Wesley Terrill, John Perry Cumalat, William T. Ford, Andrew Hart, Abbas Hassani, George Karathanasis, Nicholas Manganelli, Jannicke Pearkes, Claire Savard, Nicolas Schonbeck, Kevin Stenson, Keith Ulmer, Stephen Robert Wagner, Noah Zipper, Davide Zuolo, James Alexander, Samuel Bright-Thonney, Xuan Chen, Derek James Cranshaw, Jennet Dickinson, Jason Fan, Xingchen Fan, Shaun Hogan, Peace Kotamnives, Jose Monroy, Michael Oshiro, Juliet Ritchie Patterson, Michael Reid, Anders Ryd, Julia Thom, Peter Wittich, Rui Zou, Michael Albrow, Maral Alyari, Oz Amram, Giorgio Apollinari, Artur Apresyan, Lothar A.T. Bauerdick, Douglas Berry, Jeffrey Berryhill, Pushpalatha C. Bhat, Kevin Burkett, Joel Nathan Butler, Anadi Canepa, Giuseppe Benedetto Cerati, H.W.K. Cheung, Frank Chlebana, Grace Cum-mings, Irene Dutta, Victor Daniel Elvira, Yongbin Feng, Jim Freeman, Abhijith Gandrakota, Zoltan Gecse, Lindsey Gray, Dan Green, Aidan Grummer, Stefan Grünendahl, Daniel Guerrero, Oliver Gutsche, Robert M. Harris, Ryan Heller, Theodor Christian Herwig, James Hirschauer, Bodhitha Jayatilaka, Sergo Jindariani, Marvin Johnson, Umesh Joshi, Thomas Klijnsma, Boaz Klima, Ka Hei Martin Kwok, Stephan Lammel, Claire Lee, Don Lincoln, Ron Lipton, Tiehui Liu, Christopher Madrid, Kaori Maeshima, Cristina Mantilla, David Mason, Patricia McBride, Petra Merkel, Stephen Mrenna, Steve Nahn, Jennifer Ngadiuba, Daniel Noonan, Scarlet Norberg, Vaia Papadimitriou, Nathaniel Pastika, Kevin Pedro, Cristian Pena, Fabio Ravera, Allison Reinsvold Hall,

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2.19.1.17 Quantum information meets high-energy physics: input to the update of the European strategy for particle physics [270]

- **Authors:** Yoav Afik, Federica Fabbri, Matthew Low, Luca Marzola, Juan Antonio Aguilar-Saavedra, Mohammad Mahdi Altakach, Nedaa Alexandra Asbah, Yang Bai, Hannah Banks, Alan J. Barr, Alexander Bernal, Thomas E. Browder, Paweł Caban, J. Alberto Casas, Kun Cheng, Frédéric Dèliot, Regina Demina, Antonio Di

Domenico, Michał Eckstein, Marco Fabbrichesi, Benjamin Fuks, Emidio Gabrielli, Dorival Gonçalves, Radosław Grabarczyk, Michele Grossi, Tao Han, Timothy J. Hobbs, Paweł Horodecki, James Howarth, Shih-Chieh Hsu, Stephen Jiggins, Eleanor Jones, Andreas W. Jung, Andrea Helen Knue, Steffen Korn, Theodota Lagouri, Priyanka Lamba, Gabriel T. Landi, Haifeng Li, Qiang Li, Ian Low, Fabio Maltoni, Josh McFayden, Navin McGinnis, Roberto A. Morales, Jesús M. Moreno, Juan Ramón Muñoz de Nova, Giulia Negro, Davide Pagani, Giovanni Pelliccioli, Michele Pinamonti, Laura Pintucci, Baptiste Ravina, Alim Ruzi, Kazuki Sakurai, Ethan Simpson, Maximiliano Sioli, Shufang Su, Sokratis Trifinopoulos, Sven E. Vahsen, Sofia Vallecorsa, Alessandro Vicini, Marcel Vos, Eleni Vryonidou, Chris D. White, Martin J. White, Andrew J. Wildridge, Tong Arthur Wu, Laura Zani, Yulei Zhang, Knut Zoch

- **Posted on arXiv:** 31 March 2025
- **Published in Eur.Phys.J.Plus:** 09 September 2025

2.19.1.18 Colliders are Testing neither Locality via Bell’s Inequality nor Entanglement versus Non-Entanglement [271]

- **Authors:** Steven A. Abel, Herbi K. Dreiner, Rhitaja Sengupta, Lorenzo Ubaldi
- **Posted on arXiv:** 21 July 2025

2.19.2 Crosslists

2.19.2.1 Quantum entanglement and Bell inequality violation at colliders [7]

- **Authors:** Alan J. Barr, Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 12 February 2024
- **Published in Prog.Part.Nucl.Phys.:** 19 July 2024

2.20 Track Reconstruction

2.20.1 Quantum Algorithms - Grover’s Search Algorithm

2.20.1.1 Quantum Associative Memory in HEP Track Pattern Recognition [272]

- **Authors:** Illya Shapoval, Paolo Calafiura
- **Posted on arXiv:** 29 January 2019
- **Published in EPJ Web of Conferences 214 (2019) 01012:** 2019

2.20.1.2 Quantum speedup for track reconstruction in particle accelerators [273]

- **Authors:** Duarte Magano, Akshat Kumar, Mārtiņš Kālis, Andris Locāns, Adam Glos, Sagar Pratapsi, Gonçalo Quinta, Maksims Dimitrijevs, Aleksander Rivošs, Pedrame Bargassa, João Seixas, Andris Ambainis, Yasser Omar
- **Posted on arXiv:** 23 April 2021
- **Published in Physical Review D 105 (2022) 076012:** 01 April 2022

2.20.1.3 Quantum pathways for charged track finding in high-energy collisions [274]

- **Authors:** Christopher Brown, Michael Spannowsky, Alexander Tapper, Simon Williams, Ioannis Xiotidis
- **Posted on arXiv:** 01 November 2023
- **Published in Front. Artif. Intell. 7:1339785 (2024):** 07 May 2024

2.20.2 Quantum Algorithms - Harrow-Hassidim-Lloyd Algorithm

2.20.2.1 A quantum algorithm for track reconstruction in the LHCb vertex detector [275]

- **Authors:** Davide Nicotra, Miriam Lucio Martinez, Jacco Andreas de Vries, Marcel Merk, Kurt Driessens, Ronald Leonard Westra, Domenica Dibenedetto, Daniel Hugo Cámpora Pérez
- **Posted on arXiv:** 01 August 2023
- **Published in JINST:** 29 November 2023

2.20.3 Quantum Computing Paradigms - Quantum Annealing

2.20.3.1 A pattern recognition algorithm for quantum annealers [276]

- **Authors:** Frédéric Bapst, Wahid Bhimji, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder
- **Posted on arXiv:** 21 February 2019
- **Published in Comput.Softw.Big Sci.:** 09 December 2019

2.20.3.2 Track clustering with a quantum annealer for primary vertex reconstruction at hadron colliders [277]

- **Authors:** Souvik Das, Andrew J. Wildridge, Andreas Jung
- **Posted on arXiv:** 21 March 2019

2.20.3.3 Charged particle tracking with quantum annealing optimization [278]

- **Authors:** Alexander Zlokapa, Abhishek Anand, Jean-Roch Vlimant, Javier M. Duarte, Joshua Job, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 12 August 2019
- **Published in Quantum Mach. Intell. 3, 27 (2021):** 02 November 2021

2.20.3.4 Quantum annealing algorithms for track pattern recognition [279]

- **Authors:** Masahiko Saito, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder, Yasuyuki Okumura, Ryu Sawada, Alex Smith, Junichi Tanaka, Koji Terashi
- **Published in EPJ Web Conf.:** 2020

2.20.3.5 Particle track classification using quantum associative memory [280]

- **Authors:** Gregory Quiroz, Lauren Ice, Andrea Delgado, Travis S. Humble
- **Posted on arXiv:** 23 November 2020
- **Published in Nucl.Instrum.Meth.A:** 15 June 2021

2.20.3.6 Quantum-Annealing-Inspired Algorithms for Track Reconstruction at High-Energy Colliders [281]

- **Authors:** Hideki Okawa, Qing-Guo Zeng, Xian-Zhe Tao, Man-Hong Yung
- **Posted on arXiv:** 22 February 2024
- **Published in Comput.Softw.Big Sci.:** 28 August 2024

2.20.4 Quantum Machine Learning - Supervised Methods

2.20.4.1 Reconstructing charged particle track segments with a quantum-enhanced support vector machine [282]

- **Authors:** Philippa Duckett, Gabriel Facini, Marcin Jastrzebski, Sarah Malik, Tim Scanlon, Sebastien Rettie
- **Posted on arXiv:** 14 December 2022
- **Published in Phys.Rev.D:** 01 March 2024

2.20.5 Quantum Machine Learning - Variational Quantum Algorithms

2.20.5.1 Quantum convolutional neural networks for high energy physics data analysis [33]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

2.20.5.2 Hybrid quantum classical graph neural networks for particle track reconstruction [283]

- **Authors:** Cenk Tüysüz, Carla Rieger, Kristiane Novotny, Bilge Demirköz, Daniel Dobos, Karolos Potamianos, Sofia Vallecorsa, Jean-Roch Vlimant, Richard Forster
- **Posted on arXiv:** 26 September 2021
- **Published in Quantum Machine Intelligence:** 01 December 2021

2.20.5.3 Studying quantum algorithms for particle track reconstruction in the LUXE experiment [284]

- **Authors:** Lena Funcke, Tobias Hartung, Beate Heinemann, Karl Jansen, Annabel Kropf, Stefan Kühn, Federico Meloni, David Spataro, Cenk Tüysüz, Yee Chinn Yap
- **Posted on arXiv:** 14 February 2022
- **Published in nan:** 2023

2.20.5.4 Particle track reconstruction with noisy intermediate-scale quantum computers [285]

- **Authors:** Tim Schwägerl, Cigdem Issever, Karl Jansen, Teng Jian Khoo, Stefan Kühn, Cenk Tüysüz, Hannsjörg Weber
- **Posted on arXiv:** 23 March 2023
- **Published in J.Phys.Conf.Ser.:** 2026

2.20.5.5 Charged particle reconstruction for future high energy colliders with Quantum Approximate Optimization Algorithm [286]

- **Authors:** Hideki Okawa
- **Posted on arXiv:** 16 October 2023

2.20.6 Crosslists

2.20.6.1 Hybrid Quantum-Classical Graph Convolutional Network [30]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 15 January 2021

2.20.6.2 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.20.6.3 Quantum Computing Applications in Future Colliders [4]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

2.20.6.4 Quantum Systems for Enhanced High Energy Particle Physics Detectors [123]

- **Authors:** M. Doser, E. Auffray, F.M. Brunbauer, I. Frank, H. Hillemanns, G. Orlandini, G. Kornakov
- **Published in nan:** 24 June 2022

2.20.6.5 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borrás, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3 Quantum Information Science in Nuclear and Particle Physics

3.1 Reviews and Whitepapers and Proceedings

3.1.1 Reviews

3.1.1.1 Lattice gauge theory simulations in the quantum information era [1]

- **Authors:** M. Dalmonte, S. Montangero
- **Posted on arXiv:** 11 February 2016
- **Published in Contemporary Physics 57 388 (2016):** 09 March 2016

3.1.1.2 Search for New Physics with Atoms and Molecules [2]

- **Authors:** M.S. Safronova, D. Budker, D. DeMille, Derek F. Jackson Kimball, A. Derevianko, C.W. Clark
- **Posted on arXiv:** 04 October 2017
- **Published in Reviews of Modern Physics, 90, 025008 (2018):** 30 June 2018

3.1.1.3 Quantum Machine Learning in High Energy Physics [3]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

3.1.1.4 Quantum Computing Applications in Future Colliders [4]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

3.1.1.5 Review on Quantum Computing for Lattice Field Theory [5]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn
- **Posted on arXiv:** 01 February 2023
- **Published in nan:** 01 February 2023

3.1.1.6 Machine learning for anomaly detection in particle physics [6]

- **Authors:** Vasilis Belis, Patrick Odagiu, Thea Klaeboe Aarrestad
- **Posted on arXiv:** 20 December 2023
- **Published in Reviews in Physics, vol. 12, 2024, 100091:** 18 January 2024

3.1.1.7 Quantum entanglement and Bell inequality violation at colliders [7]

- **Authors:** Alan J. Barr, Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 12 February 2024
- **Published in Prog.Part.Nucl.Phys.:** 19 July 2024

3.1.2 Whitepapers and Proceedings

3.1.2.1 Quantum Sensing for High Energy Physics [8]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi
- **Posted on arXiv:** 29 March 2018

3.1.2.2 Mechanical quantum sensing in the search for dark matter [9]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao
- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

3.1.2.3 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [10]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

3.1.2.4 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [11]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

3.1.2.5 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [12]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

3.1.2.6 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [13]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borrás, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

3.1.2.7 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.1.2.8 New Horizons: Scalar and Vector Ultralight Dark Matter [15]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibbrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou
- **Posted on arXiv:** 28 March 2022

3.1.2.9 Quantum Networks for High Energy Physics [16]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie
- **Posted on arXiv:** 31 March 2022

3.1.2.10 Quantum Simulation for High-Energy Physics [17]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

3.1.2.11 Quantum computing hardware for HEP algorithms and sensing [18]

- **Authors:** M. Sohaib Alam, Sergey Belomestnykh, Nicholas Bornman, Gustavo Canelo, Yu-Chiu Chao, Mattia Checchin, Vinh San Dinh, Anna Grassellino, Erik J. Gustafson, Roni Harnik, Corey Rae Harrington McRae, Ziwen Huang, Keshav Kapoor, Taeyoon Kim, James B. Kowalkowski, Matthew J. Kramer, Yulia Krasnikova, Prem Kumar, Doga Murat Kurkcuoglu, Henry Lamm, Adam L. Lyon, Despina Milathianaki, Akshay Murthy, Josh Mutus, Ivan Nekrashevich, JinSu Oh, A. BarışÖzgüler, Gabriel Nathan Perdue, Matthew Reagor, Alexander Romanenko, James A. Sauls, Leandro Stefanazzi, Norm M. Tubman, Davide Venturelli, Changqing Wang, Xinyuan You, David M.T. van Zanten, Lin Zhou, Shaojiang Zhu, Silvia Zorzetti
- **Posted on arXiv:** 18 April 2022

3.1.2.12 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [19]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

3.1.2.13 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borrás, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico

Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.1.2.14 Quantum Sensors for High Energy Physics [21]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cencelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

3.2 Quantum Algorithms - Grover's Search Algorithm

3.2.1 Beyond the Standard Model

3.2.1.1 Application of a Quantum Search Algorithm to High- Energy Physics Data at the Large Hadron Collider [28]

- **Authors:** Anthony E. Armenakas, Oliver K. Baker
- **Posted on arXiv:** 01 October 2020

3.2.2 Event Classification

3.2.2.1 Implementation and analysis of quantum computing application to Higgs boson reconstruction at the large Hadron Collider [132]

- **Authors:** Anthony Alexiades Armenakas, Oliver K. Baker
- **Published in Sci.Rep.:** 24 November 2021

3.2.3 Event Generation

3.2.3.1 Quantum algorithm for Feynman loop integrals [149]

- **Authors:** Selomit Ramírez-Urbe, Andr  s E. Renter  a-Olivo, Germ  n Rodrigo, German F.R. Sborlini, Luiz Vale Silva
- **Posted on arXiv:** 18 May 2021
- **Published in JHEP:** 16 May 2022

3.2.3.2 Loop Feynman integration on a quantum computer [150]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Michele Grossi, Sofia Vallecorsa, Germán Rodrigo
- **Posted on arXiv:** 05 January 2024
- **Published in Phys. Rev. D 110, 074031 (2024):** 01 October 2024

3.2.4 Jet Reconstruction

3.2.4.1 Quantum Algorithms for Jet Clustering [167]

- **Authors:** Annie Y. Wei, Preksha Naik, Aram W. Harrow, Jesse Thaler
- **Posted on arXiv:** 23 August 2019
- **Published in Phys. Rev. D 101, 094015 (2020):** 15 May 2020

3.2.4.2 Quantum Algorithms in Particle Physics [164]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

3.2.5 Track Reconstruction

3.2.5.1 Quantum Associative Memory in HEP Track Pattern Recognition [272]

- **Authors:** Illya Shapoval, Paolo Calafiura
- **Posted on arXiv:** 29 January 2019
- **Published in EPJ Web of Conferences 214 (2019) 01012:** 2019

3.2.5.2 Quantum speedup for track reconstruction in particle accelerators [273]

- **Authors:** Duarte Magano, Akshat Kumar, Mārtiņš Kālis, Andris Locāns, Adam Glos, Sagar Pratapsi, Gonçalo Quinta, Maksims Dimitrijevs, Aleksander Rivošs, Pedrame Bargassa, João Seixas, Andris Ambainis, Yasser Omar
- **Posted on arXiv:** 23 April 2021
- **Published in Physical Review D 105 (2022) 076012:** 01 April 2022

3.2.5.3 Quantum pathways for charged track finding in high-energy collisions [274]

- **Authors:** Christopher Brown, Michael Spannowsky, Alexander Tapper, Simon Williams, Ioannis Xiotidis
- **Posted on arXiv:** 01 November 2023
- **Published in Front. Artif. Intell. 7:1339785 (2024):** 07 May 2024

3.2.6 Crosslists

3.2.6.1 Quantum clustering and jet reconstruction at the LHC [176]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 13 April 2022
- **Published in Physical Review D 106, 036021 (2022):** 01 August 2022

3.3 Quantum Algorithms - Harrow-Hassidim-Lloyd Algorithm

3.3.1 Track Reconstruction

3.3.1.1 A quantum algorithm for track reconstruction in the LHCb vertex detector [275]

- **Authors:** Davide Nicotra, Miriam Lucio Martinez, Jacco Andreas de Vries, Marcel Merk, Kurt Driessens, Ronald Leonard Westra, Domenica Dibenedetto, Daniel Hugo Cámpora Pèrez
- **Posted on arXiv:** 01 August 2023
- **Published in JINST:** 29 November 2023

3.3.2 Crosslists

3.3.2.1 Quantum-centric Supercomputing for Physics Research [124]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

3.4 Quantum Algorithms - Quantum Phase Estimation

3.4.1 Nuclear Many-Body Physics - Nuclear Reactions and Response

3.4.1.1 Dynamic linear response quantum algorithm [239]

- **Authors:** Alessandro Roggero, Joseph Carlson
- **Posted on arXiv:** 04 April 2018
- **Published in Phys. Rev. C 100, 034610 (2019):** 14 September 2019

3.5 Quantum Algorithms - Quantum Simulations

3.5.1 Dark Matter - Particle-like Dark Matter

3.5.1.1 Quantum simulations of dark sector showers [36]

- **Authors:** So Chigusa, Masahito Yamazaki
- **Posted on arXiv:** 26 April 2022
- **Published in Phys.Lett.B:** 26 September 2022

3.5.2 Effective Field Theories

3.5.2.1 Simulating Collider Physics on Quantum Computers Using Effective Field Theories [125]

- **Authors:** Christian W. Bauer, Marat Freytsis, Benjamin Nachman
- **Posted on arXiv:** 09 February 2021
- **Published in Phys.Rev.Lett.:** 18 November 2021

3.5.3 Event Generation

3.5.3.1 Quantum Algorithm for High Energy Physics Simulations [151]

- **Authors:** Christian W. Bauer, Wibe A. de Jong, Benjamin Nachman, Davide Provasoli
- **Posted on arXiv:** 05 April 2019
- **Published in Phys. Rev. Lett. 126, 062001 (2021):** 11 February 2021

3.5.3.2 Towards a quantum computing algorithm for helicity amplitudes and parton showers [152]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 13 October 2020
- **Published in Phys. Rev. D 103, 076020 (2021):** 27 April 2021

3.5.3.3 Improving quantum simulation efficiency of final state radiation with dynamic quantum circuits [153]

- **Authors:** Plato Deliyannis, James Sud, Diana Chamaki, Zoë Webb-Mack, Christian W. Bauer, Benjamin Nachman
- **Posted on arXiv:** 18 March 2022
- **Published in Physical Review D 106, 036007 (2022):** 01 August 2022

3.5.3.4 Collider events on a quantum computer [154]

- **Authors:** Gösta Gustafson, Stefan Prestel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 21 July 2022
- **Published in JHEP:** 07 November 2022

3.5.3.5 Quantum parton shower with kinematics [155]

- **Authors:** Christian W. Bauer, So Chigusa, Masahito Yamazaki
- **Posted on arXiv:** 30 October 2023
- **Published in Phys.Rev.A:** 26 March 2024

3.5.4 Lattice Field Theories - Scalar/Fermion Theories

3.5.4.1 Quantum Computation of Scattering in Scalar Quantum Field Theories [178]

- **Authors:** Stephen P. Jordan, Keith S.M. Lee, John Preskill
- **Posted on arXiv:** December 2011

3.5.4.2 Quantum Algorithms for Fermionic Quantum Field Theories [179]

- **Authors:** Stephen P. Jordan, Keith S. M. Lee, John Preskill
- **Posted on arXiv:** 28 April 2014

3.5.4.3 Digitization of scalar fields for quantum computing [180]

- **Authors:** Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 30 August 2018
- **Published in Phys. Rev. A 99, 052335 (2019):** 23 May 2019

3.5.4.4 Scalar Quantum Field Theories as a Benchmark for Near-Term Quantum Computers [181]

- **Authors:** Kubra Yeter-Aydeniz, Eugene F. Dumitrescu, Alex J. McCaskey, Ryan S. Bennink, Raphael C. Pooser, George Siopsis
- **Posted on arXiv:** 29 November 2018
- **Published in Phys. Rev. A 99, 032306 (2019):** 04 March 2019

3.5.4.5 Real-Time Scattering on Quantum Computers via Hamiltonian Truncation [182]

- **Authors:** James Ingoldby, Michael Spannowsky, Timur Sypchenko, Simon Williams, Matthew Wingate
- **Posted on arXiv:** 06 May 2025

3.5.5 Lattice Field Theories - Gauge Theories

3.5.5.1 Simulating lattice gauge theories on a quantum computer [186]

- **Authors:** Tim Byrnes, Yoshihisa Yamamoto
- **Posted on arXiv:** October 2005
- **Published in Phys.Rev.A:** 2006

3.5.5.2 Ultracold Quantum Gases and Lattice Systems: Quantum Simulation of Lattice Gauge Theories [187]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 07 May 2013
- **Published in Annalen Phys.:** 2013

3.5.5.3 Towards Quantum Simulating QCD [188]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 25 September 2014
- **Published in Nucl.Phys.A:** 28 September 2014

3.5.5.4 Quantum Simulations of Lattice Gauge Theories using Ultracold Atoms in Optical Lattices [189]

- **Authors:** Erez Zohar, J. Ignacio Cirac, Benni Reznik
- **Posted on arXiv:** 08 March 2015
- **Published in Rept.Prog.Phys.:** 18 December 2015

3.5.5.5 Real-time dynamics of lattice gauge theories with a few-qubit quantum computer [190]

- **Authors:** E.A. Martinez, C.A. Muschik, P. Schindler, D. Nigg, A. Erhard, M. Heyl, P. Hauke, M. Dalmonte, T. Monz, P. Zoller, R. Blatt
- **Posted on arXiv:** 15 May 2016
- **Published in Nature:** 22 June 2016

3.5.5.6 General Methods for Digital Quantum Simulation of Gauge Theories [191]

- **Authors:** Henry Lamm, Scott Lawrence, Yukari Yamauchi
- **Posted on arXiv:** 19 March 2019
- **Published in Phys. Rev. D 100, 034518 (2019):** 29 August 2019

3.5.5.7 SU(2) non-Abelian gauge field theory in one dimension on digital quantum computers [192]

- **Authors:** Natalie Klco, Jesse R. Stryker, Martin J. Savage
- **Posted on arXiv:** 19 August 2019
- **Published in Phys. Rev. D 101, 074512 (2020):** 22 April 2020

3.5.5.8 Simulating Lattice Gauge Theories within Quantum Technologies [193]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

3.5.5.9 Toward scalable simulations of Lattice Gauge Theories on quantum computers [194]

- **Authors:** Simon V. Mathis, Guglielmo Mazzola, Ivano Tavernelli
- **Posted on arXiv:** 20 May 2020
- **Published in Phys. Rev. D 102, 094501 (2020):** 04 November 2020

3.5.5.10 Role of boundary conditions in quantum computations of scattering observables [195]

- **Authors:** Raúl A. Briceño, Juan V. Guerrero, Maxwell T. Hansen, Alexandru M. Sturzu
- **Posted on arXiv:** 01 July 2020
- **Published in Phys. Rev. D 103, 014506 (2021):** 07 January 2021

3.5.5.11 Quantum simulation of gauge theory via orbifold lattice [196]

- **Authors:** Alexander J. Buser, Hrant Gharibyan, Masanori Hanada, Masazumi Honda, Junyu Liu
- **Posted on arXiv:** 12 November 2020
- **Published in JHEP 2109 (2021) 034:** 06 September 2021

3.5.5.12 Lattice renormalization of quantum simulations [197]

- **Authors:** Marcela Carena, Henry Lamm, Ying-Ying Li, Wanqiang Liu
- **Posted on arXiv:** 02 July 2021
- **Published in Phys.Rev.D:** 01 November 2021

3.5.5.13 Lattice Quantum Chromodynamics and Electrodynamics on a Universal Quantum Computer [198]

- **Authors:** Angus Kan, Yunseong Nam
- **Posted on arXiv:** 27 July 2021

3.5.5.14 Efficient representation for simulating U(1) gauge theories on digital quantum computers at all values of the coupling [199]

- **Authors:** Christian W. Bauer, Dorota M. Grabowska
- **Posted on arXiv:** 15 November 2021
- **Published in Phys.Rev.D:** 01 February 2023

3.5.5.15 Quantum error correction with gauge symmetries [200]

- **Authors:** Abhishek Rajput, Alessandro Roggero, Nathan Wiebe
- **Posted on arXiv:** 09 December 2021
- **Published in npj Quantum Inf.:** 25 April 2023

3.5.5.16 Preparations for quantum simulations of quantum chromodynamics in $1+1$ dimensions. I. Axial gauge [201]

- **Authors:** Roland C. Farrell, Ivan A. Chernyshev, Sarah J.M. Powell, Nikita A. Zemlevskiy, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 04 July 2022
- **Published in Phys. Rev. D 107, 054512 (2023):** 01 March 2023

3.5.5.17 Measurement-based quantum simulation of Abelian lattice gauge theories [202]

- **Authors:** Hiroki Sueno, Takuya Okuda
- **Posted on arXiv:** 19 October 2022
- **Published in SciPost Phys. 14, 129 (2023):** 25 May 2023

3.5.5.18 General quantum algorithms for Hamiltonian simulation with applications to a non-Abelian lattice gauge theory [203]

- **Authors:** Zohreh Davoudi, Alexander F. Shaw, Jesse R. Stryker
- **Posted on arXiv:** 28 December 2022
- **Published in Quantum 7, 1213 (2023):** 20 December 2023

3.5.5.19 Simulating $\mathbb{Z}_2$ lattice gauge theory on a quantum computer [204]

- **Authors:** Clement Charles, Erik J. Gustafson, Elizabeth Hardt, Florian Herren, Norman Hogan, Henry Lamm, Sara Starecheski, Ruth S. Van de Water, Michael L. Wagman
- **Posted on arXiv:** 03 May 2023
- **Published in Phys.Rev.E:** 26 January 2024

3.5.5.20 Quantum simulation of fundamental particles and forces [205]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 09 April 2024
- **Published in Nature Rev.Phys.:** 21 June 2023

3.5.5.21 Toward QCD on quantum computer: orbifold lattice approach [206]

- **Authors:** Georg Bergner, Masanori Hanada, Enrico Rinaldi, Andreas Schafer
- **Posted on arXiv:** 22 January 2024
- **Published in JHEP:** 20 May 2024

3.5.5.22 Scattering wave packets of hadrons in gauge theories: Preparation on a quantum computer [207]

- **Authors:** Zohreh Davoudi, Chung-Chun Hsieh, Saurabh V. Kadam
- **Posted on arXiv:** 01 February 2024
- **Published in Quantum:** 11 November 2024

3.5.5.23 Quantum Simulation of $SU(3)$ Lattice Yang-Mills Theory at Leading Order in Large- N_c Expansion [208]

- **Authors:** Anthony N. Ciavarella, Christian W. Bauer
- **Posted on arXiv:** 15 February 2024
- **Published in Phys.Rev.Lett.:** 09 September 2024

3.5.5.24 Scalable cold-atom quantum simulator of a $3 + 1D$ $U(1)$ lattice gauge theory with dynamical matter [209]

- **Authors:** Simone Orlando, Guo-Xian Su, Bing Yang, Jad C. Halimeh
- **Posted on arXiv:** 07 January 2026

3.5.6 Lattice Field Theories - Schwinger Model

3.5.6.1 Quantum simulation of a lattice Schwinger model in a chain of trapped ions [213]

- **Authors:** Philipp Hauke, David Marcos, Marcello Dalmonte, Peter Zoller
- **Posted on arXiv:** 10 June 2013
- **Published in Phys.Rev.X:** 22 November 2013

3.5.6.2 Quantum Algorithms for Simulating the Lattice Schwinger Model [217]

- **Authors:** Alexander F. Shaw, Pavel Lougovski, Jesse R. Stryker, Nathan Wiebe
- **Posted on arXiv:** 25 February 2020
- **Published in Quantum 4, 306 (2020):** 10 August 2020

3.5.6.3 Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits [214]

- **Authors:** Roland C. Farrell, Marc Illa, Anthony N. Ciavarella, Martin J. Savage
- **Posted on arXiv:** 08 August 2023
- **Published in PRX Quantum 5 (2024) 2, 020315:** 01 April 2024

3.5.6.4 Steps toward quantum simulations of hadronization and energy loss in dense matter [215]

- **Authors:** Roland C. Farrell, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 10 May 2024
- **Published in Phys.Rev.C 111 (2025) 1, 015202:** 14 January 2025

3.5.6.5 Enhancing quantum field theory simulations on NISQ devices with Hamiltonian truncation [218]

- **Authors:** James Ingoldby, Michael Spannowsky, Timur Sypchenko, Simon Williams
- **Posted on arXiv:** 26 July 2024
- **Published in Phys. Rev. D 110 (2024), 096016:** 01 November 2024

3.5.7 Nuclear Many-Body Physics - Nuclear Structure

3.5.7.1 Symmetry assisted preparation of entangled many-body states on a quantum computer [223]

- **Authors:** D. Lacroix
- **Posted on arXiv:** 11 June 2020
- **Published in Phys.Rev.Lett.:** 03 December 2020

3.5.7.2 Quantum computation of nuclear observables involving linear combinations of unitary operators [224]

- **Authors:** Pooja Siwach, P. Arumugam
- **Posted on arXiv:** 16 June 2022
- **Published in Phys.Rev.C:** 28 June 2022

3.5.8 Nuclear Many-Body Physics - Nuclear Reactions and Response

3.5.8.1 Quantum Computing for Neutrino-Nucleus Scattering [128]

- **Authors:** Alessandro Roggero, Andy C.Y. Li, Joseph Carlson, Rajan Gupta, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019
- **Published in Phys. Rev. D 101, 074038 (2020):** 28 April 2020

3.5.8.2 Quantum simulation of nuclear inelastic scattering [240]

- **Authors:** Weijie Du, James P. Vary, Xingbo Zhao, Wei Zuo
- **Posted on arXiv:** 01 June 2020
- **Published in Phys. Rev. A 104, 012611 (2021):** 23 July 2021

3.5.8.3 Imaginary-time propagation on a quantum chip [241]

- **Authors:** Francesco Turro, Alessandro Roggero, Valentina Amitrano, Piero Luchi, Kyle A. Wendt, Jonathan L. DuBois, Sofia Quaglioni, Francesco Pederiva
- **Posted on arXiv:** 24 February 2021
- **Published in Phys.Rev.A:** 28 February 2022

3.5.8.4 Nuclear two point correlation functions on a quantum computer [242]

- **Authors:** Alessandro Baroni, Joseph Carlson, Rajan Gupta, Andy C.Y. Li, Gabriel N. Perdue, Alessandro Roggero
- **Posted on arXiv:** 04 November 2021
- **Published in Phys.Rev.D:** 01 April 2022

3.5.8.5 A quantum algorithm for the linear response of nuclei [243]

- **Authors:** Nifeeya Singh, P. Arumugam
- **Posted on arXiv:** 17 October 2022
- **Published in Indian J.Phys.:** 31 May 2025

3.5.8.6 Faster spectral density calculation using energy moments [244]

- **Authors:** Jeremy Hartse, Alessandro Roggero
- **Posted on arXiv:** 01 November 2022
- **Published in Eur.Phys.J.A:** 09 March 2023

3.5.8.7 Quantum error mitigation for Fourier moment computation [245]

- **Authors:** Oriel Kiss, Michele Grossi, Alessandro Roggero
- **Posted on arXiv:** 23 January 2024
- **Published in Phys. Rev. D 111, 034504 (2025):** 01 February 2025

3.5.8.8 Nuclear scattering via quantum computing [246]

- **Authors:** Peiyan Wang, Weijie Du, Wei Zuo, James P. Vary
- **Posted on arXiv:** 30 January 2024
- **Published in Phys.Rev.C 109 6, 064623 (2024):** 26 June 2024

3.5.8.9 Solving reaction dynamics with quantum computing algorithms [247]

- **Authors:** Ronen Weiss, Alessandro Baroni, Joseph Carlson, Ionel Stetcu
- **Posted on arXiv:** 29 March 2024
- **Published in Physical Review C 111, 064004 (2025):** 25 June 2025

3.5.8.10 Quantum computing for extracting nuclear resonances [248]

- **Authors:** Hantao Zhang, Dong Bai, Zhongzhou Ren
- **Posted on arXiv:** 10 September 2024
- **Published in Phys.Lett.B:** 12 December 2024

3.5.8.11 Pathfinding quantum simulations of neutrinoless double- decay [216]

- **Authors:** Ivan A. Chernyshev, Roland C. Farrell, Marc Illa, Martin J. Savage, Andrii Maksymov, Felix Tripier, Miguel Angel Lopez-Ruiz, Andrew Arrasmith, Yvette de Sereville, Aharon Brodutch, Claudio Grotto, Ananth Kaushik, Martin Roetteler
- **Posted on arXiv:** 06 June 2025
- **Published in Nature Commun.:** 23 January 2026

3.5.9 Quantum Chromodynamics

3.5.9.1 Quantum simulation of quantum field theory in the light-front formulation [250]

- **Authors:** Michael Kreshchuk, William M. Kirby, Gary Goldstein, Hugo Beauchemin, Peter J. Love
- **Posted on arXiv:** 10 February 2020
- **Published in Phys. Rev. A 105, 032418 (2022):** 08 March 2022

3.5.9.2 Quantum simulation of colour in perturbative quantum chromodynamics [251]

- **Authors:** Herschel A. Chawdhry, Mathieu Pellen
- **Posted on arXiv:** 08 March 2023
- **Published in SciPost Phys. 15, 205 (2023):** 24 November 2023

3.5.9.3 Quantum simulation of scattering amplitudes and interferences in perturbative QCD [252]

- **Authors:** Herschel A. Chawdhry, Mathieu Pellen, Simon Williams
- **Posted on arXiv:** 09 July 2025

3.5.9.4 Quantum simulating multi-particle processes in high energy nuclear physics: dijet production and color (de)coherence [253]

- **Authors:** João Barata, Meijian Li, Wenyang Qian, Carlos A. Salgado, João M. Silva
- **Posted on arXiv:** 13 April 2026
- **Published in JHEP:** 08 July 2026

3.5.10 Crosslists

3.5.10.1 Lattice gauge theory simulations in the quantum information era [1]

- **Authors:** M. Dalmonte, S. Montangero
- **Posted on arXiv:** 11 February 2016
- **Published in Contemporary Physics 57 388 (2016):** 09 March 2016

3.5.10.2 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [10]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

3.5.10.3 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [11]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

3.5.10.4 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [13]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borrás, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

3.5.10.5 Quantum Simulation for High-Energy Physics [17]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

3.5.10.6 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [19]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

3.5.10.7 Review on Quantum Computing for Lattice Field Theory [5]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn
- **Posted on arXiv:** 01 February 2023
- **Published in nan:** 01 February 2023

3.5.10.8 Quantum data learning for quantum simulations in high-energy physics [163]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

3.5.10.9 Simulating quantum field theories on continuous-variable quantum computers [183]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 15 March 2024
- **Published in Phys. Rev. A 110 (2024) 1, 012607:** 08 July 2024

3.5.10.10 Real-time scattering processes with continuous-variable quantum computers [184]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 03 February 2025
- **Published in Phys. Rev. A 112 (2025), 012614:** 16 July 2025

3.6 Quantum Algorithms - Quantum Walks

3.6.1 Event Generation

3.6.1.1 Quantum walk approach to simulating parton showers [156]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 28 September 2021
- **Published in Phys. Rev. D 106 (2022) 056002:** 01 September 2022

3.7 Quantum Computing Paradigms - Continuous Variable Quantum Computing

3.7.1 Anomaly Detection

3.7.1.1 Unsupervised event classification with graphs on classical and photonic quantum computers [22]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

3.7.2 Detector Technologies and Simulations

3.7.2.1 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [118]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

3.7.3 Event Classification

3.7.3.1 Continuous-variable photonic quantum extreme learning machines for fast collider-data selection [133]

- **Authors:** Benedikt Maier, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 15 October 2025
- **Published in EPJ Quant.Technol.:** 27 April 2026

3.7.4 Lattice Field Theories - Scalar/Fermion Theories

3.7.4.1 Simulating quantum field theories on continuous-variable quantum computers [183]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 15 March 2024
- **Published in Phys. Rev. A 110 (2024) 1, 012607:** 08 July 2024

3.7.4.2 Real-time scattering processes with continuous-variable quantum computers [184]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 03 February 2025
- **Published in Phys. Rev. A 112 (2025), 012614:** 16 July 2025

3.7.4.3 Qumode Tensor Networks for False Vacuum Decay in Quantum Field Theory [185]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 20 June 2025

3.7.5 Crosslists

3.7.5.1 Subatomic Many-Body Physics Simulations on a Quantum Frequency Processor [131]

- **Authors:** Hsuan-Hao Lu, Natalie Klco, Joseph M. Lukens, Titus D. Morris, Aaina Bansal, Andreas Ekström, Gaute Hagen, Thomas Papenbrock, Andrew M. Weiner, Martin J. Savage, Pavel Lougovski
- **Published in nan:** 2019

3.8 Quantum Computing Paradigms - Quantum Annealing

3.8.1 Anomaly Detection

3.8.1.1 A Quantum Algorithm for Model-Independent Searches for New Physics [23]

- **Authors:** Konstantin T. Matchev, Prasanth Shyamsundar, Jordan Smolinsky
- **Posted on arXiv:** 04 March 2020
- **Published in LHEP:** 09 April 2023

3.8.2 Detector Technologies and Simulations

3.8.2.1 Unfolding measurement distributions via quantum annealing [119]

- **Authors:** Kyle Cormier, Riccardo Di Sipio, Peter Wittek
- **Posted on arXiv:** 22 August 2019
- **Published in JHEP:** 22 November 2019

3.8.3 Event Classification

3.8.3.1 Solving a Higgs optimization problem with quantum annealing for machine learning [134]

- **Authors:** Alex Mott, Joshua Job, Jean Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Published in Nature:** 18 October 2017

3.8.3.2 Quantum adiabatic machine learning by zooming into a region of the energy surface [135]

- **Authors:** Alexander Zlokapa, Alex Mott, Joshua Job, Jean-Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 13 August 2019
- **Published in Phys. Rev. A 102, 062405 (2020):** 05 December 2020

3.8.3.3 Restricted Boltzmann Machines for galaxy morphology classification with a quantum annealer [136]

- **Authors:** João Caldeira, Joshua Job, Steven H. Adachi, Brian Nord, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019

3.8.3.4 Quantum algorithm for the classification of supersymmetric top quark events [34]

- **Authors:** Pedrame Bargassa, Timothée Cabos, Samuele Cavinato, Artur Cordeiro Oudot Choi, Timothée Hessel
- **Posted on arXiv:** 31 May 2021
- **Published in Phys. Rev. D 104 (2021) 096004:** 01 November 2021

3.8.3.5 Completely quantum neural networks [137]

- **Authors:** Steve Abel, Juan C. Criado, Michael Spannowsky
- **Posted on arXiv:** 23 February 2022
- **Published in Phys.Rev.A:** 01 August 2022

3.8.4 Jet Reconstruction

3.8.4.1 Adiabatic quantum algorithm for multijet clustering in high energy physics [168]

- **Authors:** Diogo Pires, Yasser Omar, João Seixas
- **Posted on arXiv:** 28 December 2020
- **Published in Phys.Lett.B:** 13 June 2023

3.8.4.2 Leveraging Quantum Annealer to identify an Event-topology at High Energy Colliders [169]

- **Authors:** Minho Kim, Pyungwon Ko, Jae-hyeon Park, Myeonghun Park
- **Posted on arXiv:** 15 November 2021

3.8.4.3 Quantum annealing for jet clustering with thrust [170]

- **Authors:** Andrea Delgado, Jesse Thaler
- **Posted on arXiv:** 05 May 2022
- **Published in Phys.Rev.D:** 01 November 2022

3.8.4.4 Degeneracy engineering for classical and quantum annealing: A case study of sparse linear regression in collider physics [171]

- **Authors:** Eric R. Anschuetz, Lena Funcke, Patrick T. Komiske, Serhii Kryhin, Jesse Thaler
- **Posted on arXiv:** 20 May 2022
- **Published in Phys. Rev. D 106, 056008 (2022):** 01 September 2022

3.8.5 Lattice Field Theories - Gauge Theories

3.8.5.1 A regression algorithm for accelerated lattice QCD that exploits sparse inference on the D-Wave quantum annealer [210]

- **Authors:** Nga T.T. Nguyen, Garrett T. Kenyon, Boram Yoon
- **Posted on arXiv:** 14 November 2019
- **Published in Sci Rep 10, 10915 (2020):** 02 July 2020

3.8.5.2 SU(2) lattice gauge theory on a quantum annealer [211]

- **Authors:** Sarmed A Rahman, Randy Lewis, Emanuele Mendicelli, Sarah Powell
- **Posted on arXiv:** 15 March 2021
- **Published in Phys. Rev. D 104, 034501 (2021):** 01 August 2021

3.8.6 Nuclear Many-Body Physics - Nuclear Structure

3.8.6.1 Quasiparticle pairing encoding of atomic nuclei for quantum annealing [225]

- **Authors:** Emanuele Costa, Axel Pèrez-Obiol, Javier Menéndez, Arnau Rios, Artur García-Sáez, Bruno Juliá-Díaz
- **Posted on arXiv:** 11 October 2025
- **Published in Phys.Lett.B:** 30 November 2025

3.8.7 Track Reconstruction

3.8.7.1 A pattern recognition algorithm for quantum annealers [276]

- **Authors:** Frédéric Bapst, Wahid Bhimji, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder
- **Posted on arXiv:** 21 February 2019
- **Published in Comput.Softw.Big Sci.:** 09 December 2019

3.8.7.2 Track clustering with a quantum annealer for primary vertex reconstruction at hadron colliders [277]

- **Authors:** Souvik Das, Andrew J. Wildridge, Andreas Jung
- **Posted on arXiv:** 21 March 2019

3.8.7.3 Charged particle tracking with quantum annealing optimization [278]

- **Authors:** Alexander Zlokapa, Abhishek Anand, Jean-Roch Vlimant, Javier M. Duarte, Joshua Job, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 12 August 2019
- **Published in Quantum Mach. Intell. 3, 27 (2021):** 02 November 2021

3.8.7.4 Quantum annealing algorithms for track pattern recognition [279]

- **Authors:** Masahiko Saito, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder, Yasuyuki Okumura, Ryu Sawada, Alex Smith, Junichi Tanaka, Koji Terashi
- **Published in EPJ Web Conf.:** 2020

3.8.7.5 Particle track classification using quantum associative memory [280]

- **Authors:** Gregory Quiroz, Lauren Ice, Andrea Delgado, Travis S. Humble
- **Posted on arXiv:** 23 November 2020
- **Published in Nucl.Instrum.Meth.A:** 15 June 2021

3.8.7.6 Quantum-Annealing-Inspired Algorithms for Track Reconstruction at High-Energy Colliders [281]

- **Authors:** Hideki Okawa, Qing-Guo Zeng, Xian-Zhe Tao, Man-Hong Yung
- **Posted on arXiv:** 22 February 2024
- **Published in Comput.Softw.Big Sci.:** 28 August 2024

3.8.8 Crosslists

3.8.8.1 Quantum Algorithms for Jet Clustering [167]

- **Authors:** Annie Y. Wei, Preksha Naik, Aram W. Harrow, Jesse Thaler
- **Posted on arXiv:** 23 August 2019
- **Published in Phys. Rev. D 101, 094015 (2020):** 15 May 2020

3.8.8.2 Quantum Machine Learning in High Energy Physics [3]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

3.8.8.3 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.8.8.4 Quantum Computing Applications in Future Colliders [4]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

3.8.8.5 Preparations for quantum simulations of quantum chromodynamics in $1+1$ dimensions. I. Axial gauge [201]

- **Authors:** Roland C. Farrell, Ivan A. Chernyshev, Sarah J.M. Powell, Nikita A. Zemlevskiy, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 04 July 2022
- **Published in Phys. Rev. D 107, 054512 (2023):** 01 March 2023

3.8.8.6 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.9 Quantum Entanglement and Bell Inequalities

3.9.1 Beyond the Standard Model

3.9.1.1 Flavor patterns of fundamental particles from quantum entanglement? [29]

- **Authors:** Jesse Thaler, Sokratis Trifinopoulos
- **Posted on arXiv:** 30 October 2024
- **Published in Phys.Rev.D:** 01 March 2025

3.9.2 Effective Field Theories

3.9.2.1 Minimal entanglement and emergent symmetries in low-energy QCD [126]

- **Authors:** Qiaofeng Liu, Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 October 2022
- **Published in Phys.Rev.C 107, 025204 (2023):** 14 February 2023

3.9.3 Quantum Chromodynamics

3.9.3.1 Symmetry from entanglement suppression [254]

- **Authors:** Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 April 2021
- **Published in Phys.Rev.D:** 01 October 2021

3.9.4 Quantum Correlations at Colliders

3.9.4.1 Entanglement and quantum tomography with top quarks at the LHC [255]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 04 March 2020
- **Published in Eur. Phys. J. Plus (2021) 136:907:** 03 September 2021

3.9.4.2 Testing Bell Inequalities at the LHC with Top-Quark Pairs [256]

- **Authors:** M. Fabbrichesi, R. Floreanini, G. Panizzo
- **Posted on arXiv:** 23 February 2021
- **Published in Phys.Rev.Lett. 127 (2021) 16, 161801:** 15 October 2021

3.9.4.3 Quantum tops at the LHC: from entanglement to Bell inequalities [257]

- **Authors:** Claudio Severi, Cristian Degli Esposti Boschi, Fabio Maltoni, Maximiliano Sioli
- **Posted on arXiv:** 19 October 2021
- **Published in Eur.Phys.J.C:** 02 April 2022

3.9.4.4 Quantum SMEFT tomography: Top quark pair production at the LHC [130]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 10 March 2022
- **Published in Phys.Rev.D:** 01 September 2022

3.9.4.5 Quantum information with top quarks in QCD [258]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 10 March 2022
- **Published in Quantum:** 29 September 2022

3.9.4.6 Improved tests of entanglement and Bell inequalities with LHC tops [259]

- **Authors:** J.A. Aguilar-Saavedra, J.A. Casas
- **Posted on arXiv:** 01 May 2022
- **Published in Eur.Phys.J.C:** 03 August 2022

3.9.4.7 Constraining new physics in entangled two-qubit systems: top-quark, tau-lepton and photon pairs [260]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli
- **Posted on arXiv:** 24 August 2022
- **Published in Eur.Phys.J.C:** 19 February 2023

3.9.4.8 Quantum Discord and Steering in Top Quarks at the LHC [261]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 08 September 2022
- **Published in Phys. Rev. Lett. 130, 221801 (2023):** 31 May 2023

3.9.4.9 Bell inequalities and quantum entanglement in weak gauge boson production at the LHC and future colliders [262]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 01 February 2023
- **Published in Eur.Phys.J.C:** 15 September 2023

3.9.4.10 Three-Body Entanglement in Particle Decays [263]

- **Authors:** Kazuki Sakurai, Michael Spannowsky
- **Posted on arXiv:** 02 October 2023
- **Published in Phys.Rev.Lett.:** 09 April 2024

3.9.4.11 Observation of quantum entanglement with top quarks at the ATLAS detector [264]

- **Authors:** Georges Aad, Braden Keim Abbott, Kira Abeling, Nils Julius Abicht, Haider Abidi, Asmaa Aboulhorma, Halina Abramowicz, Henso Abreu, Yiming Abulaiti, Bobby Samir Acharya, Claire Adam Bourdarios, Leszek Adamczyk, Sagar Addepalli, Matt Addison, Jahred Adelman, Aytul Adiguzel, Tim Adye, A.A. Affolder, Tony Affolder, Yoav

Afik, Merve Nazlim Agaras, Jinky Agarwala, Anamika Aggarwal, Catalin Agheorghiesei, Ammara Ahmad, Faig Ahmadov, Waleed Syed Ahmed, Sudha Ahuja, Xiaocong Ai, Giulio Aielli, Arya Aikot, Malak Ait Tamlihat, Brahim Aitbenchikh, Iakov Aizenberg, Melike Akbiyik, Torsten Akesson, Andrei Akimov, Daiya Akiyama, Nilima Nilesk Akolkar, Sina Aktas, Konie Al Khoury, Gian Luigi Alberghi, Justin Albert, Pietro Albicocco, Guillaume Lucas Albouy, Sara Alderweireldt, Zackary Lee Alegria, Martin Aleksa, I.N. Aleksandrov, Calin Alexa, Theodoros Alexopoulos, Fabrizio Alfonsi, Malte Algren, Muhammad Alhroob, Babar Ali, H.M. J. Ali, Shahzad Ali, Samuel William Alibocus, Malik Aliev, Gianluca Alimonti, Wael Alkakhi, Corentin Allaire, Benedict Allbrooke, Julia Frances Allen, Cristian Andres Allendes Flores, Philip Patrick Allport, Alberto Aloisio, Francisco Alonso, Cristiano Alpigiani, Manuel Alvarez Estevez, Adrian Alvarez Fernandez, Mario Alves Cardoso, Mariagrazia Alviggi, Mohamed Aly, Yara Do Amaral Coutinho, Alessandro Ambler, Christoph Amelung, Maximilian Amerl, Christoph Ames, Dante Amidei, Susana Patricia Amor Dos Santos, Kieran Robert Amos, Viktor Ananiev, Christos Anastopoulos, Timothy Robert Andeen, John Kenneth Anders, Stefio Yosse Andrean, Attilio Andreazza, Stylianos Angelidakis, Aaron Angerami, Alexey Anisenkov, Alberto Annovi, Claire Antel, Matthew Thomas Anthony, Egor Antipov, Mario Antonelli, Fabio Anulli, Masato Aoki, Takumi Aoki, Javier Alberto Aparisi Pozo, Marco Aparo, Ludovica Aperio Bella, Christian Appelt, Aram Apyan, Sergio Javier Arbiol Val, Chiara Arcangeletti, Ayana Tamu Arce, Eloisa Arena, Jean-Francois Arguin, Spyros Argyropoulos, Jan-Hendrik Arling, Olivier Arnaez, Hannah Arnold, Giacomo Artoni, Haruka Asada, Kanae Asai, Shoji Asai, Nedaa Alexandra Asbah, Ketevi Adikle Assamagan, Robert Astalos, S. Atashi, Rose Atashi, Ryan Justin Atkin, Markus Julian Atkinson, Hicham Atmani, Prachi Atmasiddha, Kamil Augsten, Silvia Auricchio, Adrien Auriol, Volker Andreas Austrup, Giuseppe Avolio, Konstantinos Axiotis, Georges Azuelos, Dominik Babal, Henri Bachacou, Konstantinos Bachas, Alexander Bachi, Karl Filip Backman, Anthony Badea, Tamas Marton Baer, Paolo Bagnaia, Marzieh Bahmani, Daniel Bahner, Adam Bailey, Virginia Bailey, John Baines, Luke Baines, Keith Baker, Evelin Bakos, Debottam Bakshi Gupta, Veena Balakrishnan, Rahul Balasubramanian, Evgenii Baldin, Petr Balek, Eric Ballabene, Fabrice Balli, Lisa Marie Baltes, William Keaton Balunas, Johannes Balz, Elzbieta Banas, Marilena Bandieramonte, Anjishnu Bandyopadhyay, Shubham Bansal, Liron Barak, Marawan Barakat, Elisabetta Barberio, Dario Barberis, Marlon Benoit Barbero, Marten Zefanja Barel, Kevin Nicholas Barends, Teresa Barillari, Martin Barisits, Tim Barklow, Petr Baron, D.A. Baron Moreno, A. Baroncelli, Toni Baroncelli, Gaetano Barone, Alan Barr, Jackson Barr, Fernando Barreiro Alonso, Joao Barreiro Guimaraes Da Costa, Uriel Barron, M.G. Barros Teixeira, Sergey Barsov, Falk Bartels, Rainer Bartoldus, Adam Edward Barton, Pavol Bartos, Alexander Basan, Marta Baselga Bacardit, Ahmed Bassalat, Matthew Joseph Basso, Candice Ruth Basson, Richard Bates, Souad Batlamous, Richard Batley, Binish Batool, Marco Battaglia, Daariimaa Battulga, Matteo Bause, Michael Bauer, Patrick Bauer, L.T. Bazzano Hurrell, James Beacham, Tristan

Beau, Jean Yves Beaucamp, Pierre-Hugues Beauchemin, Philip Bechtle, Hans Peter Beck, Kathrin Becker, Andrew Beddall, Vadim Bednyakov, Chris Bee, Lars Beemster, Thomas Beermann, Marcia Begalli, Michael Begel, Arabinda Behera, Janna Katharina Behr, Joshua Falco Beirer, Florian Beisiegel, Mohamed Belfkir, Gideon Bella, Lorenzo Bellagamba, Alain Bellerive, Panagiotis Bellos, Konstantin Beloborodov, Driss Benchekroun, Fatima Bendebba, Yan Benhammou, Lydia Audrey Beresford, Matteo Mario Beretta, Elin Bergeaas Kuutmann, Nicolas Berger, Benedikt Ludwig Bergmann, Juerg Beringer, Gregorio Bernardi, Catrin Bernius, Florian Urs Bernlochner, Florent Bernon, Adrian Berrocal Guardia, Tracey Berry, Peter Berta, Anne-Sophie Berthold, Iain Bertram, Siegfried Bethke, Alessandra Betti, Adrian Bevan, Naman Kumar Bhalla, Muaaz Bhamjee, Somadutta Bhatta, Deb Sankar Bhattacharya, Prajita Bhattarai, Kartik Deepak Bhide, Vallary Shashikant Bhopatkar, Riccardo Maria Bianchi, Gianluca Bianco, Otmar Biebel, Rafal Bielski, Michela Biglietti, Sully Billingsley, Marcello Bindi, Ahmet Bingul, Cesare Bini, Alessandro Biondini, Callum Jacob Birch-Sykes, Gareth Adam Bird, Mattias Birman, Marek Biros, Stanislav Biryukov, Tobias Bisanz, Emanuele Bisceglie, Jyoti Prakash Biswal, Diptaparna Biswas, Kristian Bjoerke, Ingo Bloch, Andrew James Blue, Ulla Blumenschein, Julian Blumenthal, Gerjan Bobbink, Viktor Bobrovnikov, Michael Boehler, B. Boehm, Burkhard Bohm, Danijela Bogavac, Alexander Bogdanchikov, Christian Bohm, Veronique Boisvert, Petar Bokan, Tomasz Bold, Marco Bomben, Marcella Bona, Maarten Boonekamp, Callum Dale Booth, Albert Gyorgy Borbely, Iurii Bordulev, Hanna Maria Borecka-Bielska, Guennadi Borissov, Daniela Bortoletto, Davide Boscherini, Martine Fernandez-Bosman, J.D. Bossio Sola, Khalil Bouaouda, Naseem Bouchhar, Joseph Boudreau, E.V. Bouhova-Thacker, Djamel Eddine Boumediene, Romain Bouquet, Antonio Boveia, Jamie Boyd, Diallo Boye, Igor Boyko, Juraj Bracinik, Nihal Brahimi, Gerhard Immanuel Brandt, Oleg Brandt, Fried Erik Braren, Benjamin Paul Brau, J.E. Brau, Roy Schimmel Brener, Lydia Brenner, Richard Brenner, Shikma Bressler, David Britton, Daniel Britzger, Ian Brock, Raymond Brock, Gustaaf Brooijmans, William King Brooks, Elizabeth Brost, Leesa Brown, Laura Elaine Bruce, Tim Lukas Bruckler, Pawel Bruckman De Renstrom, Ben Bruers, Alessia Bruni, Graziano Bruni, Marco Bruschi, Nello Bruscinio, Trygve Buanes, Quentin Buat, Daniel Alexander Buchin, Andy Buckley, Oleg Bulekov, Brendon Bullard, Sergey Burdin, Carsten Burgard, Angela Maria Burger, Blake Oliver Burghgrave, Oleksandr Burlayenko, J.T. P. Burr, Charles Burton, Jackson Carl Burzynski, Elena Laura Busch, Volker Buescher, Peter John Bussey, John Mark Butler, Craig Macleod Buttar, Jonathan Butterworth, Will Buttinger, Carlos Josue Buxo Vazquez, Alexey Buzykaev, Susana Cabrera Urban, Luca Cadamuro, Davide Caforio, Huacheng Cai, Yuchen Cai, Yizhou Cai, Valentina Cairo, Orhan Cakir, Noemi Calace, Paolo Calafiura, Giovanni Calderini, Philippe Calfayan, Giuseppe Callea, Luiz Pereira Caloba, David Calvet, Samuel Calvet, Milene Calvetti, Reina Coromoto Camacho Toro, Stefano Camarda, Daniel Camarero Munoz, Paolo Camarri, Maria Teresa Camerlingo, David Cameron, Clement Camincher, Mario Campanelli, Alessandra Camplani, Vin-

cenzo Canale, Josu Cantero Garcia, Yumeng Cao, Francesca Capocasa, Marcella Capua, Antonio Carbone, Roberto Cardarelli, Juan Carlos Cardenas, Fabio Cardillo, Giovandomenico Carducci, Tancredi Carli, Giampaolo Carlino, Juan Ignacio Carlotto, B.T. Carlson, Evan Michael Carlson, Leonardo Carminati, Alberto Carnelli, Maria Carnesale, Sascha Caron, Edson Carquin Lopez, Sonia Carra, Giuseppe Carratta, Anthony Murel Carroll, J.W. S. Carter, Thomas Michael Carter, Pilar Casado Lechuga, Maximilian Caspar, Florencia Luciana Castillo, Lucia Castillo Garcia, Victoria Castillo Gimenez, Nuno Castro, Andrea Catinaccio, James Catmore, Tom Cavaliere, Viviana Cavaliere, Noemi Cavalli, Vincenzo Cavasinni, Yusuf Can Cekmecelioglu, Emre Celebi, Federico Celli, Martino Salomone Centonze, Vilius Cepaitis, Karel Cerny, Augusto Santiago Cerqueira, Alex Cerri, Lucio Cerrito, Fabio Cerutti, Beatrice Cervato, Alberto Cervelli, Gianmario Cesarini, Serkant Cetin, Dhiman Chakraborty, Jay Chan, Wai Yuen Chan, John Derek Chapman, Emilien Chapon, Bakar Chargeishvili, Dave Charlton, Meghranjana Chatterjee, Chainika Chauhan, Yimin Che, Sergei Chekanov, Sergey Chekulaev, G.A. Chelkov, Andy Chen, Boping Chen, Charlie Chen, Huirun Chen, Hucheng Chen, Jing Chen, Jiayi Chen, Maggie Chen, Shion Chen, Shenjian Chen, Xiang Chen, Xin Chen, Ye Chen, Alkaid Cheng, Hok Chuen Tom Cheng, Sanha Cheong, Alexander Cheplakov, Evgeniya Cheremushkina, Elizaveta Cherepanova, R.Cherkaoui El Moursli, Elliott Cheu, Kingman Cheung, Laurent Chevalier, Vitaliano Chiarella, Giorgio Chiarelli, Nemer Chiedde, Gabriele Chiodini, Andrew Stephen Chisholm, Adrian Chitan, Mariam Chitishvili, Mihail Chizhov, Kyungeon Choi, Yuan-Tang Chou, E.Y. S. Chow, Michael Kwok Lam Chu, Ming Chung Chu, Xiaotong Chu, Jiri Chudoba, Janusz Chwastowski, Davide Cieri, Krzysztof Ciesla, Vladimir Cindro, Alessandra Ciocio, Francesco Ciotto, Z.H. Citron, Mauro Citterio, Dan Andrei Ciubotaru, Allan Clark, Philip Clark, Cameron Clarry, Jose Manuel Clavijo Columbie, Savannah Clawson, Christophe Clement, Joshua Clercx, Yann Coadou, Marina Cobal, Andrea Coccaro, R.F. Coelho Barrue, Rafael Coelho Lopes De Sa, Simone Coelli, Brian Andrew Cole, Johann Collot, Patricia Conde Muino, Matt Connell, Simon Connell, Ian Allan Connelly, Eimear Isobel Conroy, Francesco Conventi, Harry Cooke, A.M. Cooper-Sarkar, Artur Cordeiro Oudot Choi, Louie Dartmoor Corpe, Massimo Corradi, Francois Corriveau, Arely Cortes-Gonzalez, Maria Jose Costa Mezquita, Francesco Costanza, Davide Costanzo, Benjamin Cote, Glen Cowan, Kyle Stuart Cranmer, Davide Cremonini, Sabine Crepe-Renaudin, Francesco Crescioli, Markus Cristinziani, Marco Cristoforetti, Vincent Alexander Croft, Jacob Edwin Crosby, Nanni Crosetti, G. Crosetti, Ana Rosario Cueto Gomez, Tula Cuhadar Donszelmann, Han Cui, Zhaoyuan Cui, W.R. Cunningham, Liam Cunningham, Francesco Curcio, Patrick Karl Czodrowski, Marta Czurylo, M.J. Da Cunha Sargedas De Sousa, Joao Victor Da Fonseca Pinto, Cinzia Da Via, Wladyslaw Dabrowski, Tomas Dado, Salah-Eddine Dahbi, Tiesheng Dai, Daniele Dal Santo, Carlo Dallapiccola, Mogens Dam, Gabriele D'Amen, Valerio D'Amico, Johannes Frederic Damp, Jeff Dandoy, Matthias Danninger, Valerio Dao, Nanni Darbo, G. Darbo, Smita Darmora, Sruthy Jyothi Das, Saverio D'Auria, Claire David, Tomas Davidek, Benjamin Richard Davis-

Purcell, Ian Dawson, Henry Day-Hall, Kaushik De, Riccardo De Asmundis, Nicola De Biase, Stefano De Castro, Nicolo De Groot, Paul De Jong, Hector De La Torre Perez, Antonio De Maria, Alessandro De Salvo, Umberto De Sanctis, Francesco De Santis, Antonella De Santo, Jean-Baptiste De Vivie De Regie, Dmitri Dedovich, Jordy Degens, Allison Mccarn Deiana, Francesca Del Corso, Jose Del Peso, Fer Del Rio, Line Delagrange, Frederic Deliot, Chris Malena Delitzsch, Massimo Della Pietra, Domenico Della Volpe, Andrea Dell'Acqua, Lidia Dell'Asta, Marco Delmastro, Pierre Antoine Delsart, Sarah Marie Demers, Mikhail Demichev, Serguei Denisov, Louis D'Eramo, Dominik Karol Derendarz, Frederic Derue, Paul Dervan, Klaus Desch, Christopher Deutsch, Francesco Armando Di Bello, Anna Di Ciaccio, Lucia Di Ciaccio, Antonio Di Domenico, Camilla Di Donato, Alessandro Di Girolamo, Giulia Di Gregorio, Andrea Di Luca, Biagio Di Micco, Roberto Di Nardo, Magda Diamantopoulou, Flavia De Almeida Dias, T.Dias Do Vale, Marco Aurelio Diaz Gutierrez, Federico Guillermo Diaz Capriles, Mariia Didenko, Edward Diehl, Leena Diehl, Sergio Diez Cornell, Carmen Diez Pardos, Christina Dimitriadi, Aleksandra Dimitrievska, Jochen Christian Dingfelder, Ioan-Mihail Dinu, Sebastian Dittmeier, Fido Dittus, Fares Djama, Tamar Djobava, Caterina Doglioni, Adriana Dohnalova, Jiri Dolejsi, Zdenek Dolezal, Kristin Dona, Marisilvia Donadelli, Binbin Dong, Julien Noce Donini, Adelina D'Onofrio, Monica D'Onofrio, Jens Dopke, Alessandra Doria, Nuno Dos Santos Fernandes, Patrick Dougan, Maria Teresa Dova, Tony Doyle, Maxence Draguet, Etienne Dreyer, Ioannis Drivas-Koulouris, Matthew Drnevich, Mariia Drozdova, Dongshuo Du, Tristan Arnoldus Du Pree, Filipp Dubinin, Michal Dubovsky, Ehud Duchovni, Guenter Duceck, Otilia Anamaria Ducu, Dominik Duda, Alexey Dudarev, Emily Rose Duden, Matteo D'Uffizi, Laurent Duflot, Michael Duehrssen-Debling, Ana Elena Dumitriu, Monica Dunford, Sascha Dungs, Katherine Elaine Dunne, Arnaud Duperrin, H.Duran Yildiz, Michael Johannes Dueren, Archil Durglishvili, Brianna Dwyer, G.I. Dyckes, Mateusz Dyndal, Bartosz Sebastian Dziedzic, Zoe Olivia Earnshaw, Gregor Hieronymus Eberwein, Barbora Eckerova, Stephen Eggebrecht, Edmar Egidio Purcino De Souza, Lukas Ehrke, Gerald Eigen, Kevin Frank Einsweiler, Tord Johan Carl Ekelof, Per Alexander Ekman, Saad El Farkh, Yassine El Ghazali, Hassnae El Jarrari, Ali El Moussaouy, Venugopal Ellajosyula, Mattias Ellert, Frank Ellinghaus, Nick Ellis, Johannes Elmsheuser, Markus Elsing, Dmitry Emelianov, Yuji Enari, Irina Ene, Shalini Epari, Paula Agnieszka Erland, Martin Errenst, Marc Escalier, Carlos Escobar Ibanez, Erez Etzion, Guiomar Gaspar De Andrade Evans, Hal Evans, Levi Evans, Meirin Oan Evans, Aleksei Ezhilov, Sanae Ezzarqtouni, Federica Fabbri, Laura Fabbri, Gabriel Facini, Vitaliy Fadeyev, Rinat Fakhrutdinov, Dionysios Fakoudis, Speranza Falciano, L.F. Falda Ulhoa Coelho, Peter Johannes Falke, Jana Faltova, Cunwei Fan, Yunyun Fan, Yaquan Fang, Marcello Fanti, Mohammed Faraj, Zahra Farazpay, Amir Farbin, Ada Farilla, Trisha Farooque, Sinead Farrington, Farida Fassi, Dimitris Fassouliotis, Michele Faucci Giannelli, William James Fawcett, Louis Fayard, Pavol Federic, Pavla Federicova, Oleg Fedin, Gleb Fedotov, Matthew Feickert, Lorenzo Feligioni, Deion Elgin Fellers, Cunfeng Feng, Minyu Feng, Zhuoran Feng, Michael James

Fenton, Lars Ferencz, Ruby Alice Molly Ferguson, Sergio Ivan Fernandez Luengo, Pablo Fernandez Martinez, M.J. V. Fernoux, James Ferrando, Arnaud Ferrari, Pamela Ferrari, Roberto Ferrari, Didier Ferrere, Claudio Ferretti, Frank Fiedler, Petr Fiedler, Andrej Filipcic, Emily Filmer, Frank Filthaut, Miguel Castro Nunes Fiolhais, Luca Fiorini, Wade Cameron Fisher, Tobias Fitschen, Peter Michael Fitzhugh, Ivor Fleck, Philipp Fleischmann, Tobias Flick, Marvin Flores, Luis Roberto Flores Castillo, Leyre Flores Sanz De Acedo, Francesco Maria Follega, Nikolai Fomin, Joel Hengwei Foo, Andrea Formica, Alessandra Forti, Etienne Marie Fortin, Anne Winifred Fortman, Maria Giovanna Foti, Leonidas Fountas, Daniel Fournier, Harald Fox, Paolo Francavilla, Simone Francescato, Stefano Franchellucci, Matteo Franchini, Silvia Franchino, David Francis, Luca Franco, Vinicius Franco Lima, Laura Franconi, Melissa Franklin, Guglielmo Frattari, Arran Charles Freegard, Werner Spolidoro Freund, Yuval Yitzhak Frid, Johnny Friend, Nick Fritzsche, Alexander Froch, Daniel Froidevaux, James Frost, Yao Fu, Sebastian Julio Fuenzalida Garrido, Minori Fujimoto, Kin Yip Fung, Eduardo Furtado De Simas Filho, Marin Furukawa, Juan Fuster Verdu, Alessandro Gabrielli, Andrea Gabrielli, Philipp Gadow, Guido Gagliardi, Louis-Guillaume Gagnon, Elizabeth Gallas, Bruce Joseph Gallop, Kock Kiam Gan, Sanmay Ganguly, Yanyan Gao, Francisca Garay Walls, Berenice Garcia, Carmen Garcia, Andrea Garcia Alonso, Arianna Gemma Garcia Caffaro, Jose Enrique Garcia Navarro, Maurice Garcia-Sciveres, Gwen Llewellyn Gardner, Robert William Gardner, Nicoletta Garelli, Diksha Garg, Rocky Bala Garg, Jack Gargan, Christopher Andrew Garner, Cameron Michael Garvey, Philipp Do Nascimento Gaspar, Veritas Katharina Gassmann, Gabriella Gaudio, Viveka Gautam, Paolo Gauzzi, Igor Gavrilenko, Alexander Gavriliuk, Colin Warren Gay, Goetz Gaycken, Evangelos Gazis, Andrei Alexandru Geanta, Carolyn Gee, Alex Gekow, Claudia Gemme, Marie-Helene Genest, Simonetta Gentile, Andrew Donald Gentry, Simon George, William Frederick George, Theodoros Geralis, Paul Gessinger, Marvin Emin Geyik, Mohammed Ghani, Mazuza Mohammad A. Ghneimat, Keanu Ghorbanian, Arpan Ghosal, Aishik Ghosh, Anindya Ghosh, Benedetto Giacobbe, Stefano Giagu, Tommaso Giani, Paola Giannetti, Antonio Giannini, Stephen Gibson, Matthew Gignac, Damian Gil, Alexander Kevin Gilbert, Benjamin Jacob Gilbert, Dag Gillberg, Geoffrey Gilles, Louis Ginabat, Doug Gingrich, Mapo Giordani, Pierre-Francois Giraud, Gilberto Giugliarelli, Danilo Giugni, Francesco Giuli, Ioannis Gkialas, Leonid Gladilin, Claudia Glasman, Galen Rhodes Gledhill, Gediminas Glemza, Marija Glisic, Ivan Gnesi, Yeonju Go, Maximilian Emanuel Goblirsch-Kolb, Benedikt Gocke, Dominique Godin, Berare Gokturk, Steven Goldfarb, Tobias Golling, Mpho Gift Doctor Gololo, Dmitry Golubkov, Jason Peter Gombas, Agostinho Da Silva Gomes, G.Gomes Da Silva, Antonio Jesus Gomez Delegido, Ricardo Jose Morais Silva Goncalo, Laura Gonella, Alexi Gongadze, Francesco Gonnella, Julia Lynne Gonski, Richards Gonzalez, Santiago Gonzalez De La Hoz, Ricardo Gonzalez Lopez, Cesar Gonzalez Renteria, Marcus Vinicius Gonzalez Rodrigues, Rebeca Gonzalez Suarez, Sergio Gonzalez Sevilla, Galo Rafael Gonzalvo Rodriguez, Luc Goossens, Benedetto Gorini, Edoardo Gorini, Andrej Gorisek, Thomas Christopher

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berto Ocariz, Atsuhiko Ochi, Ines Ochoa, Serhat Ordek, Jan Tuzlic Offermann, Agnieszka Ewa Ogrodnik, Alexander Oh, Christian Ohm, Hideyuki Oide, Reiyo Oishi, Martina Laura Ojeda, Yasuyuki Okumura, L.F. Oleiro Seabra, Sebastian Andres Olivares Pino, Denis Oliveira Damazio, Dayane Oliveira Goncalves, Jason Oliver, Ö. O. Öncel, Aaron Paul O'Neill, Antonio Onofre, P.U. E. Onyisi, Mark Oreglia, Gonzalo Enrique Orellana, Domizia Orestano, Nicola Orlando, Robert Orr, Val O'Shea, Lauren Melissa Osojnak, Rustem Ospanov, Gustavo Otero Y Garzon, Hidetoshi Otono, Philipp Sebastian Ott, Gregory James Ottino, Mohamed Ouchrif, Farid Ould-Saada, Mark Andrew Owen, Rhys Edward Owen, Kaan Yuksel Oyulmaz, Erkan Ozcan, Ferhat Ozturk, Nurcan Ozturk, Sertac Ozturk, Holly Pacey, Andreu Pacheco Pages, Cristobal Padilla Aranda, Giovanni Padovano, Simone Pagan Griso, Gabriel David Palacino Caviedes, Alessandra Palazzo, Jingjing Pan, Tong Pan, D.K. Panchal, Carlo Pandini, J.G. Panduro Vazquez, Hitarthi Deepak Pandya, Hao Pang, Priscilla Pani, Giancarlo Panizzo, Lorenzo Paolozzi, Santosh Parajuli, Alexander Paramonov, Christos Paraskevopoulos, Daniela Katherinne Paredes Hernandez, Ki Ryeong Park, Tae Hyoun Park, Andy Parker, Fabrizio Parodi, Elliot Parrish, Victoria Alexis Parrish, John Parsons, Ulrich Parzefall, B.Pascual Dias, Luis Pascual Dominguez, Enrico Pasqualucci, Stefano Passaggio, Francesca Pastore, Pragati Patel, Utsav Mukesh Patel, J.R. Pater, Thilo Pauly, Jannicke Andree Pearkes, Maiken Pedersen, Rute Pedro, Sergey Peleganchuk, Ondrej Penc, Emily Alexandra Pender, Gregory Douglas Penn, Katrin Elisabeth Penski, Maksim Penzin, Bernardo Sotto-Maior Peralva, A.P. Pereira Peixoto, Laura Pereira Sanchez, Dennis Perepelitsa, Estel Perez Codina, Maria Perganti, Heinz Pernegger, Oceane Perrin, Krisztian Peters, Reinhild Peters, Brian Petersen, Troels Petersen, Elisabeth Petit, Vlasios Petousis, Chariclia Petridou, Alexey Petrukhin, Mariel Pettee, Nora Emilia Pettersson, Aleksandr Petukhov, K. Petukhova, Raquel Pezoa Rivera, Lorenzo Pezzotti, Gianantonio Pezzullo, Minh Tuan Pham, Peter Phillips, Giacinto Piacquadio, Elisabetta Pianori, Federica Piazza, Ricardo Piegaia, Dorel Pietreanu, Andrew Pilkington, Michele Pinamonti, James Pinfold, B.C. Pinheiro Pereira, Andres Eloy Pinto Pinoargote, Laura Pintucci, Katie Piper, Antti Pirttikoski, Dylan Pizzi, Luca Pizzimento, Alessio Pizzini, Marc-Andre Pleier, Vladislavs Plesanovs, Vojtech Pleskot, Elena Plotnikova, Gitanjali Poddar, R. Poettgen, Luc Poggioli, Ishan Pokharel, Stanislav Polacek, Giacomo Polesello, Anne-Luise Poley, Alessandro Polini, Chris Pollard, Zachary Pollock, Elena Pompa Pacchi, Daniil Ponomarenko, Ludovico Pontecorvo, Stefan Popa, Gabriel Popeneciu, Aleksandra Poreba, Dilia Maria Portillo Quintero, Stanislav Pospisil, Michael Antony Postill, Petronel Postolache, Karolos Potamianos, Patrycja Anna Potepa, Igor Potrap, Harish Potti, Trine Poulsen, J. Poveda, Eukeni Pozo Astigarraga, Alberto Prades Ibanez, Jose Pretel, Darren Price, Margherita Primavera, Miguel Angel Principe Martin, Radek Privara, Tomasz Procter, Mason Proffitt, Nadezhda Proklova, Kirill Prokofiev, Giorgia Proto, James Proudfoot, Mariusz Przybycien, Witold Wojciech Przygoda, Andreas Psallidas, Joshua Eldon Puddefoot, Dennis Pudzha, Diana Pyatiizbyantseva, Jianming Qian, Qichen Dong, Y. Qin, Tong Qiu, Ar-

nulf Quadt, Michaela Queitsch-Maitland, Guillaume Quetant, Ryan Quinn, Gabriel Rabanal Bolanos, Dimbiniaina Rafanoharana, Francesco Ragusa, Lacey Rainbolt, Johnny Raine, Srini Rajagopalan, Ekaterina Ramakoti, Ian Alejandro Ramirez-Berend, Kunlin Ran, Hammad Rasheed, Valentina Raskina, Damir Fabrice Rassloff, Angira Rastogi, Stefan Rave, Baptiste Ravina, Ilia Ravinovich, Michel Raymond, Alexander Lincoln Read, Nathan Peter Readioff, Daniela Rebuzzi, George Redlinger, Alice Reed, Kendall Reeves, Joshua Aaron Reidelsturz, David Reikher, Amartya Rej, Christoph Rembser, Michele Renda, Marian Benedikt Rendel, Frederic Renner, Adam Rennie, Alberto Rescia, Silvia Resconi, Martina Ressegotti, Sebastien Rettie, Jose Gabriel Reyes Rivera, Elliot Reynolds, Olga Rezanova, Pavel Reznicek, Neza Ribaric, Ester Ricci, Robert Richter, Stefan Richter, Elzbieta Richter-Was, Melissa Ridel, Selaiman Ridouani, Patrick Rieck, Petra Riedler, Ellen Maria Riefel, Oliver Rieger, Michael Rijssenbeek, Adele Rimoldi, Marco Rimoldi, Lorenzo Rinaldi, Timothy Thomas Rinn, Maximilian Paul Rinnagel, Giulia Ripellino, Imma Riu, Pablo Andres Rivadeneira Bracho, Flera Rizatdinova, Eram Syed Rizvi, Bryn Arthur Roberts, Ryan Roberts, Steven Robertson, Dave Robinson, Carolina Michel Robles Gajardo, Marisol Robles Manzano, Aidan Robson, Alessandro Rocchi, Chiara Roda, Sergi Rodriguez Bosca, Yohany Rodriguez Garcia, Arturo Rodriguez Rodriguez, Ana Maria Rodriguez Vera, Shaun Roe, Jonas Till Roemer, Amber Roepe-Gier, Jens Roggel, Ole Rohne, Rimsky Alejandro Rojas Caballero, Christophe Roland, Jennifer Roloff, Anatoli Romaniouk, Emanuele Romano, Marino Romano, Nikolaos Rompotis, Lydia Roos, Stefano Rosati, Benjamin John Rosser, Leonardo Rossi, Lorenzo Rossini, Rachel Christine Rosten, Marina Rotaru, Benjamin Rottler, Charline Rougier, David Rousseau, David Rouso, Avik Roy, Sebastien Roy-Garand, Alexandre Rozanov, Z.M. A. Rozario, Yoram Rozen, Adrian Rubio Jimenez, Adam James Ruby, Victor Hugo Ruelas Rivera, Tristan Andrew Ruggeri, Alessandro Ruggiero, Arantxa Ruiz Martinez, Andre Rummler, Zuzana Rurikova, Nikolai Rusakovich, Heather Russell, Graziella Russo, John P. Rutherford, Sebastian Rutherford Colmenares, Katherine Amy Rybacki, Martin Rybar, Eli Baverfjord Rye, Andrey Ryzhov, Jorge Andres Sabater Iglesias, Paolo Sabatini, Hartmut Sadrozinski, Francesco Safai Tehrani, Batool Safarzadeh Samani, Murtaza Safdari, Shreya Saha, Merve Sahinsoy, Andrej Saibel, Matthias Saimpert, Masahiko Saito, Tomoyuki Saito, Dalila Salamani, Andy Salnikov, Jose Salt, Adrian Salvador Salas, Daniela Salvatore, Fabrizio Salvatore, Andreas Salzburger, Dirk Sammel, Dimos Sampsonidis, Despoina Sampsonidou, Francisco Javier Sanchez Martinez, Victoria Sanchez Sebastian, Heidi Sandaker, Christian Sander, Jay Ajitbhai Sandesara, Marisa Sandhoff, Carlos Sandoval Usme, D.P. C. Sankey, Takane Sano, Andrea Sansoni, Lorenzo Santi, Claudio Santoni, Helena Santos, Arka Santra, Kamal Saoucha, Joao Gentil Mendes Saraiva, Jad Mathieu Sardain, Osamu Sasaki, Koji Sato, Christof Sauer, Frank Sauerburger, Emmanuel Sauvan, Pierre Savard, Ryu Sawada, Craig Sawyer, Lee Sawyer, Ivan Sayago Galvan, Carla Sbarra, Antonio Sbrizzi, Timothy Paul Scanlon, Jana Schaarschmidt, Uli Schaefer, Dorothee Schaile, Robert Dean Schamberger, Christian Scharf, Meinrad Moritz

Schefer, V.A. Schegelsky, Daniel Scheirich, Ferdinand Schenck, Michael Schernau, Chris Scheulen, Carlo Schiavi, Enrico Schioppa, Marco Schioppa, Bastian Schlag, Katharina Schleicher, Stefan Schlenker, Jonas Schmeing, Mustafa Andre Schmidt, Kristof Schmieden, Christian Schmitt, Niklas Schmitt, Stefan Schmitt, Laurent Olivier Schoeffel, Andre Schoening, Patrick Scholer, Elisabeth Schopf, Matthias Schott, Jaroslava Schovancova, Steven Schramm, Tomke Schroer, Hans-Christian Schultz-Coulon, Markus Schumacher, Bruce Andrew Schumm, Ph. Schune, Alexander Joseph Schuy, Hava Rhian Schwartz, Ariel Gustavo Schwartzman, Thomas Andrew Schwarz, Ph. Schwemling, Reinhard Schwienhorst, Andrea Sciandra, Gabriella Sciolla, Fabrizio Scuri, Cristiano Sebastiani, Kevin Sedlaczek, Pienpen Seema, Sally Seidel, Abraham Seiden, Blair Daniel Seidlitz, Claudia Seitz, Jose Seixas, Givi Sekhniaidze, Luka Selem, Nicola Semprini Cesari, Debajyoti Sengupta, Varsha Senthilkumar, Laurent Serin, Leonid Serkin, Marco Sessa, Horst Severini, Federico Sforza, Anna Sfyrla, Elizaveta Shabalina, Rabia Shaheen, Jeff Shahinian, Dan Shaked Renous, Lianyou Shan, Marjorie Shapiro, Punit Sharma, P.B. Shatalov, Kate Shaw, Savanna Shaw, A. Shcherbakova, Qiuping Shen, Damian Joseph Sheppard, Peter Sherwood, Liaoshan Shi, Xin Shi, Chase Owen Shimmin, James David Shinner, I.P. J. Shipsey, Shohei Shirabe, Mariya Shiyakova, Jonathan Shlomi, M.J. Shochet, Jafar Shojaii, David Richard Shope, Bijay Shrestha, Suyog Shrestha, Esra Mohammed Shrif, Maheyar Jamshed Shroff, Petr Sicho, Anne Marie Sickles, Elias Sideras Haddad, Antonio Sidoti, Frank Siegert, Dj. Sijacki, Francisco Sili, Marcos Vini-
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 lakshmi Sindhu, Pekka Sinervo, Sahibjeet Singh, Maximiliano Sioli, Ismet Siral, Eliza-
 veta Sitnikova, Serguei Sivoklov, Joergen Sjoelin, Ali Skaf, Eleni Skorda, Patrick Skubic, Magdalena Slawinska, Vladimir Smakhtin, Ben Harry Smart, Serge Smirnov, Yury Smirnov, Lidia Smirnova, Oxana Smirnova, Andrew Caldon Smith, Emily Ann Smith, Hayden Alexander Smith, James Smith, Rachel Emma Clarke Smith, Maria Smizanska, Karel Smolek, Andrei Snegarev, Sean Robert Snider, Hella Snoek, Scott Snyder, Randy Sobie, Abi Soffer, Evgeny Soldatov, Urmila Soldevila Serrano, A.A. Solodkov, Shalu Solomon, Aleksey Soloshenko, Ksenia Solovieva, Oleg Solovyanov, Vic-
 tor Solovyev, Philip Sommer, Anil Sonay, Wen Yi Song, Andre Sopczak, Alex So-
 pio, Filomena Sopkova, Josef Daniel Sorenson, Isai Roberto Sotarriva Alvarez, Var-
 siha Sothilingam, Orlando Javier Soto Sandoval, Simone Sottocornola, Rachik Soualah, Zainab Soumairi, David South, Nathalie Soybelman, Stefania Spagnolo, Margherita Spalla, Dennis Sperlich, Giancarlo Spigo, Sebastiano Spinali, Dwayne Patrick Spiteri, Martin Spousta, Ezekiel Staats, Rainer Stamen, Alexios Stampeki, Mark Standke, Ewa Stanecka, Max Vincent Stange, Beojan Stanislaus, Marcel Stanitzki, Birgit Sylvia Stapf, E.A. Starchenko, Giordon Holtsberg Stark, Jan Stark, Pavel Staroba, Pavel Starovoitov, Steffen Staerz, Rafal Staszewski, George Stavropoulos, Jonas Steentoft, Pe-
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Gabriel Stoicea, Marcin Stolarski, Stefan Stonjek, Arno Straessner, Jonas Strandberg, Sara Strandberg, Maren Stratmann, Mike Strauss, Thomas Strebler, Pavol Strizenec, Raimund Strohmer, David Strom, Ryszard Stroynowski, Antonia Strubig, Stefania Antonia Stucci, Bjarne Stugu, John Stupak, Nicholas Styles, Dong Su, Shixiang Su, Wanyun Su, Xiaowen Su, Kaito Sugizaki, Vladimir Sulin, Matthew James Sullivan, D.M. S. Sultan, Laily Sultanaliyeva, S. Sultansoy, Toshi Sumida, O.Sunneborn Gudnadottir, Nairit Sur, Mark Sutton, Hisanori Suzuki, Michal Svatos, Maximilian J. Swiatlowski, Thorben Swirski, Ivan Sykora, Martin Sykora, Tomas Sykora, Duc Ta, Kerstin Tackmann, Anyes Taffard, Reda Tafirout, Juan Salvador Tafoya Vargas, Yosuke Takubo, Mossadek Talby, Alexei Talyshev, Kai Chung Tam, Nadav Michael Tamir, Aoto Tanaka, Junichi Tanaka, Rei Tanaka, Martino Tanasini, Zhengcheng Tao, Sebastian Tapia Araya, Stefan Tapprogge, A.Tarek Abouelfadl Mohamed, Shlomit Tarem, Khuram Tariq, Grigore Tarna, Francesco Tartarelli, Petr Tas, Marek Tasevsky, Enrico Tassi, Aric Tate, Gen Tateno, Yahya Tayalati, Geoffrey Norman Taylor, Wendy Jane Taylor, A.S. Tee, Rafael Teixeira De Lima, Pedro Teixeira-Dias, Jia Jian Teoh, Koji Terashi, Juan Terron Cuadrado, Stefano Terzo, Marianna Testa, Richard Teuscher, Alexander Thaler, Ondrej Theiner, Neofytos Themistokleous, Timothee Theveneaux-Pelzer, Oliver Thielmann, David William Thomas, Juergen Thomas, Emily Anne Thompson, Paul Thompson, Evelyn Jean Thomson, Yusong Tian, Vladimir Tikhomirov, Yu. A. Tikhonov, Sergei Timoshenko, Denys Timoshyn, Edmund Xiang Lin Ting, Paul Louis Tipton, Humphry Tlou, Abdellah Tnourji, Kazuki Todome, S. Todorova-Nova, Stefanie Todt, Manabu Togawa, Junji Tojo, Stano Tokar, Katsuo Tokushuku, O. Toldaiev, Rupert Tombs, Makoto Tomoto, Lauren Alexandra Tompkins, Kacper Wojciech Topolnicki, Eric Torrence, Heberth Torres, Emma Torro Pastor, Mariana Toscani, Cecilia Toscirci, Marc Tost, Daniel Tovey, Are Sivertsen Traeet, Iulia-Stefania Trandafir, Thomas Trefzger, Alessandro Tricoli, Isabel Trigger, Sophie Trincaz-Duvoid, Dominique Trischuk, Benjamin Trocme, Clara Troncon, Thi Ngoc Loan Truong, Maciej Trzebinski, Adam Trzupek, Fang-Ying Tsai, Meng-Ju Tsai, Angelos Tsiamis, Pavel Tsiareshka, Stergios Tsigaridas, Apostolos Tsirigotis, Vakhtang Tsiskaridze, Edisher Tskhadadze, Maria-Evanthia Tsopoulou, Yoshiaki Tsujikawa, Ilia Tsukerman, Vakhtang Tsulaia, Soshi Tsuno, Kimu Tsuru, Tsybychev Dmitri Tsybyshev, Yanjun Tu, Alexandra Tudorache, Valentina Tudorache, Alexander Naip Tuna, Semen Turchikhin, Ilkay Turk Cakir, Ruggero Turra, Tulgaa Turtuvshin, Spyros Tzamaras, Polyneikis Tzanis, Eftychia Tzovara, Fumihiko Ukegawa, Pablo Augusto Ulloa Poblete, Ejiro Naomi Umaka, Guillaume Unal, Mesut Unal, Alexander Undrus, Gokhan Unel, Josef Urban, Phillip Urquijo, Pedro Urrejola, Giulio Usai, Risa Ushioda, Muhammad Usman, Zekeriya Uysal, Vic Vacek, Brigitte Vachon, Knut Oddvar Hoie Vadla, Theodoros Vafeiadis, Andrius Vaitkus, Chrysostomos Valderanis, Eduardo Valdes Santurio, Marco Valente, Sara Valentinetti, Alberto Valero Biot, Enrique Valiente Moreno, Alexis Vallier, Juan Valls Ferrer, Dylan Remberto Van Arneman, Tal Roelof Van Daalen, Aaron Van Der Graaf, Peter Van Gemmeren, Milou Van Rijnbach, Samuel Van Stroud, Ivo Van Vulpen, Marco Vanadia,

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Xinmeng Ye, Yoran Yeh, Ivan Yeletsikh, Beom Ki Yeo, Melissa Yexley, Pengqi Yin, Kohei Yorita, Sulman Younas, Christopher Young, Charlie Young, Chengjun Yu, Yi Yu, Man Yuan, Rui Yuan, Luzhan Yue, Mohamed Zaazoua, Bartłomiej Henryk Zabinski, Estifa’A Zaid, Zuzanna Katarzyna Zak, Tamar Zakareishvili, Nataliia Zakharchuk, Stefano Zambito, Jilberto Antonio Zamora Saa, Jiaqi Zang, Daniele Zanzi, Ota Zaplatilek, Christian Zeitnitz, Hao Zeng, Jiancong Zeng, Todd Zenger, Oleg Zenin, Tibor Zenis, Seth Zenz, Soufiane Zerradi, Dirk Zerwas, Mingjie Zhai, Dengfeng Zhang, Jie Zhang, Jinlong Zhang, Kaili Zhang, Lei Zhang, Peng Zhang, Rui Zhang, Shuzhou Zhang, Si-jing Zhang, Tingyu Zhang, Xiangke Zhang, Xueyao Zhang, Yulei Zhang, Yuwen Ebony Zhang, Yixiang Zhang, Zhicai Zhang, Zhiqing Philippe Zhang, Haoran Zhao, Tongbin Zhao, Yuzhan Zhao, Zhengguo Zhao, Alexey Zhemchugov, Jinchao Zheng, Kai Zheng, Xiangxuan Zheng, Zhi Zheng, Dewen Zhong, Bing Zhou, Hao Zhou, Ning Zhou, Yong Zhou, You Zhou, Chengguang Zhu, Junjie Zhu, Yifan Zhu, Yingchun Zhu, Xuai Zhuang, Konstantin Zhukov, Nikolai Zimine, Joachim Zinsser, Michal Ziolkowski, Lidija Zivkovic, Antonio Zoccoli, Knut Zoch, Theodore Zorbas, Olga Zormpa, Wenkai Zou, Lukasz Zwalinski

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3.9.4.12 Can Bell inequalities be tested via scattering cross-section at colliders ? [265]

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3.9.4.13 Quantum detection of new physics in top-quark pair production at the LHC [266]

- **Authors:** Fabio Maltoni, Claudio Severi, Simone Tentori, Eleni Vryonidou
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3.9.4.14 Full quantum tomography of top quark decays [267]

- **Authors:** J.A. Aguilar-Saavedra
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3.9.4.15 Observation of quantum entanglement in top quark pair production in proton–proton collisions at $\sqrt{s} = 13$ TeV [268]

- **Authors:** Aram Hayrapetyan, Armen Tumasyan, Wolfgang Adam, Janik Walter Andrejkovic, Thomas Bergauer, Suman Chatterjee, Konstantinos Damanakis, Marko Dragicevic, Andre Hoang, Priya Sajid Hussain, Manfred Jeitler, Natascha Krammer, Ang Li, Dietrich Liko, Ivan Mikulec, Jochen Schieck, Robert Schöffbeck, Dennis Schwarz, Mangesh Sonawane, Sebastian Templ, Wolfgang Waltenberger, Claudia-Elisabeth Wulz, Tahys Janssen, Thomas Van Laer, Pierre Van Mechelen, Nordin Breugelmans, Jorgen D’Hondt, Soumya Dansana, Alexandre De Moor, Martin Delcourt, Felix Heyen, Steven Lowette, Inna Makarenko, Denise Müller, Stefaan Tavernier, Michael Tytgat, Gerrit Patrick Van Onsem, Senne Van Putte, David Vannerom, Bugra Bilin, Barbara Clerbaux, Alope Kumar Das, Gilles De Lentdecker, Hugues Evard, Laurent Favart, Paraskevas Gianneios, Johny Jaramillo, Ali Khalilzadeh, Fakhri Alam Khan, Kyeongpil Lee, Mostafa Mahdavihorrani, Andrea Malara, Santiago Paredes, Muhammad Aamir Shahzad, Laurent Thomas, Max Vanden Bemden, Catherine Vander Velde, Pascal Vanlaer, Maarten De Coen, Didar Dobur, Gul Gokbulut, Yanwen Hong, Joscha Knolle, Luka Lambrecht, David Marckx, Kevin Mota Amarilo, Amrutha Samalan, Kirill Skovpen, Niels Van Den Bossche, Jan van der Linden, Liam Wezenbeek, Anna Bencke, Agni Bethani, Giacomo Bruno, Claudio Caputo, Jerome De Favereau De Jeneret, Christophe Delaere, Izzeddin Suat Donertas, Andrea Giammanco, Ahmet Oguz Guzel, Sandhya Jain, Vincent Lemaître, Jindrich Lidrych, Paola Mastrapasqua, Tu Thong Tran, Sèbastien Wertz, Gilvan Alves, Miguel Alves Gallo Pereira, Eduardo Coelho, Gilson Correia Silva, Carsten Hensel, Thales Menezes De Oliveira, Clemencia Mora Herrera, Arthur Moraes, Patricia Rebello Teles, Mariana Soeiro, Antonio Vilela Pereira, Walter Luiz Aldá Júnior, Mapse Barroso Ferreira Filho, Helena Brandao Malbouisson, Wagner Carvalho, Jose Chinellato, Eliza Melo Da Costa, Gustavo Gil Da Silveira, Dilson De Jesus Damiao, Sandro Fonseca De Souza, Raphael Gomes De Souza, Matheus Macedo, Jordan Martins, Luiz Mundim, Helio Nogima, Joao Pedro Pinheiro, Alberto Santoro, Andre Sznajder, Mauricio Thiel, Cesar Augusto Bernardes, Luigi Calligaris, Thiago Tomei, Eduardo De Moraes Gregores, Bruno Lopes Da Costa, Isabela Maietto Silverio, Pedro G. Mercadante, Sergio F. Novaes, Breno Orzari, Sandra Padula, Aleksandar Aleksandrov, Georgy Antchev, Roumyana Hadjiiska, Plamen Iaydjiev, Milena Misheva, Mariana Shopova, Georgi Sultanov, Anton Dimitrov, Leander Litov, Borislav Pavlov, Peicho Petkov, Anton Petrov, Elton Shumka, Sumit Keshri, David Nicolas Laroze Navarrete, Shalini Thakur, Tongguang Cheng, Tahir Javaid, Li Yuan, Zhen Hu, Zhengchen Liang, Jinfeng Liu, Kai Yi, Guo-Ming Chen, He-Sheng Chen, Mingshui Chen, Fabio Iemmi, Chun-Hua Jiang, Anshul Kapoor, Hongbo Liao, Zhenan Liu, Ramkrishna Sharma, Jia-ning Song, Junquan Tao, Chu Wang, Jin Wang, Zebing Wang, Huaqiao Zhang, Jingzhou Zhao, Antonis Agapitos, Yong Ban, Sen Deng, Botao Guo, Chuqiao Jiang, Andrew Levin, Congqiao Li, Qiang Li, Yajun Mao, Sitian

Qian, Si-Jin Qian, Xuelong Qin, Xiaohu Sun, Dayong Wang, Heng Yang, Licheng Zhang, Yuzhe Zhao, Chen Zhou, Shuai Yang, Zhengyun You, Khawla Jaffel, Nan Lu, Gerry Bauer, Bolin Li, Jingqing Zhang, Xuyang Gao, Yuji Li, Zhen Lin, Chenfeng Lu, Meng Xiao, Carlos Avila, Diego Andres Barbosa Trujillo, Andr s Cabrera, Carlos Florez, Jorge Fraga, Jose Antonio Reyes Vega, Felipe Ramirez, C sar Rend n, Manuel Rodriguez, Anderson Alexis Ruales Barbosa, Jos  David Ruiz Alvarez, Duje Giljanovic, Nikola Godinovic, Damir Lelas, Ana Sculac, Marko Kovac, Andro Petkovic, Toni Sculac, Pedrame Bargassa, Vuko Brigljevic, Bhakti Kanulal Chitroda, Dinko Ferencek, Kresimir Jakovcic, Saswat Mishra, Andrei Starodumov, Tatjana Susa, Alexandros Attikis, Konstantinos Christoforou, Andreas Hadjiagapiou, Christos Leonidou, Jehad Mousa, Charalambos Nicolaou, Leonidas Paizanos, Fotios Ptochos, Panos A. Razis, Hans Rykaczewski, Halil Saka, Anton Stepennov, Miroslav Finger, Michael Finger, Antonin Kveton, Edgar Carrera Jarrin, Yasser Assran, Basma El-mahdy, Sherif Elgammal, Muhammad Abdullah Abdulfattah Muhammad, Mohammed Mahmoud, Karl Ehataht, Mario Kadastik, Torben Lange, Saswati Nandan, Christine Nielsen, Joosep Pata, Martti Raidal, Laurits Tani, Christian Veelken, Henning Kirschenmann, Kenneth Osterberg, Mikko Voutilainen, Shudhashil Bharthuar, Nurfikri Bin Norjoharuddeen, Erik Br cken, Francisco Ignacio Garcia Fuentes, Patin Inkaew, Kimmo Tuomas Samuli Kallonen, Tapio Lamp n, Kati Lassila-Perini, Sami Lehti, Tomas Lind n, Laura Martikainen, Mikael Myllym ki, Milla-maarit Rantanen, Hannu Siikonen, Jorma Tuominiemi, Panja-Riina Luukka, Henri Petrow, Marc Besancon, Fabrice Couderc, Marc Dejardin, Daniel Denegri, Jean-Louis Faure, Federico Ferri, Serguei Ganjour, Philippe Gras, Gautier Hamel de Monchenault, Mintu Kumar, Victor Lohezic, Julie Malcles, Francesco Orlandi, Louis Portales, Andr  Rosowsky, Mehmet  zg r Sahin, Aurore Savoy-Navarro, Polina Simkina, Maksym Titov, Marta Tornago, Florian Beaudette, Giacomo Boldrini, Philippe Busson, Alessandra Cappati, Claude Charlot, Marco Chiusi, Florian Damas, Olivier Davignon, Adinda De Wit, Isaac Telford Ehle, Bruno Afonso Fontana Santos Alves, Shamik Ghosh, Andrew Gilbert, Raphael Granier de Cassagnac, Alexandre Hakimi, Bharadwaj Harikrishnan, Lida Kalipoliti, Geliang Liu, Matthew Nguyen, Christophe Ochando, Roberto Salerno, Jean-Baptiste Sauvan, Yves Sirois, Lourdes Urda G mez, Elena Vernazza, Alexandre Zabi, Amina Zghiche, Jean-Laurent Agram, Jeremy Andrea, Dylan Apparu, Daniel Bloch, Jean-Marie Brom, Eric Christian Chabert, Caroline Collard, Saskia Falke, Ulrich Goerlach, Raphael Haeberle, Anne-Catherine Le Bihan, Meena Meena, Oceane Poncet, Gourab Saha, Mario Antoine Sessini, Pierre Van Hove, Paul Vaucelle, Adriano Di Florio, David Amram, Stephanie Beauceron, Benjamin Blancon, Gaelle Boudoul, Nicolas Chanon, Didier Contardo, Pierre Depasse, Candan Dozen, Houmani El Mamouni, Jean Fay, Susan Gascon, Maxime Gouzevitch, Christopher Greenberg, G rald Grenier, Bernard Ille, Elise Jourd'huy, Imad Baptiste Laktineh, Morgan Lethuillier, Laurent Mirabito, Stephane Perries, Arnab Purohit, Muriel Vander Donckt, Patrice Verdier, Jie Xiao, Arsen Khvedelidze, Irakli Lomidze, Zviad Tsamalaidze, Valeria Botta, Sandra Consuegra Rodr guez, Lutz Feld, Katja

Klein, Martin Lipinski, Danilo Meuser, Alexander Pauls, Danyer Pèrez Adán, Nicolas Röwert, Marius Teroerde, Svenja Diekmann, Alena Dodonova, Niclas Eich, Dmitry Eliseev, Frederic Engelke, Johannes Erdmann, Martin Erdmann, Peter Fackeldey, Benjamin Fischer, Thomas Hebbeker, Kerstin Hoepfner, Francesco Ivone, Alexander Jung, Ming-yan Lee, Florian Mausolf, Markus Merschmeyer, Arnd Meyer, Swagata Mukherjee, Dennis Noll, Fabian Nowotny, Andrey Pozdnyakov, Yannik Rath, Wahid Redjeb, Florian Rehm, Hans Reithler, Valentina Sarkisovi, Alexander Schmidt, Archana Sharma, Jan Lukas Spah, Annika Stein, Felipe Torres Da Silva De Araujo, Sebastian Wiedenbeck, Shawn Zaleski, Christian Dziwok, Günter Flügge, Thomas Kress, Andreas Nowack, Oliver Pooth, Achim Stahl, Tim Ziemons, Alexander Zotz, Henriette Aarup Petersen, Maria Aldaya Martin, Juliette Alimena, Simone Amoroso, Ying An, Jorn Bach, Samuel Baxter, Maryam Bayatmakou, Hugo Becerril Gonzalez, Olaf Behnke, Alberto Belvedere, Freya Blekman, Kerstin Borrás, Alan Campbell, Andrea Cardini, Chun Cheng, Federica Colombina, Malinda De Silva, Guenter Eckerlin, Doris Eckstein, Luis Ignacio Estevez Banos, Oleg Filatov, Elisabetta Gallo, Achim Geiser, Valentina Guglielmi, Moritz Guthoff, Andreas Hinzmann, Laurids Jeppe, Benno Kaech, Matthias Kasemann, Claus Kleinwort, Roman Kogler, Matthias Komm, Dirk Krücker, Wolfgang Lange, Daina Leyva Pernia, Katerina Lipka, Wolfgang Lohmann, Florian Lorkowski, Rainer Mankel, Isabell-Alissandra Melzer-Pellmann, Mikel Mendizabal Morentin, Andreas Bernhard Meyer, Gabriele Milella, Keila Moral Figueroa, Andreas Mussgiller, Lakshmi Priya Nair, Jeremi Niedziela, Andreas Nürnberg, Younes Otari, Jiwon Park, Evan Ranken, Alexei Raspereza, Daniil Rastorguev, Jonas Rübenach, Lovisa Rygaard, Alessia Saggio, Moritz Scham, Simon Schnake, Paul Schütze, Christian Schwanenberger, Daria Selivanova, Konstantin Sharko, Mykyta Shchedrolosiev, Dominic Stafford, Federico Vazzoler, Ana Ventura Barroso, Roberval Walsh, Di Wang, Qun Wang, Yiwen Wen, Katarzyna Wichmann, Lucas Wiens, Christoph Wissing, Yewon Yang, André Zimmermann Castro Santos, Anna Albrecht, Steffen Albrecht, Massimiliano Antonello, Samuel Bein, Lisa Benato, Sven Bollweg, Matteo Bonanomi, Patrick Connor, Karim El Morabit, Yannick Fischer, Erika Garutti, Alexander Grohsjean, Johannes Haller, Henrik Ronald Jabusch, Gregor Kasieczka, Philip Keicher, Robert Klanner, William Korcari, Tobias Kramer, Chin-chia Kuo, Viktor Kutzner, Finn Labe, Johannes Lange, Artur Lobanov, Christopher Matthies, Louis Moureaux, Malte Mrowietz, Aliya Nigamova, Yuval Nissan, Alexander Paasch, Karla Josefina Pena Rodriguez, Tobias Quadfasel, Bianca Raciti, Marcel Rieger, Daniel Savoiu, Jörg Schindler, Peter Schleper, Matthias Schröder, Joern Schwandt, Manuel Sommerhalder, Hartmut Stadie, Georg Steinbrück, Alexandra Tews, Moritz Wolf, Sebastian Brommer, Maximilian Burkart, Erik Butz, Thorsten Chwalek, Alexander Dierlamm, Alexander Droll, Umut Elicabuk, Nils Faltermann, Manuel Giffels, Artur Gottmann, Frank Hartmann, Robin Hofsaess, Maximilian Horzela, Ulrich Husemann, Jan Kieseler, Markus Klute, Roland Koppenhöfer, Jay Mathew Lawhorn, Marco Link, Adelina Lintuluoto, Benedikt Maier, Stefan Maier, Soureek Mitra, Michele Mormile, Thomas Müller, Max Neukum, Minseok Oh, Emanuel

Pfeffer, Matteo Presilla, Gunter Quast, Klaus Rabbertz, Brendan Regnery, Nikita Shadskiy, Ivan Shvetsov, Hans-Jürgen Simonis, Lars Sowa, Lea Stockmeier, Komal Taukeer, Maria Toms, Nicolò Trevisani, Ralf Florian Von Cube, Michael Wassmer, Sebastian Wieland, Florian Wittig, Roger Wolf, Xunwu Zuo, Georgios Anagnostou, Georgios Daskalakis, Aristotelis Kyriakis, Alkiviadis Papadopoulos, Anna Stakia, Pantelis Kontaxakis, Georgios Melachroinos, Zacharias Painesis, Ioanna Papavergou, Ioannis Paraskevas, Niki Saoulidou, Konstantinos Theofilatos, Eirini Tziaferi, Konstantinos Vellidis, Ilias Zisopoulos, Georgios Bakas, Theodoros Chatzistavrou, Georgia Karapostoli, Konstantinos Kousouris, Ioannis Papakrivopoulos, Eirini Siamarkou, Georgios Tsipolitis, Anna Zacharopoulou, Kosmas Adamidis, Ioannis Bestintzanos, Ioannis Evangelou, Costas Foudas, Christos Kamtsikis, Panagiotis Katsoulis, Panagiotis Kokkas, Polidamas Georgios Kosmoglou Kioseoglou, Nikolaos Manthos, Ioannis Papadopoulos, John Strologas, Csaba Hajdu, Dezso Horvath, Krisztina Márton, Attila József Rádl, Ferenc Sikler, Viktor Veszpremi, Mate Csanad, Krisztián Farkas, Anna Fehérkúti, Mahmod Moussa Abdelkhalek Gadallah, Ármín Kadlecsek, Péter Major, Gabriella Pasztor, Gabor Istvan Veres, Balázs Ujvári, Gyula Zilizi, György Bencze, Sandor Czellar, József Molnár, Zoltan Szillasi, Frigyes Nemes, Tamas Novak, Sunil Bansal, Suman Bala Beri, Vipin Bhatnagar, Geetanjali Chaudhary, Sushil Chauhan, Nitish Dhingra, Amandeep Kaur, Anureet Kaur, Harjot Kaur, Manjit Kaur, Sunil Kumar, Kaur Sandeep, Tanvi Sheokand, Jasbir Singh, Aayushi Singla, Asar Ahmed, Ashutosh Bhardwaj, Arjun Chhetri, Brajesh C. Choudhary, Ashok Kumar, Arun Kumar, Mohammad Naimuddin, Kirti Ranjan, Mahesh Kumar Saini, Saumya Saumya, Sweta Baradia, Soumyadip Barman, Satyaki Bhattacharya, Suman Das Gupta, Shubham Dutta, Suchandra Dutta, Subir Sarkar, Mohammad Mobassir Ameen, Prafulla Kumar Behera, Subash Chandra Behera, Sayan Chatterjee, Ganapati Dash, Pranati Jana, Pritam Kalbhor, Samadhan Kamble, Jyotsna Rani Komaragiri, Deepak Kumar, Prabhat Ranjan Pujahari, Nihar Ranjan Saha, Ashish Sharma, Anup Kumar Sikdar, Rohit Kumar Singh, Pranjal Verma, Sadhana Verma, Anusree Vijay, Shashikant Dugad, Gagan Bihari Mohanty, Bibhuti Parida, Mukund Shelake, Pruthvi Suryadevara, Anirban Bala, Sudeshna Banerjee, Rajdeep Mohan Chatterjee, Monoranjan Guchait, Shilpi Jain, Atul Jaiswal, Sanjeev Kumar, Gobinda Majumder, Kajari Mazumdar, Shubhi Parolia, Aravind Thachayath, Seema Bahinipati, Chandiprasad Kar, Dipak Maity, Prolay Mal, Tribeni Mishra, Vinaya Krishnan Muralaeddharan Nair Bindhu, Kousik Naskar, Aruna Nayak, Samarendra Nayak, Kuldeep Pal, Priyanka Sadangi, Sanjay Kumar Swain, Sanu Varghese, Diwakar Vats, Shriniketan Acharya, Alpana Alpana, Sourabh Dube, Bhawna Gomber, Prachurjya Hazarika, Bhumika Kansal, Arnab Laha, Bisnupriya Sahu, Seema Sharma, Kumar Yash Vaish, Hamed Bakhshiansohi, Abideh Jafari, Maryam Zeinali, Sima Bashiri, Shirin Chenarani, Seyed Mohsen Etesami, Yasaman Hosseini, Mohsen Khakzad, Elham Khazaie, Mojtaba Mohammadi Najafabadi, Seddigheh Tizchang, Marta Felcini, Martin Grunewald, Marcello Abbrescia, Anna Colaleo, Donato Creanza, Brunella D'Anzi, Nicola De Filippis, Mauro De Palma, Walaa Elmetenawee, Luigi Fiore, Giuseppe Iaselli, Luigi Longo,

Magdy Louka, Giorgio Maggi, Marcello Maggi, Ilirjan Margjeka, Vincenzo Mastrapasqua, Salvatore My, Salvatore Nuzzo, Antonello Pellecchia, Alexis Pompili, Gabriella Pugliese, Raffaella Radogna, Dayron Ramos, Antonio Ranieri, Lucia Silvestris, Federica Maria Simone, Ümit Sözbilir, Anna Stamerra, Donato Troiano, Rosamaria Venditti, Piet Verwilligen, Angela Zaza, Giovanni Abbiendi, Carlo Battilana, Daniele Bonacorsi, Paolo Capiluppi, Andrea Castro, Francesca Romana Cavallo, Marco Cuffiani, Gaetano-Marco Dallavalle, Tommaso Diotallevi, Fabrizio Fabbri, Alessandra Fanfani, Daniele Fasanella, Paolo Giacomelli, Luca Giommi, Claudio Grandi, Luigi Guiducci, Sergio Lo Meo, Marco Lorusso, Leonardo Lunerti, Stefano Marcellini, Gianni Masetti, Francesco Navarra, Giulia Paggi, Andrea Perrotta, Federica Primavera, Antonio Rossi, Simone Rossi Tisbeni, Tiziano Rovelli, Gian Piero Siroli, Salvatore Costa, Alessandro Di Mattia, Alessandro Lapertosa, Renato Potenza, Alessia Tricomi, Cristina Tuve, Panagiotis Assiouras, Giuseppe Barbagli, Giulio Bardelli, Benedetta Camaiani, Antonio Cassese, Rudy Ceccarelli, Vitaliano Ciulli, Carlo Civinini, Raffaello D'Alessandro, Ettore Focardi, Tomas Kello, Giuseppe Latino, Piergiulio Lenzi, Mattia Lizzo, Marco Meschini, Simone Paoletti, Alkis Papanastassiou, Giacomo Sguazzoni, Lorenzo Viliani, Luigi Benussi, Stefano Bianco, Sabino Meola, Davide Piccolo, Pierre Chatagnon, Fabrizio Ferro, Enrico Robutti, Silvano Tosi, Andrea Benaglia, Francesco Brivio, Flavia Cetorelli, Federico De Guio, Mauro Emanuele Dinardo, Paolo Dini, Simone Gennai, Raffaele Gerosa, Alessio Ghezzi, Pietro Govoni, Luca Guzzi, Marco Toliman Lucchini, Martina Malberti, Sandra Malvezzi, Andrea Massironi, Dario Menasce, Luigi Moroni, Marco Paganoni, Simona Palluotto, Daniele Pedrini, Aurora Perego, Bianca Sofia Pinolini, Giorgio Pizzati, Stefano Ragazzi, Tommaso Tabarelli de Fatis, Salvatore Buontempo, Antimo Cagnotta, Francesco Carnevali, Nicola Cavallo, Francesco Fabozzi, Alberto Orso Maria Iorio, Luca Lista, Pierluigi Paolucci, Biagio Rossi, Rocco Ardino, Patrizia Azzi, Nicola Bacchetta, Pierluigi Bortignon, Gabriele Bortolato, Alberto Bragagnolo, Andrea Claudio Maria Bulla, Roberto Carlin, Paolo Checchia, Tommaso Dorigo, Ugo Gasparini, Franco Gonella, Andrea Gozzelino, Enrico Lusiani, Martino Margoni, Anna Teresa Meneguzzo, Matteo Migliorini, Jacopo Pazzini, Paolo Ronchese, Roberto Rossin, Franco Simonetto, Mia Tosi, Andrea Triossi, Sandro Ventura, Marco Zanetti, Pierluigi Zotto, Alberto Zucchetta, Gianni Zumerle, Chiara Aimè, Alessandro Braghieri, Simone Calzaferri, Davide Fiorina, Paolo Montagna, Valerio Re, Cristina Riccardi, Paola Salvini, Ilaria Vai, Paolo Vitulo, Sehar Ajmal, Maria Elena Ascioti, Gian Mario Bilei, Costanza Carrivale, Diego Ciangottini, Livio Fanò, Matteo Magherini, Valentina Mariani, Mauro Menichelli, Francesco Moscatelli, Alessandro Rossi, Attilio Santocchia, Daniele Spiga, Tommaso Tedeschi, Cristina Andreea Alexe, Patrick Asenov, Paolo Azzurri, Giuseppe Bagliesi, Rajarshi Bhattacharya, Lorenzo Bianchini, Tommaso Boccali, Edoardo Bossini, Davide Bruschini, Rino Castaldi, Maria Agnese Ciocci, Marco Cipriani, Valeria D'Amante, Roberto Dell'Orso, Silvio Donato, Alessandro Giassi, Franco Ligabue, Andrea Carlo Marini, Diego Matos Figueiredo, Alberto Messineo, Marco Musich, Fabrizio Palla, Andrea Rizzi, Gigi Rolandi, Suvankar Roy Chowdhury, Tanmay Sarkar, Angelo Scribano,

Paolo Spagnolo, Roberto Tenchini, Guido Tonelli, Nicola Turini, Francesco Vaselli, Andrea Venturi, Piero Giorgio Verdini, Cristian Baldenegro Barrera, Patrizia Barria, Chiara Basile, Mattia Campana, Francesca Cavallari, Leticia Cunqueiro Mendez, Daniele Del Re, Emanuele Di Marco, Marcella Diemoz, Filippo Errico, Egidio Longo, Jelena Mijuskovic, Giovanni Organtini, Francesco Pandolfi, Riccardo Paramatti, Claudio Quaranta, Shahram Rahatlou, Chiara Rovelli, Francesco Santanastasio, Livia Soffi, Nicola Amapane, Roberta Arcidiacono, Stefano Argiro, Michele Arneodo, Nazar Bartosik, Riccardo Bellan, Andrea Bellora, Cristina Biino, Cecilia Borca, Nicolo Cartiglia, Marco Costa, Roberto Covarelli, Natale Demaria, Linda Finco, Michael Grippo, Bilal Kiani, Federica Legger, Fabio Luongo, Chiara Mariotti, Lazar Markovic, Silvia Maselli, Alberto Mecca, Luca Menzio, Paolo Meridiani, Ernesto Migliore, Marco Monteno, Roberto Mulargia, Maria Margherita Obertino, Giacomo Ortona, Luca Pacher, Nadia Pastrone, Mario Pelliccioni, Marta Ruspa, Federico Siviero, Valentina Sola, Ada Solano, Amedeo Staiano, Cristiano Tarricone, Daniele Trocino, Giulio Umoret, Robert White, Jyoti Babbar, Stefano Belforte, Vieri Candelise, Massimo Casarsa, Fabio Cossutti, Ksenia De Leo, Giuseppe Della Ricca, Sunil Dogra, Jieun Hong, Changgi Huh, Bobae Kim, Jiwoong Kim, Dongyub Lee, Hakseong Lee, Seh Wook Lee, Chang-Seong Moon, Young Do Oh, Min Sang Ryu, Sezen Sekmen, Bongho Tae, Yu Chul Yang, Min Suk Kim, Gyeonghwan Bak, Piljun Gwak, Hyunchul Kim, Dong Ho Moon, Ece Asilar, Jieun Choi, Donghyun Kim, Tae Jeong Kim, Jeremie Alexandre Merlin, Yeonsu Ryou, Suyong Choi, Seyoung Han, Byung-Sik Hong, Kisoo Lee, Kyong Sei Lee, Soohwan Lee, Jaehyeok Yoo, Junghwan Goh, Seungjin Yang, Hyunsoo Kim, Yongsun Kim, Seungjun Lee, John Almond, Ji Hwan Bhyun, Junho Choi, Jin Choi, Won Jun, Junho Kim, Sanghyun Ko, Hyejin Kwon, Haneol Lee, Joonbin Lee, Jeongeun Lee, Byung Hun Oh, Sung Bin Oh, Hyonsan Seo, Unki Yang, Inseok Yoon, Woojin Jang, Da Young Kang, Yechan Kang, Seulgi Kim, Byeonghak Ko, Jason Sang Hun Lee, Yunjae Lee, Inkyu Park, Youn Roh, Ian Watson, Seungkyu Ha, Hwi Dong Yoo, Minuk Choi, Mi Ran Kim, Heewon Lee, Younghoon Lee, Intae Yu, Taha Beyrouthy, Yousef Gharbia, Karlis Dreimanis, Antra Gaile, Conrado Munoz Diaz, Dace Osite, Guntis Pikurs, Andris Potrebko, Markus Seidel, Dimitrios Sidiropoulos Kontos, Normunds Ralfs Strautnieks, Marijus Ambrozias, Andrius Juodagalvis, Aurelijus Rinkevicius, Gintautas Tamulaitis, Imran Yusuff, Zukhaimira Zolkapli, Jose F. Benitez, Alfredo Castaneda Hernandez, Hedwin Aaron Encinas Acosta, Luis Gabriel Gallegos Maríñez, Moisés León Coello, Javier Alberto Murillo Quijada, Ashish Sehrawat, Lizardo Valencia Palomo, Gabriel Ayala, Heriberto Castilla-Valdez, Horacio Crotte Ledesma, Eduard De La Cruz-Burelo, Ivan Heredia-De La Cruz, Ricardo Lopez-Fernandez, Jhovanny Mejia Guisao, Cesar Atzin Mondragon Herrera, Alberto Sánchez Hernández, Cristina Oropeza Barrera, Dalia Lucero Ramirez Guadarrama, Mateo Ramírez García, Irais Bautista, Isabel Pedraza, Humberto Antonio Salazar Ibarguen, Cecilia Uribe Estrada, Itana Bujanja, Natasa Raicevic, Philip H. Butler, Ashfaq Ahmad, Muhammad Irfan Asghar, Ali Awais, Muhammad Imran Malik Awan, Hafeez R. Hoorani, Wajid Ali Khan, Valentina Avati, Leszek Grzanka, Maciej Malawski, He-

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bak Kim, Amina Jinzi Li, Phillip Masterson, Hualin Mei, Jeffrey Richman, Sai Neha Santpur, Ulascan Sarica, Ryan Schmitz, Francesco Setti, James Sheplock, David Stuart, Tamás Álmos Vámi, Sicheng Wang, Danyi Zhang, Soham Bhattacharya, Adolf Bornheim, Olmo Cerri, Anthony Latorre, Jiajing Mao, Harvey B. Newman, Guillermo Reales Gutiérrez, Maria Spiropulu, Jean-Roch Vlimant, Christina Wang, Si Xie, Ren-Yuan Zhu, John Alison, Sitong An, Patrick Bryant, Matteo Cremonesi, Valentina Dutta, Thomas Ferguson, Tirso Alejandro Gómez Espinosa, Abhirami Harilal, Aleesha Kallil Tharayil, Chuyuan Liu, Tanmay Mudholkar, Sindhu Murthy, Pritam Palit, Kyungmin Park, Manfred Paulini, Andrew Roberts, Alejandro Sanchez, Wesley Terrill, John Perry Cumalat, William T. Ford, Andrew Hart, Abbas Hassani, George Karathanasis, Nicholas Manganelli, Jannicke Parkes, Claire Savard, Nicolas Schonbeck, Kevin Stenson, Keith Ulmer, Stephen Robert Wagner, Noah Zipper, Davide Zuolo, James Alexander, Samuel Bright-Thonney, Xuan Chen, Derek James Cranshaw, Jason Fan, Xingchen Fan, Shaun Hogan, Peace Kotamnives, Jose Monroy, Michael Oshiro, Juliet Ritchie Patterson, Michael Reid, Anders Ryd, Julia Thom, Peter Wittich, Rui Zou, Michael Albrow, Maral Alyari, Oz Amram, Giorgio Apollinari, Artur Apresyan, Lothar A.T. Bauerdick, Douglas Berry, Jeffrey Berryhill, Pushpalatha C. Bhat, Kevin Burkett, Joel Nathan Butler, Anadi Canepa, Giuseppe Benedetto Cerati, Harry Cheung, Frank Chlebana, Grace Cummings, Jennet Dickinson, Irene Dutta, Victor Daniel Elvira, Yongbin Feng, Jim Freeman, Abhijith Gandrakota, Zoltan Gece, Lindsey Gray, Dan Green, Aidan Grummer, Stefan Grünendahl, Daniel Guerrero, Oliver Gutsche, Robert M. Harris, Ryan Heller, Theodor Christian Herwig, James Hirschauer, Bodhitha Jayatilaka, Sergo Jindariani, Marvin Johnson, Umesh Joshi, Thomas Klijnsma, Boaz Klima, Ka Hei Martin Kwok, Stephan Lammel, Don Lincoln, Ron Lipton, Tiehui Liu, Christopher Madrid, Kaori Maeshima, Cristina Mantilla, David Mason, Patricia McBride, Petra Merkel, Stephen Mrenna, Steve Nahn, Jennifer Ngadiuba, Daniel Noonan, Scarlet Norberg, Vaia Papadimitriou, Nathaniel Pastika, Kevin Pedro, Cristian Pena, Fabio Ravera, Allison Reinsvold Hall, Luciano Ristori, Murtaza Safdari, Elizabeth Sexton-Kennedy, Nicholas Smith, Aron Soha, Leonard Spiegel, Stoyan Stoynev, James Strait, Lucas Taylor, Slawek Tkaczyk, Nhan Viet Tran, Lorenzo Uplegger, Eric Wayne Vaandering, Irene Zoi, Caterina Aruta, Paul Avery, Dimitri Bourilkov, Philip Chang, Vladimir Cherepanov, Richard D. Field, Evan Koenig, Marina Kolosova, Jacobo Konigsberg, Andrey Korytov, Konstantin Matchev, Nicholas Menendez, Guenakh Mitselmakher, Kelci Mohrman, Arun Muthirakalayil Madhu, Neha Rawal, Suzanne Rosenzweig, Yuta Takahashi, Jian Wang, Todd Adams, Ali Al Kadhimi, Andrew Askew, Steffi Bower, Vasken Hagopian, Rafey Hashmi, Ryan Sangwoo Kim, Suho Kim, Ted Kolberg, German Martinez, Harrison Prosper, Purbita Rahman Prova, Meutia Wulansatiti, Rachel Yohay, Jingyu Zhang, Bandar Alsufyani, Marc M. Baarmand, Stephen Butalla, Souvik Das, Tamer Elkafrawy, Marcus Hohlmann, Erick Yanes, Mark Raymond Adams, Austin Baty, Clayton Bennett, Richard Cavanaugh, Ricardo Escobar Franco, Olga Evdokimov, Cecilia Elena Gerber, Michelle Hawksworth, Abhishek Hingrajiya, David Jonathan Hof-

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Babaev, Vladimir Blinov, Edouard Boos, Vladislav Borshch, Dzmitry Budkouski, Viacheslav Bunichev, Vladimir Chekhovsky, Ruslan Chistov, Mikhail Danilov, Alexander Dermenev, Tatyana Dimova, Dmitry Druzhkin, Mikhail Dubinin, Lev Dudko, Alexander Ershov, Genady Gavrilov, Vladimir Gavrilov, Sergei Gninenko, Victor Golovtsov, Nikolai Golubev, Igor Golutvin, Ilya Gorbunov, Yury Ivanov, Vassili Kachanov, Vladimir Karjavine, Anton Karneyeu, Victor Kim, Martin Kirakosyan, Dmitrii Kirpichnikov, Mikhail Kirsanov, Vyacheslav Klyukhin, Dmitri Konstantinov, Vladimir Korenkov, Aleksei Kozyrev, Nikolai Krasnikov, Alexander Lanev, Petr Levchenko, Natalia Lychkovskaya, Vladimir Makarenko, Alexander Malakhov, Anastasia Markina, Viktor Matveev, Victor Murzin, Alexander Nikitenko, Stepan Obraztsov, Vadim Oreshkin, Vladimir Palichik, Victor Perelygin, Maxim Perfilov, Sergey Polikarpov, Vladimir Popov, Olesia Radchenko, Maria Savina, Viktor Savrin, Vladislav Shalaev, Sergey Shmatov, Siarhei Shulha, Yuri Skovpen, Sergei Slabospitskii, Vitaly Smirnov, Dmitry Sosnov, Valentin Sulimov, Adel Terkulov, Oleg Teryaev, Irina Tlisova, Alexander Toropin, Lev Uvarov, Andrey Uzunian, Petr Volkov, Alexey Vorobyev, George Vorotnikov, Nikolay Voytishin, Bekhzod S. Yuldashev, Anatoli Zarubin, Ilia Zhizhin, Alexander Zhokin

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3.9.4.16 Measurements of polarization and spin correlation and observation of entanglement in top quark pairs using $\ell^+\ell^- \rightarrow t\bar{t} + \text{jets}$ events from proton-proton collisions at $\sqrt{s} = 13.6$ TeV

- **Authors:** Aram Hayrapetyan, Armen Tumasyan, Wolfgang Adam, Janik Walter Andrejkovic, Lisa Benato, Thomas Bergauer, Sandhya Jain, Muhammad Abdullah Abdulfattah Muhammad, Thomas Müller, Sayan Chatterjee, Konstantinos Damanakis, Marko Dragicevic, Priya Sajid Hussain, Manfred Jeitler, Natascha Krammer, Ang Li, Dietrich Liko, Ivan Mikulec, Jochen Schieck, Robert Schöfbeck, Dennis Schwarz, Mangesh Sonawane, Wolfgang Waltenberger, Claudia-Elisabeth Wulz, Tahys Janssen, Thomas Van Laer, Pierre Van Mechelen, Nordin Breugelmans, Jorgen D'Hondt, Soumya Dansana, Alexandre De Moor, Martin Delcourt, Felix Heyen, Steven Lowette, Inna Makarenko, Denise Müller, Stefaan Tavernier, Michael Tytgat, Gerrit Patrick Van Onsem, Senne Van Putte, David Vannerom, Bugra Bilin, Barbara Clerbaux, Alope Kumar Das, Isabelle De Bruyn, Gilles De Lentdecker, Hugues Evard, Laurent Favart, Paraskevas Giannaios, Johny Jaramillo, Ali Khalilzadeh, Fakhri Alam Khan, Kyeongpil Lee, Andrea Malara, Santiago Paredes, Muhammad Aamir Shahzad, Laurent Thomas, Max Vanden Bemden, Catherine Vander Velde, Pascal Vanlaer, Maarten De Coen, Didar Dobur, Gul Gokbulut, Yanwen Hong, Joscha Knolle, Luka Lambrecht, David Marckx, Kevin Mota Amarilo, Kirill Skovpen, Niels Van Den Bossche, Jan van der Linden, Liam Wezenbeek, Anna Benecke, Agni Bethani, Giacomo Bruno, Claudio Caputo, Jerome De Favereau

De Jeneret, Christophe Delaere, Izzeddin Suat Donertas, Andrea Giammanco, Ahmet Oguz Guzel, Sa. Jain, Vincent Lemaitre, Jindrich Lidrych, Paola Mastrapasqua, Tu Thong Tran, Semra Turkcapar, Gilvan Alves, Eduardo Coelho, Gilson Correia Silva, Carsten Hensel, Thales Menezes De Oliveira, Clemencia Mora Herrera, Patricia Rebello Teles, Mariana Soeiro, Edmilson José Tonelli Manganote, Antonio Vilela Pereira, Walter Luiz Aldá Júnior, Mapse Barroso Ferreira Filho, Helena Brandao Malbouisson, Wagner Carvalho, Jose Chinellato, Eliza Melo Da Costa, Gustavo Gil Da Silveira, Dilson De Jesus Damiao, Sandro Fonseca De Souza, Raphael Gomes De Souza, Tulio Laux Kuhn, Matheus Macedo, Jordan Martins, Luiz Mundim, Helio Nogima, Joao Pedro Pinheiro, Alberto Santoro, Andre Sznajder, Mauricio Thiel, Cesar Augusto Bernardes, Luigi Calligaris, T.R. Fernandez Perez Tomei, Eduardo De Moraes Gregores, Isabela Maietto Silverio, Pedro G. Mercadante, Sergio F. Novaes, Breno Orzari, Sandra S. Padula, Aleksandar Aleksandrov, Georgy Antchev, Roumyana Hadjiiska, Plamen Iaydjiev, Milena Misheva, Mariana Shopova, Georgi Sultanov, Anton Dimitrov, Leander Litov, Borislav Pavlov, Peicho Petkov, Anton Petrov, Elton Shumka, Sumit Keshri, David Nicolas Laroze Navarrete, Shalini Thakur, Tongguang Cheng, Tahir Javaid, Li Yuan, Zhen Hu, Zhengchen Liang, Jinfeng Liu, Guo-Ming Chen, He-Sheng Chen, Mingshui Chen, Fabio Iemmi, Chun-Hua Jiang, Anshul Kapoor, Hongbo Liao, Zhenan Liu, Ramkrishna Sharma, Jia-ning Song, Junquan Tao, Shilpi Jain, Hakseong Lee, Yongsun Kim, Young Wan Kim, Heewon Lee, Christina Wang, Jin Wang, Drew Baden, Zhangqier Wang, Huaqiao Zhang, Jingzhou Zhao, Antonis Agapitos, Yong Ban, Sen Deng, Botao Guo, Chuqiao Jiang, Andrew Levin, Congqiao Li, Qiang Li, Yajun Mao, Sitian Qian, Si-Jin Qian, Xuelong Qin, Xiaohu Sun, Dayong Wang, Heng Yang, Licheng Zhang, Yuzhe Zhao, Chen Zhou, Shuai Yang, Zhengyun You, Khawla Jaffel, Nan Lu, Gerry Bauer, Bolin Li, Kai Yi, Jingqing Zhang, Yuji Li, Zhen Lin, Chenfeng Lu, Meng Xiao, Carlos Avila, Diego Andres Barbosa Trujillo, Andrés Cabrera, Carlos Florez, Jorge Fraga, Jose Antonio Reyes Vega, Felipe Ramirez, Cèsar Rendón, Manuel Rodriguez, Anderson Alexis Ruales Barbosa, José David Ruiz Alvarez, Duje Giljanovic, Nikola Godinovic, Damir Lelas, Ana Sculac, Marko Kovac, Andro Petkovic, Toni Sculac, Pedrame Bargassa, Vuko Brigljevic, Bhakti Kanulal Chitroda, Dinko Ferencek, Kresimir Jakovcic, Andrei Starodumov, Tatjana Susa, Alexandros Attikis, Konstantinos Christoforou, Andreas Hadjiagapiou, Christos Leonidou, Jihad Mousa, Charalambos Nicolaou, Leonidas Paizanos, Fotios Ptochos, Panos A. Razis, Hans Rykaczewski, Halil Saka, Anton Stepenov, Miroslav Finger, Michael Finger, Antonin Kveton, Edgar Carrera Jarrin, Yasser Assran, Basma El-mahdy, Sherif Elgammal, M. Abdullah Al-Mashad, Mohammed Mahmoud, Karl Ehataht, Mario Kadastik, Torben Lange, Saswati Nandan, Christine Nielsen, Joosep Pata, Martti Raidal, Laurits Tani, Christian Veelken, Henning Kirschenmann, Kenneth Osterberg, Mikko Voutilainen, Shudhashil Bharthuar, Nurfikri Bin Norjoharuddeen, Erik Brücken, Francisco Ignacio Garcia Fuentes, Patin Inkaew, Kimmo Tuomas Samuli Kallonen, Tapio Lampèn, Kati Lassila-Perini, Sami Lehti, Tomas Lindèn, Mikael Myllymäki, Milla-maarit Rantanen, Hannu Siikonen,

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- **Posted on arXiv:** 17 September 2024

- **Published in Phys. Rev. D 110 (2024) 112016:** 01 December 2024

3.9.4.17 Quantum information meets high-energy physics: input to the update of the European strategy for particle physics [270]

- **Authors:** Yoav Afik, Federica Fabbri, Matthew Low, Luca Marzola, Juan Antonio Aguilar-Saavedra, Mohammad Mahdi Altakach, Nedaa Alexandra Asbah, Yang Bai, Hannah Banks, Alan J. Barr, Alexander Bernal, Thomas E. Browder, Paweł Caban, J. Alberto Casas, Kun Cheng, Frédéric Dèliot, Regina Demina, Antonio Di Domenico, Michał Eckstein, Marco Fabbrichesi, Benjamin Fuks, Emidio Gabrielli, Dorival Gonçalves, Radosław Grabarczyk, Michele Grossi, Tao Han, Timothy J. Hobbs, Paweł Horodecki, James Howarth, Shih-Chieh Hsu, Stephen Jiggins, Eleanor Jones, Andreas W. Jung, Andrea Helen Knue, Steffen Korn, Theodota Lagouri, Priyanka Lamba, Gabriel T. Landi, Haifeng Li, Qiang Li, Ian Low, Fabio Maltoni, Josh McFayden, Navin McGinnis, Roberto A. Morales, Jesús M. Moreno, Juan Ramón Muñoz de Nova, Giulia Negro, Davide Pagani, Giovanni Pelliccioli, Michele Pinamonti, Laura Pintucci, Baptiste Ravina, Alim Ruzi, Kazuki Sakurai, Ethan Simpson, Maximiliano Sioli, Shufang Su, Sokratis Trifinopoulos, Sven E. Vahsen, Sofia Vallecorsa, Alessandro Vicini, Marcel Vos, Eleni Vryonidou, Chris D. White, Martin J. White, Andrew J. Wildridge, Tong Arthur Wu, Laura Zani, Yulei Zhang, Knut Zoch

- **Posted on arXiv:** 31 March 2025

- **Published in Eur.Phys.J.Plus:** 09 September 2025

3.9.4.18 Colliders are Testing neither Locality via Bell’s Inequality nor Entanglement versus Non-Entanglement [271]

- **Authors:** Steven A. Abel, Herbi K. Dreiner, Rhitaja Sengupta, Lorenzo Ubaldi

- **Posted on arXiv:** 21 July 2025

3.9.5 Crosslists

3.9.5.1 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [11]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle

- **Posted on arXiv:** 14 March 2022

3.9.5.2 Quantum entanglement and Bell inequality violation at colliders [7]

- **Authors:** Alan J. Barr, Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola

- **Posted on arXiv:** 12 February 2024

- **Published in Prog.Part.Nucl.Phys.:** 19 July 2024

3.10 Quantum Error Correction and Mitigation

3.10.1 Crosslists

3.10.1.1 Quantum error mitigation for Fourier moment computation [245]

- **Authors:** Oriel Kiss, Michele Grossi, Alessandro Roggero
- **Posted on arXiv:** 23 January 2024
- **Published in Phys. Rev. D 111, 034504 (2025):** 01 February 2025

3.11 Quantum Machine Learning - Supervised Methods

3.11.1 Anomaly Detection

3.11.1.1 Unravelling physics beyond the standard model with classical and quantum anomaly detection [24]

- **Authors:** Julian Schuhmacher, Laura Boggia, Vasilis Belis, Ema Puljak, Michele Grossi, Maurizio Pierini, Sofia Vallecorsa, Francesco Tacchino, Panagiotis Barkoutsos, Ivano Tavernelli
- **Posted on arXiv:** 25 January 2023
- **Published in Mach.Learn.Sci.Tech.:** 16 November 2023

3.11.2 Event Classification

3.11.2.1 Quantum Support Vector Machines for Continuum Suppression in B Meson Decays [138]

- **Authors:** Jamie Heredge, Charles Hill, Lloyd Hollenberg, Martin Sevier
- **Posted on arXiv:** 22 March 2021
- **Published in Comput.Softw.Big Sci.:** 30 November 2021

3.11.2.2 Application of quantum machine learning using the quantum kernel algorithm on high energy physics analysis at the LHC [139]

- **Authors:** Sau Lan Wu, Shaojun Sun, Wen Guan, Chen Zhou, Jay Chan, Chi Lung Cheng, Tuan Pham, Yan Qian, Alex Zeng Wang, Rui Zhang, Miron Livny, Jennifer Glick, Panagiotis Kl. Barkoutsos, Stefan Woerner, Ivano Tavernelli, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 11 April 2021
- **Published in Phys. Rev. Research 3, 033221 (2021):** 08 September 2021

3.11.2.3 Higgs analysis with quantum classifiers [140]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in nan:** 2021

3.11.2.4 Enforcing exact permutation and rotational symmetries in the application of quantum neural networks on point cloud datasets [141]

- **Authors:** Zhelun Li, Lento Nagano, Koji Terashi
- **Posted on arXiv:** 17 May 2024
- **Published in Phys.Rev.Res.:** 10 October 2024

3.11.2.5 From Reachability to Learnability: Geometric Design Principles for Quantum Neural Networks [142]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky
- **Posted on arXiv:** 03 March 2026

3.11.3 Jet Reconstruction

3.11.3.1 Quantum-inspired machine learning on high-energy physics data [172]

- **Authors:** Timo Felser, Marco Trenti, Lorenzo Sestini, Alessio Gianelle, Davide Zuliani, Donatella Lucchesi, Simone Montangero
- **Posted on arXiv:** 28 April 2020
- **Published in npj Quantum Inf.:** 15 July 2021

3.11.3.2 Quantum-inspired event reconstruction with Tensor Networks: Matrix Product States [173]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 15 June 2021
- **Published in JHEP 08 (2021) 112:** 23 August 2021

3.11.3.3 Classical versus quantum: Comparing tensor-network-based quantum circuits on Large Hadron Collider data [174]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 21 February 2022
- **Published in Phys. Rev. A 106 (Dec, 2022) 062423:** 19 December 2022

3.11.4 Lattice Field Theories - Gauge Theories

3.11.4.1 Quantum data learning for quantum simulations in high-energy physics [163]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

3.11.5 Track Reconstruction

3.11.5.1 Reconstructing charged particle track segments with a quantum-enhanced support vector machine [282]

- **Authors:** Philippa Duckett, Gabriel Facini, Marcin Jastrzebski, Sarah Malik, Tim Scanlon, Sebastien Rettie
- **Posted on arXiv:** 14 December 2022
- **Published in Phys.Rev.D:** 01 March 2024

3.11.6 Crosslists

3.11.6.1 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.11.6.2 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borrás, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico

Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.11.6.3 Quantum-centric Supercomputing for Physics Research [124]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

3.12 Quantum Machine Learning - Unsupervised Methods

3.12.1 Anomaly Detection

3.12.1.1 Quantum anomaly detection in the latent space of proton collision events at the LHC [25]

- **Authors:** Vasilis Belis, Kinga Anna Woźniak, Ema Puljak, Panagiotis Barkoutsos, Günther Dissertori, Michele Grossi, Maurizio Pierini, Florentin Reiter, Ivano Tavernelli, Sofia Vallecorsa
- **Posted on arXiv:** 25 January 2023
- **Published in Commun.Phys.:** 14 October 2024

3.12.2 Detector Technologies and Simulations

3.12.2.1 Dual-Parameterized Quantum Circuit GAN Model in High Energy Physics [120]

- **Authors:** Su Yeon Chang, Steven Herbert, Sofia Vallecorsa, Elías F. Combarro, Ross Duncan
- **Posted on arXiv:** 29 March 2021
- **Published in EPJ Web Conf.:** 2021

3.12.2.2 Running the Dual-PQC GAN on noisy simulators and real quantum hardware [121]

- **Authors:** Su Yeon Chang, Edwin Agnew, Elías F. Combarro, Michele Grossi, Steven Herbert, Sofia Vallecorsa
- **Posted on arXiv:** 30 May 2022
- **Published in nan:** 2023

3.12.2.3 Precise image generation on current noisy quantum computing devices [122]

- **Authors:** Florian Rehm, Sofia Vallecorsa, Kerstin Borras, Dirk Krücker, Michele Grossi, Valle Varo
- **Posted on arXiv:** 11 July 2023
- **Published in IOP Quantum Science and Technology (October 2023):** 30 October 2023

3.12.3 Event Classification

3.12.3.1 Guided graph compression for quantum graph neural networks [143]

- **Authors:** Mikel Casals, Vasilis Belis, Elias F. Combarro, Eduard Alarcón, Sofia Vallecorsa, Michele Grossi
- **Posted on arXiv:** 11 June 2025
- **Published in Mach.Learn.Sci.Tech.:** 12 September 2025

3.12.4 Event Generation

3.12.4.1 Style-based quantum generative adversarial networks for Monte Carlo events [157]

- **Authors:** Carlos Bravo-Prieto, Julien Baglio, Marco Cè, Anthony Francis, Dorota M. Grabowska, Stefano Carrazza
- **Posted on arXiv:** 13 October 2021
- **Published in Quantum 6, 777 (2022):** 17 August 2022

3.12.4.2 Quantum integration of elementary particle processes [158]

- **Authors:** Gabriele Agliardi, Michele Grossi, Mathieu Pellen, Enrico Prati
- **Posted on arXiv:** 05 January 2022
- **Published in Phys.Lett.B:** 07 June 2022

3.12.4.3 Generative invertible quantum neural networks [159]

- **Authors:** Armand Rousselot, Michael Spannowsky
- **Posted on arXiv:** 24 February 2023
- **Published in SciPost Phys. 16, 146 (2024):** 04 June 2024

3.12.4.4 Quantum-centric Supercomputing for Physics Research [124]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

3.12.5 Jet Reconstruction

3.12.5.1 A Digital Quantum Algorithm for Jet Clustering in High-Energy Physics [175]

- **Authors:** Diogo Pires, Pedrame Bargassa, João Seixas, Yasser Omar
- **Posted on arXiv:** 11 January 2021

3.12.5.2 Quantum clustering and jet reconstruction at the LHC [176]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 13 April 2022
- **Published in Physical Review D 106, 036021 (2022):** 01 August 2022

3.12.6 Crosslists

3.12.6.1 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [118]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

3.12.6.2 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.12.6.3 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.12.6.4 Machine learning for anomaly detection in particle physics [6]

- **Authors:** Vasilis Belis, Patrick Odagiu, Thea Klæboe Aarrestad
- **Posted on arXiv:** 20 December 2023
- **Published in Reviews in Physics, vol. 12, 2024, 100091:** 18 January 2024

3.13 Quantum Machine Learning - Variational Quantum Algorithms

3.13.1 Anomaly Detection

3.13.1.1 Anomaly detection in high-energy physics using a quantum autoencoder [26]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky, Michihisa Takeuchi
- **Posted on arXiv:** 09 December 2021
- **Published in Phys. Rev. D 105, 095004, 2022:** 01 May 2022

3.13.1.2 Quantum anomaly detection for collider physics [27]

- **Authors:** Sulaiman Alvi, Christian W. Bauer, Benjamin Nachman
- **Posted on arXiv:** 16 June 2022
- **Published in JHEP:** 22 February 2023

3.13.2 Beyond the Standard Model

3.13.2.1 Hybrid Quantum-Classical Graph Convolutional Network [30]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 15 January 2021

3.13.3 Event Classification

3.13.3.1 Event Classification with Quantum Machine Learning in High-Energy Physics [31]

- **Authors:** Koji Terashi, Michiru Kaneda, Tomoe Kishimoto, Masahiko Saito, Ryu Sawada, Junichi Tanaka
- **Posted on arXiv:** 23 February 2020
- **Published in Comput. Softw. Big Sci. 5, 2 (2021):** 03 January 2021

3.13.3.2 Quantum Machine Learning for Particle Physics using a Variational Quantum Classifier [32]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 14 October 2020
- **Published in JHEP:** 2021

3.13.3.3 Application of quantum machine learning using the quantum variational classifier method to high energy physics analysis at the LHC on IBM quantum computer simulator and hardware with 10 qubits [144]

- **Authors:** Sau Lan Wu, Jay Chan, Wen Guan, Shaojun Sun, Alex Wang, Chen Zhou, Miron Livny, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 21 December 2020
- **Published in J.Phys.G:** 26 October 2021

3.13.3.4 Hybrid actor-critic algorithm for quantum reinforcement learning at CERN beam lines [145]

- **Authors:** Michael Schenk, Elías F. Combarro, Michele Grossi, Verena Kain, Kevin Shing Bruce Li, Mircea-Marian Popa, Sofia Vallecorsa
- **Posted on arXiv:** 22 September 2022
- **Published in Quantum Sci.Technol.:** 21 February 2024

3.13.3.5 Fitting a collider in a quantum computer: tackling the challenges of quantum machine learning for big datasets [146]

- **Authors:** Miguel Caçador Peixoto, Nuno Filipe Castro, Miguel Crispim Romão, Maria Gabriela Jordão Oliveira, Inês Ochoa
- **Posted on arXiv:** 06 November 2022
- **Published in Front. Artif. Intell. 6 (2023) 1268852:** 20 November 2023

3.13.3.6 Quantum Metric Learning for New Physics Searches at the LHC [35]

- **Authors:** A. Hammad, Kyoungchul Kong, Myeonghun Park, Soyoung Shim
- **Posted on arXiv:** 28 November 2023

3.13.3.7 Hybrid Quantum Vision Transformers for Event Classification in High Energy Physics [147]

- **Authors:** Eyup B. Unlu, Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhongtian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva
- **Posted on arXiv:** 01 February 2024
- **Published in Axioms v. 13, no 3, (2024) 187:** 13 March 2024

3.13.3.8 Quantum Vision Transformers for Quark–Gluon Classification [148]

- **Authors:** Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhongtian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva, Eyup B. Unlu
- **Posted on arXiv:** 16 May 2024
- **Published in Axioms 2024, 13(5), 323:** 13 May 2024

3.13.4 Event Generation

3.13.4.1 Partonic collinear structure by quantum computing [160]

- **Authors:** Tianyin Li, Xingyu Guo, Wai Kin Lai, Xiaohui Liu, Enke Wang, Hongxi Xing, Dan-Bo Zhang, Shi-Liang Zhu
- **Posted on arXiv:** 07 June 2021
- **Published in Phys.Rev.D:** 01 June 2022

3.13.4.2 Unsupervised quantum circuit learning in high energy physics [161]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton
- **Posted on arXiv:** 07 March 2022
- **Published in Phys.Rev.D:** 01 November 2022

3.13.4.3 Quantum Wishlist: Lessons from Parton Showers [162]

- **Authors:** Masahito Yamazaki
- **Posted on arXiv:** 25 February 2025
- **Published in nan:** 24 October 2025

3.13.5 Jet Reconstruction

3.13.5.1 Quantum Machine Learning for b-jet charge identification [177]

- **Authors:** Alessio Gianelle, Patrick Koppenburg, Donatella Lucchesi, Davide Nicotra, Eduardo Rodrigues, Lorenzo Sestini, Jacco de Vries, Davide Zuliani
- **Posted on arXiv:** 28 February 2022
- **Published in JHEP:** 01 August 2022

3.13.6 Lattice Field Theories - Gauge Theories

3.13.6.1 SU(2) hadrons on a quantum computer via a variational approach [212]

- **Authors:** Yasar Y. Atas, Jinglei Zhang, Randy Lewis, Amin Jahanpour, Jan F. Haase, Christine A. Muschik
- **Posted on arXiv:** 17 February 2021
- **Published in Nature Communications 2021:** 11 November 2021

3.13.7 Lattice Field Theories - Schwinger Model

3.13.7.1 Quantum-classical computation of Schwinger model dynamics using quantum computers [219]

- **Authors:** N. Klco, E.F. Dumitrescu, A.J. McCaskey, T.D. Morris, R.C. Pooser, M. Sanz, E. Solano, P. Lougovski, M.J. Savage
- **Posted on arXiv:** 08 March 2018
- **Published in Phys. Rev. A 98, 032331 (2018):** 29 September 2018

3.13.7.2 Quantum-Classical Simulation of Quantum Field Theory by Quantum Circuit Learning [220]

- **Authors:** Kazuki Ikeda
- **Posted on arXiv:** 27 November 2023
- **Published in Annalen Phys.:** 01 June 2025

3.13.8 Nuclear Many-Body Physics - Nuclear Structure

3.13.8.1 Cloud Quantum Computing of an Atomic Nucleus [127]

- **Authors:** E.F. Dumitrescu, A.J. McCaskey, G. Hagen, G.R. Jansen, T.D. Morris, T. Papenbrock, R.C. Pooser, D.J. Dean, P. Lougovski
- **Posted on arXiv:** 11 January 2018
- **Published in Phys. Rev. Lett. 120, 210501 (2018):** 24 May 2018

3.13.8.2 Improving Hamiltonian encodings with the Gray code [129]

- **Authors:** Olivia Di Matteo, Anna McCoy, Peter Gysbers, Takayuki Miyagi, R.M. Woloshyn, Petr Navrátil
- **Posted on arXiv:** 11 August 2020
- **Published in Phys. Rev. A 103, 042405 (2021):** 03 April 2021

3.13.8.3 Lipkin model on a quantum computer [226]

- **Authors:** Michael J. Cervia, A.B. Balantekin, S.N. Coppersmith, Calvin W. Johnson, Peter J. Love, C. Poole, K. Robbins, M. Saffman
- **Posted on arXiv:** 08 November 2020
- **Published in Phys. Rev. C 104, 024305 (2021):** 03 August 2021

3.13.8.4 Variational approaches to constructing the many-body nuclear ground state for quantum computing [227]

- **Authors:** I. Stetcu, A. Baroni, J. Carlson
- **Posted on arXiv:** 12 October 2021
- **Published in Phys. Rev. C 105, 064308 (2022):** 21 June 2022

3.13.8.5 Simulating excited states of the Lipkin model on a quantum computer [228]

- **Authors:** Manqoba Q. Hlatshwayo, Yinu Zhang, Herlik Wibowo, Ryan LaRose, Denis Lacroix, Elena Litvinova
- **Posted on arXiv:** 02 March 2022
- **Published in Phys.Rev.C:** 18 August 2022

3.13.8.6 Solving nuclear structure problems with the adaptive variational quantum algorithm [229]

- **Authors:** A.M. Romero, J. Engel, Ho Lun Tang, Sophia E. Economou
- **Posted on arXiv:** 03 March 2022
- **Published in Phys. Rev. C 105, 064317 (2022):** 27 June 2022

3.13.8.7 Quantum computing of the ${}^6\text{Li}$ nucleus via ordered unitary coupled cluster [230]

- **Authors:** Oriel Kiss, Michele Grossi, Pavel Lougovski, Federico Sanchez, Sofia Vallecorsa, Thomas Papenbrock
- **Posted on arXiv:** 02 May 2022
- **Published in Phys. Rev. C 106, 034325 (2022):** 29 September 2022

3.13.8.8 QCSH: A full quantum computer nuclear shell-model package [231]

- **Authors:** Peng Lv, Shi-Jie Wei, Hao-Nan Xie, Gui-Lu Long
- **Posted on arXiv:** 24 May 2022
- **Published in Sci.China Phys.Mech.Astron.:** 06 March 2023

3.13.8.9 Subatomic Many-Body Physics Simulations on a Quantum Frequency Processor [131]

- **Authors:** Hsuan-Hao Lu, Natalie Klco, Joseph M. Lukens, Titus D. Morris, Aaina Bansal, Andreas Ekström, Gaute Hagen, Thomas Papenbrock, Andrew M. Weiner, Martin J. Savage, Pavel Lougovski
- **Published in nan:** 2019

3.13.8.10 Nuclear shell-model simulation in digital quantum computers [232]

- **Authors:** A. Pèrez-Obiol, A.M. Romero, J. Menéndez, A. Rios, A. García-Sáez, B. Juliá-Díaz
- **Posted on arXiv:** 07 February 2023
- **Published in Sci.Rep.:** 29 July 2023

3.13.8.11 Deep quantum circuit simulations of low-energy nuclear states [233]

- **Authors:** Ang Li, Alessandro Baroni, Ionel Stetcu, Travis S. Humble
- **Posted on arXiv:** 26 October 2023
- **Published in Eur.Phys.J.A:** 14 May 2024

3.13.8.12 Exploiting symmetries in nuclear Hamiltonians for ground state preparation [234]

- **Authors:** Joe Gibbs, Zoë Holmes, Paul Stevenson
- **Posted on arXiv:** 15 February 2024
- **Published in Quantum Machine Intelligence:** 30 January 2025

3.13.8.13 Comparison of variational quantum eigensolvers in light nuclei [235]

- **Authors:** Miquel Carrasco-Codina, Emanuele Costa, Antonio Márquez Romero, Javier Menéndez, Arnau Rios
- **Posted on arXiv:** 18 July 2025
- **Published in Phys.Rev.C:** 23 February 2026

3.13.8.14 Toward scalable quantum computations of atomic nuclei [236]

- **Authors:** Chenyi Gu, Matthias Heinz, Oriel Kiss, Thomas Papenbrock
- **Posted on arXiv:** 19 July 2025
- **Published in Phys. Rev. C 113, 034321 (2026):** 24 March 2026

3.13.8.15 Bridging quantum computing and nuclear structure: Atomic nuclei on a trapped-ion quantum computer [237]

- **Authors:** Sota Yoshida, Takeshi Sato, Takumi Ogata, Masaaki Kimura
- **Posted on arXiv:** 25 September 2025
- **Published in Phys. Rev. Research 8, 013134 (2026):** 06 February 2026

3.13.9 Nuclear Many-Body Physics - Nuclear Reactions and Response

3.13.9.1 Quantum simulations of nuclear resonances with variational methods [249]

- **Authors:** Ashutosh Singh, Pooja Siwach, P. Arumugam
- **Posted on arXiv:** 15 April 2025
- **Published in Phys.Rev.C:** 25 August 2025

3.13.10 Track Reconstruction

3.13.10.1 Quantum convolutional neural networks for high energy physics data analysis [33]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

3.13.10.2 Hybrid quantum classical graph neural networks for particle track reconstruction [283]

- **Authors:** Cenk Tüysüz, Carla Rieger, Kristiane Novotny, Bilge Demirköz, Daniel Dobos, Karolos Potamianos, Sofia Vallecorsa, Jean-Roch Vlimant, Richard Forster
- **Posted on arXiv:** 26 September 2021
- **Published in Quantum Machine Intelligence:** 01 December 2021

3.13.10.3 Studying quantum algorithms for particle track reconstruction in the LUXE experiment [284]

- **Authors:** Lena Funcke, Tobias Hartung, Beate Heinemann, Karl Jansen, Annabel Kropf, Stefan Kühn, Federico Meloni, David Spataro, Cenk Tüysüz, Yee Chinn Yap
- **Posted on arXiv:** 14 February 2022
- **Published in nan:** 2023

3.13.10.4 Particle track reconstruction with noisy intermediate-scale quantum computers [285]

- **Authors:** Tim Schwägerl, Cigdem Issever, Karl Jansen, Teng Jian Khoo, Stefan Kühn, Cenk Tüysüz, Hannsjörg Weber
- **Posted on arXiv:** 23 March 2023
- **Published in J.Phys.Conf.Ser.:** 2026

3.13.10.5 Charged particle reconstruction for future high energy colliders with Quantum Approximate Optimization Algorithm [286]

- **Authors:** Hideki Okawa
- **Posted on arXiv:** 16 October 2023

3.13.11 Crosslists

3.13.11.1 Simulating Lattice Gauge Theories within Quantum Technologies [193]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

3.13.11.2 Quantum Machine Learning in High Energy Physics [3]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

3.13.11.3 Dual-Parameterized Quantum Circuit GAN Model in High Energy Physics [120]

- **Authors:** Su Yeon Chang, Steven Herbert, Sofia Vallecorsa, Elías F. Combarro, Ross Duncan
- **Posted on arXiv:** 29 March 2021
- **Published in EPJ Web Conf.:** 2021

3.13.11.4 Higgs analysis with quantum classifiers [140]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in nan:** 2021

3.13.11.5 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [13]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borrás, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

3.13.11.6 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.13.11.7 Quantum Computing Applications in Future Colliders [4]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

3.13.11.8 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borrás, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.13.11.9 Precise image generation on current noisy quantum computing devices [122]

- **Authors:** Florian Rehm, Sofia Vallecorsa, Kerstin Borrás, Dirk Krücker, Michele Grossi, Valle Varo
- **Posted on arXiv:** 11 July 2023
- **Published in IOP Quantum Science and Technology (October 2023):** 30 October 2023

3.13.11.10 Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits [214]

- **Authors:** Roland C. Farrell, Marc Illa, Anthony N. Ciavarella, Martin J. Savage
- **Posted on arXiv:** 08 August 2023
- **Published in PRX Quantum** 5 (2024) 2, 020315: 01 April 2024

3.13.11.11 Quantum Algorithms in Particle Physics [164]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

3.13.11.12 Scattering wave packets of hadrons in gauge theories: Preparation on a quantum computer [207]

- **Authors:** Zohreh Davoudi, Chung-Chun Hsieh, Saurabh V. Kadam
- **Posted on arXiv:** 01 February 2024
- **Published in Quantum:** 11 November 2024

3.13.11.13 Steps toward quantum simulations of hadronization and energy loss in dense matter [215]

- **Authors:** Roland C. Farrell, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 10 May 2024
- **Published in Phys.Rev.C** 111 (2025) 1, 015202: 14 January 2025

3.14 Quantum Measurement and Tomography

3.14.1 Nuclear Many-Body Physics - Nuclear Structure

3.14.1.1 Shadow-based quantum subspace algorithm for the nuclear shell model [238]

- **Authors:** Ruyu Yang, Tianren Wang, Bing-Nan Lu, Ying Li, Xiaosi Xu
- **Posted on arXiv:** 15 June 2023
- **Published in Phys.Rev.A:** 21 January 2025

3.15 Quantum Sensors - Atomic/Molecular/Nuclear Sensors

3.15.1 Dark Matter - Particle-like Dark Matter

3.15.1.1 Trapped Electrons and Ions as Particle Detectors [37]

- **Authors:** Daniel Carney, Hartmut Häffner, David C. Moore, Jacob M. Taylor
- **Posted on arXiv:** 12 April 2021
- **Published in Phys Rev Lett, Vol. 127, No. 6, 2021. "Editor's suggestion":** 05 August 2021

3.15.1.2 Millicharged Dark Matter Detection with Ion Traps [38]

- **Authors:** Dmitry Budker, Peter W. Graham, Harikrishnan Ramani, Ferdinand Schmidt-Kaler, Christian Smorra, Stefan Ulmer
- **Posted on arXiv:** 11 August 2021
- **Published in PRX Quantum:** 01 February 2022

3.15.2 Dark Matter - Wave-like Dark Matter

3.15.2.1 Intensity interferometry for ultralight bosonic dark matter detection [53]

- **Authors:** Hector Masia-Roig, Nataniel L. Figueroa, Ariday Bordon, Joseph A. Smiga, Yevgeny V. Stadnik, Dmitry Budker, Gary P. Centers, Alexander V. Gramolin, Paul S. Hamilton, Sami Khamis, Christopher A. Palm, Szymon Pustelny, Alexander O. Sushkov, Arne Wickenbrock, Derek F. Jackson Kimball
- **Posted on arXiv:** 05 February 2022
- **Published in Phys. Rev. D 108, 015003 (2023):** 01 July 2023

3.15.2.2 One-Electron Quantum Cyclotron as a Milli-eV Dark-Photon Detector [54]

- **Authors:** Xing Fan, Gerald Gabrielse, Peter W. Graham, Roni Harnik, Thomas G. Myers, Harikrishnan Ramani, Benedict A.D. Sukra, Samuel S.Y. Wong, Yawen Xiao
- **Posted on arXiv:** 12 August 2022
- **Published in Phys. Rev. Lett. 129, 261801 (2022):** 23 December 2022

3.15.2.3 Detection of bosonovae with quantum sensors on Earth and in space [55]

- **Authors:** Jason Arakawa, Joshua Eby, Marianna S. Safronova, Volodymyr Takhistov, Muhammad H. Zaheer
- **Posted on arXiv:** 28 June 2023
- **Published in Phys.Rev.D:** 01 October 2024

3.15.2.4 Rydberg-atom-based single-photon detection for haloscope axion searches [56]

- **Authors:** Eleanor Graham, Sumita Ghosh, Yuqi Zhu, Xiran Bai, Sidney B. Cahn, Elsa Durcan, Michael J. Jewell, Danielle H. Speller, Sabrina M. Zacarias, Laura T. Zhou, Reina H. Maruyama
- **Posted on arXiv:** 23 October 2023
- **Published in Phys.Rev.D:** 01 February 2024

3.15.2.5 Quantum entanglement of ions for light dark matter detection [57]

- **Authors:** Asuka Ito, Ryuichiro Kitano, Wakutaka Nakano, Ryoto Takai
- **Posted on arXiv:** 20 November 2023
- **Published in JHEP:** 16 February 2024

3.15.2.6 Nuclear interferometer for ultralight dark matter detection [58]

- **Authors:** Hannah Banks, Elina Fuchs, Matthew McCullough
- **Posted on arXiv:** 15 July 2024
- **Published in Phys. Rev. D, 2026:** 01 July 2026

3.15.2.7 Highly excited electron cyclotron for QCD axion and dark-photon detection [59]

- **Authors:** Xing Fan, Gerald Gabrielse, Peter W. Graham, Harikrishnan Ramani, Samuel S.Y. Wong, Yawen Xiao
- **Posted on arXiv:** 07 October 2024
- **Published in Phys.Rev.D 111 (2025) 7, 075022:** 01 April 2025

3.15.2.8 Eliminating Incoherent Noise: A Coherent Quantum Approach in Multi-Sensor Dark Matter Detection [60]

- **Authors:** Jing Shu, Bin Xu, Yuan Xu
- **Posted on arXiv:** 29 October 2024

3.15.2.9 Sensitively searching for microwave dark photons with atomic ensembles [61]

- **Authors:** Suirong He, De He, Yufen Li, Li Gao, Xianing Feng, Hao Zheng, L.F. Wei
- **Posted on arXiv:** 01 December 2024
- **Published in Phys.Rev.D:** 15 September 2025

3.15.2.10 Searches for exotic spin-dependent interactions with spin sensors [62]

- **Authors:** Min Jiang, Haowen Su, Yifan Chen, Man Jiao, Ying Huang, Yuanhong Wang, Xing Rong, Xinhua Peng, Jiangfeng Du
- **Posted on arXiv:** 04 December 2024
- **Published in Rep. Prog. Phys. 88 (2025) 016401:** 13 December 2024

3.15.2.11 Spin squeezing of macroscopic nuclear spin ensembles [63]

- **Authors:** Eric Boyers, Garry Goldstein, Alexander O. Sushkov
- **Posted on arXiv:** 19 February 2025
- **Published in Phys.Rev.D:** 01 March 2025

3.15.2.12 Detecting Dark Matter Using Optically Trapped Rydberg Atom Tweezer Arrays [64]

- **Authors:** So Chigusa, Taiyo Kasamaki, Toshi Kusano, Takeo Moroi, Kazunori Nakayama, Naoya Ozawa, Yoshiro Takahashi, Atsuhiko Umemoto, Amar Vutha
- **Posted on arXiv:** 17 July 2025
- **Published in Phys. Rev. Lett. 136, 151801 (2026):** 14 April 2026

3.15.2.13 On the Speed-up of Wave-like Dark Matter Searches with Entangled Qubits [65]

- **Authors:** Arushi Bodas, Sohriti Ghosh, Roni Harnik
- **Posted on arXiv:** 13 October 2025

3.15.2.14 Symmetric Dicke States as Optimal Probes for Wave-Like Dark Matter [66]

- **Authors:** Ping He, Jing Shu, Bin Xu, Jincheng Xu
- **Posted on arXiv:** 16 December 2025

3.15.3 Detector Technologies and Simulations

3.15.3.1 Quantum Systems for Enhanced High Energy Particle Physics Detectors [123]

- **Authors:** M. Doser, E. Auffray, F.M. Brunbauer, I. Frank, H. Hillemanns, G. Orlandini, G. Kornakov
- **Published in nan:** 24 June 2022

3.15.3.2 Quantum sensing for particle physics [114]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

3.15.4 Neutrino Physics

3.15.4.1 Superradiant interactions for relic detection with entangled nuclear spins [117]

- **Authors:** Marios Galanis, Onur Hosten, Asimina Arvanitaki, Savas Dimopoulos
- **Posted on arXiv:** 28 August 2025
- **Published in Phys.Rev.A:** 10 June 2026

3.15.5 Crosslists

3.15.5.1 Dark Matter Direct Detection with Accelerometers [90]

- **Authors:** Peter W. Graham, David E. Kaplan, Jeremy Mardon, Surjeet Rajendran, William A. Terrano
- **Posted on arXiv:** 18 December 2015
- **Published in Phys.Rev.D:** 20 April 2016

3.15.5.2 Search for New Physics with Atoms and Molecules [2]

- **Authors:** M.S. Safronova, D. Budker, D. DeMille, Derek F. Jackson Kimball, A. Derevianko, C.W. Clark
- **Posted on arXiv:** 04 October 2017
- **Published in Reviews of Modern Physics, 90, 025008 (2018):** 30 June 2018

3.15.5.3 Quantum Sensing for High Energy Physics [8]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik

Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O’Brien, Roger O’Brien, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi

- **Posted on arXiv:** 29 March 2018

3.15.5.4 Simulating Lattice Gauge Theories within Quantum Technologies [193]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

3.15.5.5 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [12]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

3.15.5.6 New Horizons: Scalar and Vector Ultralight Dark Matter [15]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo,

J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O’Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.15.5.7 Quantum Networks for High Energy Physics [16]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

3.15.5.8 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [19]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

3.15.5.9 Quantum Sensors for High Energy Physics [21]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Canelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek

- **Posted on arXiv:** 03 November 2023

3.15.5.10 Superradiant interactions of the cosmic neutrino background, axions, dark matter, and reactor neutrinos [115]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Marios Galanis
- **Posted on arXiv:** 07 August 2024
- **Published in Phys.Rev.D:** 01 March 2025

3.15.5.11 Multimessenger astronomy beyond the Standard Model: New window from quantum sensors [116]

- **Authors:** Jason Arakawa, Muhammad H. Zaheer, Volodymyr Takhistov, Marianna S. Safronova, Joshua Eby, Charles Cheung
- **Posted on arXiv:** 12 February 2025
- **Published in JCAP:** 10 February 2026

3.15.5.12 Quantum-Error-Correction-Inspired Noise Mitigation for Wavelike Dark Matter Searches with Quantum Sensors [88]

- **Authors:** Hajime Fukuda, Takeo Moroi, Thanaporn Sichanugrist
- **Posted on arXiv:** 05 November 2025
- **Published in Phys.Rev.Lett.:** 08 July 2026

3.16 Quantum Sensors - Optical/Photonic Sensors

3.16.1 Dark Matter - Wave-like Dark Matter

3.16.1.1 Analysis of single-photon and linear amplifier detectors for microwave cavity dark matter axion searches [67]

- **Authors:** S.K. Lamoreaux, K.A. van Bibber, K.W. Lehnert, G. Carosi
- **Posted on arXiv:** 15 June 2013
- **Published in Phys.Rev.D:** 23 August 2013

3.16.1.2 Laser Interferometers as Dark Matter Detectors [68]

- **Authors:** Evan D. Hall, Rana X. Adhikari, Valery V. Frolov, Holger Müller, Maxim Pospelov, Rana X Adhikari
- **Posted on arXiv:** 03 May 2016
- **Published in Phys.Rev.D:** 23 October 2018

3.16.1.3 Accelerating dark-matter axion searches with quantum measurement technology [69]

- **Authors:** Huaixiu Zheng, Matti Silveri, R.T. Brierley, S.M. Girvin, K.W. Lehnert
- **Posted on arXiv:** 08 July 2016

3.16.1.4 Squeezed vacuum used to accelerate the search for a weak classical signal [70]

- **Authors:** M. Malnou, D.A. Palken, B.M. Brubaker, Leila R. Vale, Gene C. Hilton, K.W. Lehnert
- **Posted on arXiv:** 17 September 2018
- **Published in Phys.Rev.X:** 04 May 2019

3.16.1.5 A quantum-enhanced search for dark matter axions [71]

- **Authors:** K.M. Backes, D.A. Palken, S. Al Kenany, B.M. Brubaker, S.B. Cahn, A. Droster, Gene C. Hilton, Sumita Ghosh, H. Jackson, S.K. Lamoreaux, A.F. Leder, K.W. Lehnert, S.M. Lewis, M. Malnou, R.H. Maruyama, N.M. Rapidis, M. Simanovskaia, Sukhman Singh, D.H. Speller, I. Urdinaran, Leila R. Vale, E.C. van Assendelft, K. van Bibber, H. Wang
- **Posted on arXiv:** 04 August 2020
- **Published in Nature:** 10 February 2021

3.16.1.6 Searching for Dark Matter with a Superconducting Qubit [72]

- **Authors:** Akash V. Dixit, Srivatsan Chakram, Kevin He, Ankur Agrawal, Ravi K. Naik, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 28 August 2020
- **Published in Phys. Rev. Lett. 126, 141302 (2021):** 09 April 2021

3.16.1.7 Cavity Entanglement and State Swapping to Accelerate the Search for Axion Dark Matter [73]

- **Authors:** K. Wurtz, B.M. Brubaker, Y. Jiang, E.P. Ruddy, D.A. Palken, K.W. Lehnert
- **Posted on arXiv:** 08 July 2021
- **Published in PRX Quantum:** 10 December 2021

3.16.1.8 Entangled Sensor-Networks for Dark-Matter Searches [74]

- **Authors:** Anthony J. Brady, Christina Gao, Roni Harnik, Zhen Liu, Zheshen Zhang, Quntao Zhuang
- **Posted on arXiv:** 10 March 2022
- **Published in PRX Quantum 3, 030333 (2022):** 01 September 2022

3.16.1.9 Detecting Hidden Photon Dark Matter Using the Direct Excitation of Transmon Qubits [75]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 07 December 2022
- **Published in Phys.Rev.Lett.:** 22 November 2023

3.16.1.10 New results from HAYSTAC’s phase II operation with a squeezed state receiver [76]

- **Authors:** M.J. Jewell, A.F. Leder, K.M. Backes, Xiran Bai, K. van Bibber, B.M. Brubaker, S.B. Cahn, A. Droster, Maryam H. Esmat, Sumita Ghosh, Eleanor Graham, Gene C. Hilton, H. Jackson, Claire Laffan, S.K. Lamoreaux, K.W. Lehnert, S.M. Lewis, M. Malnou, R.H. Maruyama, D.A. Palken, N.M. Rapidis, E.P. Ruddy, M. Simanovskaia, Sukhman Singh, D.H. Speller, Leila R. Vale, H. Wang, Yuqi Zhu
- **Posted on arXiv:** 23 January 2023
- **Published in Phys.Rev.D:** 01 April 2023

3.16.1.11 Stimulated Emission of Signal Photons from Dark Matter Waves [77]

- **Authors:** Ankur Agrawal, Akash V. Dixit, Tanay Roy, Srivatsan Chakram, Kevin He, Ravi K. Naik, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 05 May 2023
- **Published in Phys. Rev. Lett. 132, 140801 (2024):** 03 April 2024

3.16.1.12 Quantum measurements in fundamental physics: a user’s manual [78]

- **Authors:** Jacob Beckey, Daniel Carney, Giacomo Marocco
- **Posted on arXiv:** 13 November 2023
- **Published in SciPost Phys. Rev. 2 (2026):** 2026

3.16.1.13 Quantum Enhancement in Dark Matter Detection with Quantum Computation [79]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 17 November 2023
- **Published in Phys.Rev.Lett.:** 08 July 2024

3.16.1.14 Quantum-Enhanced Sensing of Axion Dark Matter with a Transmon-Based Single Microwave Photon Counter [80]

- **Authors:** C. Braggio, L. Balembois, R. Di Vora, Z. Wang, J. Travesedo, L. Pallegoix, G. Carugno, A. Ortolan, G. Ruoso, U. Gambardella, D. D’Agostino, P. Bertet, E. Flurin
- **Posted on arXiv:** 04 March 2024
- **Published in Phys.Rev.X:** 01 April 2025

3.16.1.15 Search for QCD axion dark matter with transmon qubits and quantum circuit [81]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 29 July 2024
- **Published in Phys.Rev.D:** 01 December 2024

3.16.1.16 Axion bounds from quantum technology [82]

- **Authors:** Martin Bauer, Sreemanti Chakraborti, Guillaume Rostagni
- **Posted on arXiv:** 12 August 2024
- **Published in JHEP:** 05 May 2025

3.16.1.17 Quantum-enhanced dark matter detection with in-cavity control: mitigating the Rayleigh curse [83]

- **Authors:** Haowei Shi, Anthony J. Brady, Wojciech Górecki, Lorenzo Maccone, Roberto Di Candia, Quntao Zhuang
- **Posted on arXiv:** 06 September 2024
- **Published in npj Quantum Information 11, 48 (2025):** 17 March 2025

3.16.1.18 Transmon qubit modeling and characterization for Dark Matter search [84]

- **Authors:** Roberto Moretti, Danilo Labranca, Pietro Campana, Rodolfo Carobene, Marco Gobbo, Manuel A. Castellanos-Beltran, David Olaya, Peter F. Hopkins, Leonardo Banchi, Matteo Borghesi, Alessandro Candido, Stefano Carrazza, Hervé Atsè Corti, Alessandro D’Elia, Marco Faverzani, Elena Ferri, Angelo Nucciotti, Luca Origo, Andrea Pasquale, Alex Stephane Piedjou Komnang, Alessio Rettaroli, Simone Tocci, Claudio Gatti, Andrea Giachero
- **Posted on arXiv:** 09 September 2024
- **Published in IEEE Trans.Quantum Eng.:** 2026

3.16.1.19 Dark Matter Axion Search with HAYSTAC Phase II [85]

- **Authors:** Xiran Bai, M.J. Jewell, J. Echevers, K. van Bibber, A. Droster, Maryam H. Esmat, Sumita Ghosh, Eleanor Graham, H. Jackson, Claire Laffan, S.K. Lamoreaux, A.F. Leder, K.W. Lehnert, S.M. Lewis, R.H. Maruyama, R.D. Nath, N.M. Rapidis, E.P. Ruddy, M. Silva-Feaver, M. Simanovskaia, Sukhman Singh, D.H. Speller, Sabrina Zacarias, Yuqi Zhu
- **Posted on arXiv:** 13 September 2024
- **Published in Phys. Rev. Lett. 134, 151006 (2025):** 18 April 2025

3.16.1.20 Flux-Tunable Cavity for Dark Matter Detection [86]

- **Authors:** Fang Zhao, Ziqian Li, Akash V. Dixit, Tanay Roy, Andrei Vrajitoarea, Riju Banerjee, Alexander Anferov, Kan-Heng Lee, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 12 January 2025
- **Published in Phys.Rev.Lett.:** 13 November 2025

3.16.1.21 Quantum Enhanced Dark-Matter Search with Entangled Fock States in High-Quality Cavities [87]

- **Authors:** Benjamin Freiman, Xinyuan You, Andy C.Y. Li, Raphael Cervantes, Taeyoon Kim, Anna Grasselino, Roni Harnik, Yao Lu
- **Posted on arXiv:** 30 October 2025

3.16.1.22 Quantum-Error-Correction-Inspired Noise Mitigation for Wavelike Dark Matter Searches with Quantum Sensors [88]

- **Authors:** Hajime Fukuda, Takeo Moroi, Thanaporn Sichanugrist
- **Posted on arXiv:** 05 November 2025
- **Published in Phys.Rev.Lett.:** 08 July 2026

3.16.2 Gravitational Waves

3.16.2.1 Quantum sensor networks as exotic field telescopes for multi-messenger astronomy [112]

- **Authors:** Conner Dailey, Colin Bradley, Derek F. Jackson Kimball, Ibrahim A. Sulai, Szymon Pustelny, Arne Wickenbrock, Andrei Derevianko
- **Posted on arXiv:** 11 February 2020
- **Published in Nature Astron.:** 02 November 2020

3.16.2.2 Electromagnetic cavities as mechanical bars for gravitational waves [113]

- **Authors:** Asher Berlin, Diego Blas, Raffaele Tito D’Agnolo, Sebastian A.R. Ellis, Roni Harnik, Yonatan Kahn, Jan Schütte-Engel, Michael Wentzel
- **Posted on arXiv:** 02 March 2023
- **Published in Phys.Rev.D:** 15 October 2023

3.16.2.3 Multimessenger astronomy beyond the Standard Model: New window from quantum sensors [116]

- **Authors:** Jason Arakawa, Muhammad H. Zaheer, Volodymyr Takhistov, Marianna S. Safronova, Joshua Eby, Charles Cheung
- **Posted on arXiv:** 12 February 2025
- **Published in JCAP:** 10 February 2026

3.16.3 Neutrino Physics

3.16.3.1 Superradiant interactions of the cosmic neutrino background, axions, dark matter, and reactor neutrinos [115]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Marios Galanis
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3.16.4 Crosslists

3.16.4.1 Quantum Sensing for High Energy Physics [8]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei

Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O’Brien, Roger O’Brien, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi

- **Posted on arXiv:** 29 March 2018

3.16.4.2 Simulating Lattice Gauge Theories within Quantum Technologies [193]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller

- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

3.16.4.3 Intensity interferometry for ultralight bosonic dark matter detection [53]

- **Authors:** Hector Masia-Roig, Nataniel L. Figueroa, Ariday Bordon, Joseph A. Smiga, Yevgeny V. Stadnik, Dmitry Budker, Gary P. Centers, Alexander V. Gramolin, Paul S. Hamilton, Sami Khamis, Christopher A. Palm, Szymon Pustelny, Alexander O. Sushkov, Arne Wickenbrock, Derek F. Jackson Kimball

- **Posted on arXiv:** 05 February 2022
- **Published in Phys. Rev. D 108, 015003 (2023):** 01 July 2023

3.16.4.4 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [12]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

3.16.4.5 New Horizons: Scalar and Vector Ultralight Dark Matter [15]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibbrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O’Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.16.4.6 Quantum Networks for High Energy Physics [16]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie
- **Posted on arXiv:** 31 March 2022

3.16.4.7 Quantum computing hardware for HEP algorithms and sensing [18]

- **Authors:** M. Sohaib Alam, Sergey Belomestnykh, Nicholas Bornman, Gustavo Canelo, Yu-Chiu Chao, Mattia Checchin, Vinh San Dinh, Anna Grassellino, Erik J. Gustafson, Roni Harnik, Corey Rae Harrington McRae, Ziwen Huang, Keshav Kapoor, Taeyoon Kim, James B. Kowalkowski, Matthew J. Kramer, Yulia Krasnikova, Prem Kumar, Doga Murat Kurkcuoglu, Henry Lamm, Adam L. Lyon, Despina Milathianaki, Akshay Murthy, Josh Mutus, Ivan Nekrashevich, JinSu Oh, A. BarışÖzgüler, Gabriel Nathan Perdue, Matthew Reagor, Alexander Romanenko, James A. Sauls, Leandro Stefanazzi, Norm M. Tubman, Davide Venturelli, Changqing Wang, Xinyuan You, David M.T. van Zanten, Lin Zhou, Shaojiang Zhu, Silvia Zorzetti
- **Posted on arXiv:** 18 April 2022

3.16.4.8 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [19]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

3.16.4.9 Quantum sensing for particle physics [114]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

3.16.4.10 Detection of bosonovae with quantum sensors on Earth and in space [55]

- **Authors:** Jason Arakawa, Joshua Eby, Marianna S. Safronova, Volodymyr Takhistov, Muhammad H. Zaheer
- **Posted on arXiv:** 28 June 2023
- **Published in Phys.Rev.D:** 01 October 2024

3.16.4.11 Rydberg-atom-based single-photon detection for haloscope axion searches [56]

- **Authors:** Eleanor Graham, Sumita Ghosh, Yuqi Zhu, Xiran Bai, Sidney B. Cahn, Elsa Durcan, Michael J. Jewell, Danielle H. Speller, Sabrina M. Zacarias, Laura T. Zhou, Reina H. Maruyama
- **Posted on arXiv:** 23 October 2023
- **Published in Phys.Rev.D:** 01 February 2024

3.16.4.12 Quantum Sensors for High Energy Physics [21]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibbrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

3.16.4.13 Eliminating Incoherent Noise: A Coherent Quantum Approach in Multi-Sensor Dark Matter Detection [60]

- **Authors:** Jing Shu, Bin Xu, Yuan Xu
- **Posted on arXiv:** 29 October 2024

3.16.4.14 Symmetric Dicke States as Optimal Probes for Wave-Like Dark Matter [66]

- **Authors:** Ping He, Jing Shu, Bin Xu, Jincheng Xu
- **Posted on arXiv:** 16 December 2025

3.17 Quantum Sensors - Optomechanical Sensors

3.17.1 Dark Matter - Particle-like Dark Matter

3.17.1.1 Search for Kilogram-scale Dark Matter with Precision Displacement Sensors [39]

- **Authors:** Akio Kawasaki
- **Posted on arXiv:** 26 August 2018
- **Published in Phys. Rev. D 99, 023005 (2019):** 07 January 2019

3.17.1.2 Proposal for gravitational direct detection of dark matter [40]

- **Authors:** Daniel Carney, Sohriti Ghosh, Gordan Krnjaic, Jacob M. Taylor
- **Posted on arXiv:** 01 March 2019
- **Published in Phys.Rev.D:** 14 October 2020

3.17.1.3 Search for composite dark matter with optically levitated sensors [41]

- **Authors:** Fernando Monteiro, Gadi Afek, Daniel Carney, Gordan Krnjaic, Jiaxiang Wang, David C. Moore
- **Posted on arXiv:** 23 July 2020
- **Published in Phys. Rev. Lett. 125, 181102 (2020):** 29 October 2020

3.17.1.4 Coherent Scattering of Low Mass Dark Matter from Optically Trapped Sensors [42]

- **Authors:** Gadi Afek, Daniel Carney, David C. Moore
- **Posted on arXiv:** 05 November 2021
- **Published in Phys.Rev.Lett.:** 09 March 2022

3.17.1.5 Models of ultraheavy dark matter visible to macroscopic mechanical sensing arrays [43]

- **Authors:** Carlos Blanco, Bahaa Elshimy, Rafael F. Lang, Robert Orlando
- **Posted on arXiv:** 29 December 2021
- **Published in Phys.Rev.D:** 01 June 2022

3.17.1.6 Optomechanical dark matter instrument for direct detection [44]

- **Authors:** Christopher G. Baker, Warwick P. Bowen, Peter Cox, Matthew J. Dolan, Maxim Goryachev, Glen Harris
- **Posted on arXiv:** 16 June 2023
- **Published in Phys. Rev. D 110, 043005 (2024):** 02 August 2024

3.17.1.7 Probing levitodynamics with multi-stochastic forces and the simple applications on the dark matter detection in optical levitation experiment [45]

- **Authors:** Xi Cheng, Ji-Heng Guo, Wenyu Wang, Bin Zhu
- **Posted on arXiv:** 07 December 2023
- **Published in Nucl. Phys. B 1010 (2025) 116780:** 19 December 2024

3.17.1.8 Dark Matter Searches with Levitated Sensors [46]

- **Authors:** Eva Kilian, Markus Rademacher, Jonathan M.H. Gosling, Julian H. Iacoponi, Fiona Alder, Marko Toroš, Antonio Pontin, Chamkaur Ghag, Sougato Bose, Tania S. Monteiro, P.F. Barker
- **Posted on arXiv:** 31 January 2024
- **Published in AVS Quantum Sci. 6, 030503 (2024):** 2024

3.17.1.9 Probing light particles with optically trapped sensors through nucleon scattering [47]

- **Authors:** Bhaskar Dutta, Dilip Kumar Ghosh, Sk Jeesun
- **Posted on arXiv:** 31 January 2025
- **Published in Phys.Rev.D:** 01 January 2026

3.17.1.10 Mechanical sensors for ultraheavy dark matter searches via long-range forces [48]

- **Authors:** Juehang Qin, Dorian W.P. Amaral, Sunil A. Bhave, Erqian Cai, Daniel Carney, Rafael F. Lang, Shengchao Li, Alberto M. Marino, Giacomo Marocco, Claire Marvinnay, Jared R. Newton, Jacob M. Taylor, Christopher Tunnell
- **Posted on arXiv:** 14 March 2025
- **Published in Phys.Rev.D 112 (2025) 7, 072003:** 01 October 2025

3.17.2 Dark Matter - Wave-like Dark Matter

3.17.2.1 Sound of Dark Matter: Searching for Light Scalars with Resonant-Mass Detectors [89]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Ken Van Tilburg
- **Posted on arXiv:** 07 August 2015
- **Published in Phys.Rev.Lett.:** 22 January 2016

3.17.2.2 Dark Matter Direct Detection with Accelerometers [90]

- **Authors:** Peter W. Graham, David E. Kaplan, Jeremy Mardon, Surjeet Rajendran, William A. Terrano
- **Posted on arXiv:** 18 December 2015
- **Published in Phys.Rev.D:** 20 April 2016

3.17.2.3 Ultralight dark matter detection with mechanical quantum sensors [91]

- **Authors:** Daniel Carney, Anson Hook, Zhen Liu, Jacob M. Taylor, Yue Zhao
- **Posted on arXiv:** 13 August 2019
- **Published in New Journal of Physics, Volume 23, February 2021:** 02 March 2021

3.17.2.4 Searching for scalar dark matter with compact mechanical resonators [92]

- **Authors:** Jack Manley, Russell Stump, Dalziel Wilson, Daniel Grin, Swati Singh
- **Posted on arXiv:** 16 October 2019
- **Published in Phys. Rev. Lett. 124, 151301 (2020):** 17 April 2020

3.17.2.5 Ultrasensitive optomechanical detection of an axion-mediated force based on a sharp peak emerging in probe absorption spectrum [93]

- **Authors:** Lei Chen, Jian Liu, Kadi Zhu
- **Posted on arXiv:** 28 November 2019
- **Published in Opt.Communic.: 15 January 2021**

3.17.2.6 Searching for vector dark matter with an optomechanical accelerometer [94]

- **Authors:** Jack Manley, Mitul Dey Chowdhury, Daniel Grin, Swati Singh, Dalziel J. Wilson
- **Posted on arXiv:** 09 July 2020
- **Published in Phys. Rev. Lett. 126, 061301 (2021):** 11 February 2021

3.17.2.7 Entanglement-enhanced optomechanical sensor array with application to dark matter searches [95]

- **Authors:** Anthony J. Brady, Xin Chen, Kewen Xiao, Yi Xia, Jack Manley, Mitul Dey Chowdhury, Zhen Liu, Roni Harnik, Dalziel J. Wilson, Zheshen Zhang, Quntao Zhuang
- **Posted on arXiv:** 13 October 2022
- **Published in Commun.Phys.: 01 September 2023**

3.17.2.8 Axion detection with optomechanical cavities [96]

- **Authors:** Clara Murgui, Yikun Wang, Kathryn M. Zurek
- **Posted on arXiv:** 15 November 2022
- **Published in Phys.Lett.B: 27 June 2024**

3.17.2.9 Superfluid helium ultralight dark matter detector [97]

- **Authors:** M. Hirschel, V. Vadakkumbatt, N.P. Baker, F.M. Schweizer, J.C. Sankey, S. Singh, J.P. Davis
- **Posted on arXiv:** 14 September 2023
- **Published in Physical Review D 109, 095011 (2024):** 01 May 2024

3.17.2.10 Vector wave dark matter and terrestrial quantum sensors [98]

- **Authors:** Dorian W.P. Amaral, Mudit Jain, Mustafa A. Amin, Christopher Tunnell
- **Posted on arXiv:** 04 March 2024
- **Published in JCAP:** 21 June 2024

3.17.3 Gravitational Waves

3.17.3.1 Quantum interactions between a laser interferometer and gravitational waves [165]

- **Authors:** Belinda Pang, Yanbei Chen
- **Posted on arXiv:** 28 August 2018
- **Published in Phys. Rev. D 98, 124006 (2018):** 11 December 2018

3.17.3.2 Superconducting Levitated Detector of Gravitational Waves [166]

- **Authors:** Daniel Carney, Gerard Higgins, Giacomo Marocco, Michael Wentzel
- **Posted on arXiv:** 02 August 2024
- **Published in Carney (2025) Phys. Rev. Lett. 134, 181402:** 06 May 2025

3.17.4 Neutrino Physics

3.17.4.1 Requirements on quantum superpositions of macro-objects for sensing neutrinos [221]

- **Authors:** Eva Kilian, Marko Toroš, Frank F. Deppisch, Ruben Saakyan, Sougato Bose
- **Posted on arXiv:** 27 April 2022
- **Published in Phys. Rev. Research 5, 023012 (2023):** 07 April 2023

3.17.4.2 Searches for Massive Neutrinos with Mechanical Quantum Sensors [222]

- **Authors:** Daniel Carney, Kyle G. Leach, David C. Moore
- **Posted on arXiv:** 12 July 2022
- **Published in Phys.Rev.X:** 01 February 2023

3.17.5 Crosslists

3.17.5.1 Quantum Sensing for High Energy Physics [8]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowering, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi
- **Posted on arXiv:** 29 March 2018

3.17.5.2 Mechanical quantum sensing in the search for dark matter [9]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao
- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol.** 6 024002, 2021 (invited): 18 January 2021

3.17.5.3 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [12]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

3.17.5.4 New Horizons: Scalar and Vector Ultralight Dark Matter [15]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibbrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O’Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.17.5.5 Quantum Networks for High Energy Physics [16]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie
- **Posted on arXiv:** 31 March 2022

3.17.5.6 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [19]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

3.17.5.7 Quantum sensing for particle physics [114]

- **Authors:** Steven D. Bass, Michael Doser
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- **Published in Nature Rev.Phys.:** 16 April 2024

3.17.5.8 Quantum Sensors for High Energy Physics [21]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

3.17.5.9 Quantum measurements in fundamental physics: a user’s manual [78]

- **Authors:** Jacob Beckey, Daniel Carney, Giacomo Marocco
- **Posted on arXiv:** 13 November 2023
- **Published in SciPost Phys. Rev. 2 (2026):** 2026

3.17.5.10 Symmetric Dicke States as Optimal Probes for Wave-Like Dark Matter [66]

- **Authors:** Ping He, Jing Shu, Bin Xu, Jincheng Xu
- **Posted on arXiv:** 16 December 2025

3.18 Quantum Sensors - Solid State Sensors

3.18.1 Dark Matter - Particle-like Dark Matter

3.18.1.1 Dark matter direct detection with quantum dots [49]

- **Authors:** Carlos Blanco, Rouven Essig, Marivi Fernandez-Serra, Harikrishnan Ramani, Oren Slone
- **Posted on arXiv:** 10 August 2022
- **Published in Phys.Rev.D:** 01 May 2023

3.18.1.2 Dark Matter Induced Power in Quantum Devices [50]

- **Authors:** Anirban Das, Noah Kurinsky, Rebecca K. Leane
- **Posted on arXiv:** 17 October 2022
- **Published in Phys. Rev. Lett. 132, 121801 (2024):** 22 March 2024

3.18.1.3 Transmon Qubit constraints on dark matter-nucleon scattering [51]

- **Authors:** Anirban Das, Noah Kurinsky, Rebecca K. Leane
- **Posted on arXiv:** 30 April 2024
- **Published in JHEP:** 25 July 2024

3.18.1.4 Quantum parity detectors: A qubit-based particle-detection scheme with meV thresholds for rare-event searches [52]

- **Authors:** Karthik Ramanathan, Brandon J. Sandoval, John E. Parker, Lalit M. Joshi, Andrew D. Beyer, Pierre M. Echternach, Serge Rosenblum, Sunil R. Golwala
- **Posted on arXiv:** 27 May 2024
- **Published in APS Open Science:** 01 May 2026

3.18.2 Dark Matter - Wave-like Dark Matter

3.18.2.1 Searching for an exotic spin-dependent interaction with a single electron-spin quantum sensor [99]

- **Authors:** Xing Rong, Mengqi Wang, Jianpei Geng, Xi Qin, Maosen Guo, Man Jiao, Yijin Xie, Pengfei Wang, Pu Huang, Fazhan Shi, Yi-Fu Cai, Chongwen Zou, Jiangfeng Du
- **Posted on arXiv:** 12 June 2017
- **Published in Nature Commun.:** 21 February 2018

3.18.2.2 Axion search with quantum nondemolition detection of magnons [100]

- **Authors:** Tomonori Ikeda, Asuka Ito, Kentaro Miuchi, Jiro Soda, Hisaya Kurashige, Yutaka Shikano
- **Posted on arXiv:** 17 February 2021
- **Published in Phys. Rev. D** **105**, 102004 (2022): 15 May 2022

3.18.2.3 Proposal for the search for new spin interactions at the micrometer scale using diamond quantum sensors [101]

- **Authors:** P.-H. Chu, N. Ristoff, J. Smits, N. Jackson, Y.J. Kim, I. Savukov, V.M. Acosta
- **Posted on arXiv:** 29 December 2021
- **Published in Physical Review Research** **4**, 023162 (2022): 31 May 2022

3.18.2.4 Ultimate precision limit of noise sensing and dark matter search [102]

- **Authors:** Haowei Shi, Quntao Zhuang
- **Posted on arXiv:** 29 August 2022
- **Published in npj Quantum Inf.:** 20 March 2023

3.18.2.5 Light dark matter search with nitrogen-vacancy centers in diamonds [103]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 24 February 2023
- **Published in JHEP:** 12 March 2025

3.18.2.6 Axion detection via superfluid ^3He ferromagnetic phase and quantum measurement techniques [104]

- **Authors:** So Chigusa, Dan Kondo, Hitoshi Murayama, Risshin Okabe, Hiroyuki Sudo
- **Posted on arXiv:** 17 September 2023
- **Published in JHEP:** 26 September 2024

3.18.2.7 Nuclear spin metrology with nitrogen vacancy center in diamond for axion dark matter detection [105]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Yuichiro Matsuzaki, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 09 July 2024
- **Published in Phys. Rev. D** **111** (2025) 075028: 01 April 2025

3.18.2.8 Listening for new physics with quantum acoustics [106]

- **Authors:** Ryan Linehan, Tanner Trickle, Christopher R. Conner, Sohriti Ghosh, Tongyan Lin, Mukul Sholapurkar, Andrew N. Cleland
- **Posted on arXiv:** 22 October 2024
- **Published in Phys.Rev.D:** 01 December 2025

3.18.2.9 Entanglement-enhanced ac magnetometry in the presence of Markovian noise [107]

- **Authors:** Thanaporn Sichanugrist, Hajime Fukuda, Takeo Moroi, Kazunori Nakayama, So Chigusa, Norikazu Mizuochi, Masashi Hazumi, Yuichiro Matsuzaki
- **Posted on arXiv:** 28 October 2024
- **Published in Phys.Rev.A:** 07 April 2025

3.18.2.10 Probing Sub-eV Dark Photon, Scalar and Axion-like Particle Dark Matters with Transmon Qubits [108]

- **Authors:** Wei Chao, Yu Gao, Ming-Jie Jin, Xiao-Sheng Liu, Xi-Lei Sun
- **Posted on arXiv:** 30 December 2024
- **Published in Phys.Dark Univ. 50 (2025) 102171:** 15 November 2025

3.18.2.11 Piezoelectric bulk acoustic resonators for dark photon detection [109]

- **Authors:** Tanner Trickle
- **Posted on arXiv:** 09 January 2025
- **Published in Phys.Rev.D:** 01 September 2025

3.18.2.12 Background suppression in quantum sensing of dark matter via collective entangled-state projection [110]

- **Authors:** Shion Chen, Hajime Fukuda, Yutaro Iiyama, Yuya Mino, Takeo Moroi, Mikio Nakahara, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 02 October 2025
- **Published in Phys.Rev.D:** 01 March 2026

3.18.2.13 Hybrid-spin decoupling for noise-resilient DC quantum sensing [111]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Yuichiro Matsuzaki, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 20 November 2025

3.18.3 Crosslists

3.18.3.1 Quantum Sensing for High Energy Physics [8]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowering, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi

- **Posted on arXiv:** 29 March 2018

3.18.3.2 Ultrasensitive optomechanical detection of an axion-mediated force based on a sharp peak emerging in probe absorption spectrum [93]

- **Authors:** Lei Chen, Jian Liu, Kadi Zhu
- **Posted on arXiv:** 28 November 2019
- **Published in Opt.Commun.:** 15 January 2021

3.18.3.3 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [12]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova

- **Posted on arXiv:** 14 March 2022

3.18.3.4 New Horizons: Scalar and Vector Ultralight Dark Matter [15]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibbrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.18.3.5 Quantum Networks for High Energy Physics [16]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

3.18.3.6 Quantum Systems for Enhanced High Energy Particle Physics Detectors [123]

- **Authors:** M. Doser, E. Auffray, F.M. Brunbauer, I. Frank, H. Hillemanns, G. Orlandini, G. Kornakov
- **Published in nan:** 24 June 2022

3.18.3.7 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [19]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

3.18.3.8 Quantum sensing for particle physics [114]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

3.18.3.9 Quantum Sensors for High Energy Physics [21]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

3.18.3.10 Quantum measurements in fundamental physics: a user’s manual [78]

- **Authors:** Jacob Beckey, Daniel Carney, Giacomo Marocco
- **Posted on arXiv:** 13 November 2023
- **Published in SciPost Phys. Rev. 2 (2026):** 2026

3.18.3.11 Searches for exotic spin-dependent interactions with spin sensors [62]

- **Authors:** Min Jiang, Haowen Su, Yifan Chen, Man Jiao, Ying Huang, Yuanhong Wang, Xing Rong, Xinhua Peng, Jiangfeng Du
- **Posted on arXiv:** 04 December 2024
- **Published in Rep. Prog. Phys. 88 (2025) 016401:** 13 December 2024

3.18.3.12 Spin squeezing of macroscopic nuclear spin ensembles [63]

- **Authors:** Eric Boyers, Garry Goldstein, Alexander O. Sushkov
- **Posted on arXiv:** 19 February 2025
- **Published in Phys.Rev.D:** 01 March 2025

3.18.3.13 Symmetric Dicke States as Optimal Probes for Wave-Like Dark Matter [66]

- **Authors:** Ping He, Jing Shu, Bin Xu, Jincheng Xu
- **Posted on arXiv:** 16 December 2025

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